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United States Patent Application Publication

20250283886

Kind Code

A1

Publication Date

September 11, 2025

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METHODS AND COMPOSITIONS FOR PROTEIN SEQUENCING

Abstract

Aspects of the application provide methods of identifying and sequencing proteins, polypeptides, and amino acids, and compositions useful for the same. In some aspects, the application provides methods of obtaining data during a degradation process of a polypeptide, and outputting a sequence representative of the polypeptide. In some aspects, the application provides amino acid recognition molecules comprising a shielding element that enhances photostability in polypeptide sequencing reactions.

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Family ID: 68841224

Appl. No.: 18/955622

Filed: November 21, 2024

Related U.S. Application Data

parent US continuation 18598736 20240307 parent-grant-document US 12174196 child US 18955622

parent US continuation 16708989 20191210 parent-grant-document US 11959920 child US 18598736

parent US continuation 16686028 20191115 PENDING child US 16708989

us-provisional-application US 62768076 20181115

us-provisional-application US 62907507 20190927

Publication Classification

Int. Cl.: G01N33/58 (20060101); **C07K14/47** (20060101); **C07K19/00** (20060101); **C12Q1/6806** (20180101); **G01N1/28** (20060101); **G01N21/64** (20060101); **G01N33/68** (20060101); **G16B25/10** (20190101); **G16B30/00** (20190101); **G16B40/00** (20190101); **G16B40/10** (20190101); **G16B50/00** (20190101); **G16B50/30** (20190101)

U.S. Cl.:

CPC G01N33/581 (20130101); **C07K14/47** (20130101); **C07K19/00** (20130101); **C12Q1/6806** (20130101); **G01N1/28** (20130101); **G01N21/6428** (20130101); **G01N33/58** (20130101); **G01N33/582** (20130101); **G01N33/6821** (20130101); **G01N33/6824** (20130101); **G16B25/10** (20190201); **G16B40/00** (20190201); **G16B40/10** (20190201); **G16B50/30** (20190201); G01N2021/6439 (20130101); G01N2458/00 (20130101); G16B30/00 (20190201); G16B50/00 (20190201)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 62/907,507, filed Sep. 27, 2019, and U.S. Provisional Patent Application No. 62/768,076, filed Nov. 15, 2018, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] Proteomics has emerged as an important and necessary complement to genomics and transcriptomics in the study of biological systems. The proteomic analysis of an individual organism can provide insights into cellular processes and response patterns, which lead to improved diagnostic and therapeutic strategies. The complexity surrounding protein structure, composition, and modification present challenges in determining large-scale protein sequencing information for a biological sample.

SUMMARY

[0003] In some aspects, the application provides methods and compositions for determining amino acid sequence information from polypeptides (e.g., for sequencing one or more polypeptides). In some embodiments, amino acid sequence information can be determined for single polypeptide molecules. In some embodiments, the relative position of two or more amino acids in a polypeptide is determined, for example for a single polypeptide molecule. In some embodiments, one or more amino acids of a polypeptide are labeled (e.g., directly or indirectly) and the relative positions of the labeled amino acids in the polypeptide is determined.

[0004] In some aspects, the application provides methods comprising obtaining data during a degradation process of a polypeptide. In some embodiments, the methods further comprise analyzing the data to determine portions of the data corresponding to amino acids that are sequentially exposed at a terminus of the polypeptide during the degradation process. In some embodiments, the methods further comprise outputting an amino acid sequence representative of the polypeptide. In some embodiments, the data is indicative of amino acid identity at the terminus of the polypeptide during the degradation process. In some embodiments, the data is indicative of a signal produced by one or more amino acid recognition molecules binding to different types of terminal amino acids at the terminus during the degradation process. In some embodiments, the data is indicative of a luminescent signal generated during the degradation process. In some embodiments, the data is indicative of an electrical signal generated during the degradation process.

[0005] In some embodiments, analyzing the data further comprises detecting a series of cleavage events and determining the portions of the data between successive cleavage events. In some

embodiments, analyzing the data further comprises determining a type of amino acid for each of the individual portions. In some embodiments, each of the individual portions comprises a pulse pattern (e.g., a characteristic pattern), and analyzing the data further comprises determining a type of amino acid for one or more of the portions based on its respective pulse pattern. In some embodiments, determining the type of amino acid further comprises identifying an amount of time within a portion when the data is above a threshold value and comparing the amount of time to a duration of time for the portion. In some embodiments, determining the type of amino acid further comprises identifying at least one pulse duration for each of the one or more portions. In some embodiments, determining the type of amino acid further comprises identifying at least one interpulse duration for each of the one or more portions. In some embodiments, the amino acid sequence includes a series of amino acids corresponding to the portions.

[0006] In some aspects, the application provides systems comprising at least one hardware processor, and at least one non-transitory computer-readable storage medium storing processor-executable instructions that, when executed by the at least one hardware processor, cause the at least one hardware processor to perform a method in accordance with the application. In some aspects, the application provides at least one non-transitory computer-readable storage medium storing processor-executable instructions that, when executed by at least one hardware processor, cause the at least one hardware processor to perform a method in accordance with the application.

[0007] In some aspects, the application provides methods of polypeptide sequencing. In some embodiments, the methods comprise contacting a single polypeptide molecule with one or more terminal amino acid recognition molecules. In some embodiments, the methods further comprise detecting a series of signal pulses indicative of association of the one or more terminal amino acid recognition molecules with successive amino acids exposed at a terminus of the single polypeptide molecule while it is being degraded, thereby obtaining sequence information about the single polypeptide molecule. In some embodiments, the amino acid sequence of most or all of the single polypeptide molecule is determined. In some embodiments, the series of signal pulses is a series of real-time signal pulses.

[0008] In some embodiments, association of the one or more terminal amino acid recognition molecules with each type of amino acid exposed at the terminus produces a characteristic pattern in the series of signal pulses that is different from other types of amino acids exposed at the terminus. In some embodiments, a signal pulse of the characteristic pattern corresponds to an individual association event between a terminal amino acid recognition molecule and an amino acid exposed at the terminus. In some embodiments, the characteristic pattern corresponds to a series of reversible terminal amino acid recognition molecule binding interactions with the amino acid exposed at the terminus of the single polypeptide molecule. In some embodiments, the characteristic pattern is indicative of the amino acid exposed at the terminus of the single polypeptide molecule and an amino acid at a contiguous position (e.g., amino acids of the same type or different types).

[0009] In some embodiments, the single polypeptide molecule is degraded by a cleaving reagent that removes one or more amino acids from the terminus of the single polypeptide molecule. In some embodiments, the methods further comprise detecting a signal indicative of association of the cleaving reagent with the terminus. In some embodiments, the cleaving reagent comprises a detectable label (e.g., a luminescent label, a conductivity label). In some embodiments, the single polypeptide molecule is immobilized to a surface. In some embodiments, the single polypeptide molecule is immobilized to the surface through a terminal end distal to the terminus to which the one or more terminal amino acid recognition molecules associate. In some embodiments, the single polypeptide molecule is immobilized to the surface through a linker (e.g., a solubilizing linker comprising a biomolecule).

[0010] In some aspects, the application provides methods of sequencing a polypeptide comprising contacting a single polypeptide molecule in a reaction mixture with a composition comprising one

or more terminal amino acid recognition molecules and a cleaving reagent. In some embodiments, the methods further comprise detecting a series of signal pulses indicative of association of the one or more terminal amino acid recognition molecules with a terminus of the single polypeptide molecule in the presence of the cleaving reagent. In some embodiments, the series of signal pulses is indicative of a series of amino acids exposed at the terminus over time as a result of terminal amino acid cleavage by the cleaving reagent.

[0011] In some aspects, the application provides methods of sequencing a polypeptide comprising (a) identifying a first amino acid at a terminus of a single polypeptide molecule, (b) removing the first amino acid to expose a second amino acid at the terminus of the single polypeptide molecule, and (c) identifying the second amino acid at the terminus of the single polypeptide molecule. In some embodiments, (a)-(c) are performed in a single reaction mixture. In some embodiments, (a)-(c) occur sequentially. In some embodiments, (c) occurs before (a) and (b). In some embodiments, the single reaction mixture comprises one or more terminal amino acid recognition molecules. In some embodiments, the single reaction mixture comprises a cleaving reagent. In some embodiments, the first amino acid is removed by the cleaving reagent. In some embodiments, the methods further comprise repeating the steps of removing and identifying one or more amino acids at the terminus of the single polypeptide molecule, thereby determining a sequence (e.g., a partial sequence or a complete sequence) of the single polypeptide molecule.

[0012] In some aspects, the application provides methods of identifying an amino acid of a polypeptide comprising contacting a single polypeptide molecule with one or more amino acid recognition molecules that bind to the single polypeptide molecule. In some embodiments, the methods further comprise detecting a series of signal pulses indicative of association of the one or more amino acid recognition molecules with the single polypeptide molecule under polypeptide degradation conditions. In some embodiments, the methods further comprise identifying a first type of amino acid in the single polypeptide molecule based on a first characteristic pattern in the series of signal pulses.

[0013] In some aspects, the application provides methods of identifying a terminal amino acid (e.g., the N-terminal or the C-terminal amino acid) of a polypeptide. In some embodiments, the methods comprise contacting a polypeptide with one or more labeled affinity reagents (e.g., one or more amino acid recognition molecules) that selectively bind one or more types of terminal amino acids at a terminus of the polypeptide. In some embodiments, the methods further comprise identifying a terminal amino acid at the terminus of the polypeptide by detecting an interaction of the polypeptide with the one or more labeled affinity reagents.

[0014] In yet other aspects, the application provides methods of polypeptide sequencing by Edman-type degradation reactions. In some embodiments, Edman-type degradation reactions may be performed by contacting a polypeptide with different reaction mixtures for purposes of either detection or cleavage (e.g., as compared to a dynamic sequencing reaction, which can involve detection and cleavage using a single reaction mixture).

[0015] Accordingly, in some aspects, the application provides methods of determining an amino acid sequence of a polypeptide comprising (i) contacting a polypeptide with one or more labeled affinity reagents that selectively bind one or more types of terminal amino acids at a terminus of the polypeptide. In some embodiments, the methods further comprise (ii) identifying a terminal amino acid (e.g., the N-terminal or the C-terminal amino acid) at the terminus of the polypeptide by detecting an interaction of the polypeptide with the one or more labeled affinity reagents. In some embodiments, the methods further comprise (iii) removing the terminal amino acid. In some embodiments, the methods further comprise (iv) repeating (i)-(iii) one or more times at the terminus of the polypeptide to determine an amino acid sequence of the polypeptide.

[0016] In some embodiments, the methods further comprise, after (i) and before (ii), removing any of the one or more labeled affinity reagents that do not selectively bind the terminal amino acid. In some embodiments, the methods further comprise, after (ii) and before (iii), removing any of the

one or more labeled affinity reagents that selectively bind the terminal amino acid.

[0017] In some embodiments, removing a terminal amino acid (e.g., (iii)) comprises modifying the terminal amino acid by contacting the terminal amino acid with an isothiocyanate (e.g., phenyl isothiocyanate), and contacting the modified terminal amino acid with a protease that specifically binds and removes the modified terminal amino acid. In some embodiments cleaving a terminal amino acid (e.g., (iii)) comprises modifying the terminal amino acid by contacting the terminal amino acid with an isothiocyanate, and subjecting the modified terminal amino acid to acidic or basic conditions sufficient to remove the modified terminal amino acid.

[0018] In some embodiments, identifying a terminal amino acid comprises identifying the terminal amino acid as being one type of the one or more types of terminal amino acids to which the one or more labeled affinity reagents bind. In some embodiments, identifying a terminal amino acid comprises identifying the terminal amino acid as being a type other than the one or more types of terminal amino acids to which the one or more labeled affinity reagents bind.

[0019] In some aspects, the application provides amino acid recognition molecules comprising a shielding element, e.g., for enhanced photostability in polypeptide sequencing reactions. In some aspects, the application provides an amino acid recognition molecule of Formula (I):

A-(Y).sub.n-D (I),

wherein: A is an amino acid binding component comprising at least one amino acid recognition molecule; each instance of Y is a polymer that forms a covalent or non-covalent linkage group; n is an integer from 1 to 10, inclusive; and D is a label component comprising at least one detectable label. In some embodiments, D is less than 200 Å in diameter. In some embodiments, —(Y).sub.n— is at least 2 nm in length (e.g., at least 5 nm, at least 10 nm, at least 20 nm, at least 30 nm, at least 50 nm, or more, in length). In some embodiments, —(Y).sub.n— is between about 2 nm and about 200 nm in length (e.g., between about 2 nm and about 100 nm, between about 5 nm and about 50 nm, or between about 10 nm and about 100 nm in length). In some embodiments, each instance of Y is independently a biomolecule or a dendritic polymer (e.g., a polyol, a dendrimer). In some embodiments, the application provides a composition comprising the amino acid recognition molecule of Formula (I). In some embodiments, the amino acid recognition molecule is soluble in the composition.

[0020] In some aspects, the application provides an amino acid recognition molecule of Formula (II):

A-Y.sup.1-D (II),

wherein: A is an amino acid binding component comprising at least one amino acid recognition molecule; Y.sup.1 is a nucleic acid or a polypeptide; D is a label component comprising at least one detectable label. In some embodiments, when Y.sup.1 is a nucleic acid, the nucleic acid forms a covalent or non-covalent linkage group. In some embodiments, provided that when Y.sup.1 is a polypeptide, the polypeptide forms a non-covalent linkage group characterized by a dissociation constant (K.sub.D) of less than 50×10^{-9} M. In some embodiments, the K.sub.D is less than 1×10^{-9} M, less than 1×10^{-10} M, less than 1×10^{-11} M, or less than 1×10^{-12} M.

[0021] In some aspects, the application provides an amino acid recognition molecule comprising: a nucleic acid; at least one amino acid recognition molecule attached to a first attachment site on the nucleic acid; and at least one detectable label attached to a second attachment site on the nucleic acid, where the nucleic acid forms a covalent or non-covalent linkage group between the at least one amino acid recognition molecule and the at least one detectable label. In some embodiments, the nucleic acid comprises a first oligonucleotide strand. In some embodiments, the nucleic acid further comprises a second oligonucleotide strand hybridized with the first oligonucleotide strand.

[0022] In some aspects, the application provides an amino acid recognition molecule comprising: a

multivalent protein comprising at least two ligand-binding sites; at least one amino acid recognition molecule attached to the protein through a first ligand moiety bound to a first ligand-binding site on the protein; and at least one detectable label attached to the protein through a second ligand moiety bound to a second ligand-binding site on the protein. In some embodiments, the multivalent protein is an avidin protein.

[0023] In some embodiments, a shielded amino acid recognition molecule may be used in polypeptide sequencing methods in accordance with the application, or any method known in the art. Accordingly, in some aspects, the application provides methods of polypeptide sequencing (e.g., in an Edman-type degradation reaction, in a dynamic sequencing reaction, or other method known in the art) comprising contacting a polypeptide molecule with one or more shielded amino acid recognition molecules of the application. For example, in some embodiments, the methods comprise contacting a polypeptide molecule with at least one amino acid recognition molecule that comprises a shield or shielding element in accordance with the application, and detecting association of the at least one amino acid recognition molecule with the polypeptide molecule.

[0024] In some aspects, the application provides methods of identifying a protein of interest in a mixed sample. In some embodiments, the methods comprise cleaving a mixed protein sample to produce a plurality of polypeptide fragments. In some embodiments, the methods further comprise determining an amino acid sequence of at least one polypeptide fragment of the plurality in a method in accordance with the methods of the application. In some embodiments, the methods further comprise identifying a protein of interest in the mixed sample if the amino acid sequence is uniquely identifiable to the protein of interest.

[0025] In some embodiments, methods of identifying a protein of interest in a mixed sample comprise cleaving a mixed protein sample to produce a plurality of polypeptide fragments. In some embodiments, the methods further comprise labeling one or more types of amino acids in the plurality of polypeptide fragments with one or more different luminescent labels. In some embodiments, the methods further comprise measuring luminescence over time for at least one labeled polypeptide of the plurality. In some embodiments, the methods further comprise determining an amino acid sequence of the at least one labeled polypeptide based on the luminescence detected. In some embodiments, the methods further comprise identifying a protein of interest in the mixed sample if the amino acid sequence is uniquely identifiable to the protein of interest.

[0026] Accordingly, in some embodiments, a polypeptide molecule or protein of interest to be analyzed in accordance with the application can be of a mixed or purified sample. In some embodiments, the polypeptide molecule or protein of interest is obtained from a biological sample (e.g., blood, tissue, saliva, urine, or other biological source). In some embodiments, the polypeptide molecule or protein of interest is obtained from a patient sample (e.g., a human sample).

[0027] The details of certain embodiments of the invention are set forth in the Detailed Description of Certain Embodiments, as described below. Other features, objects, and advantages of the invention will be apparent from the Definitions, Examples, Figures, and Claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] The skilled artisan will understand that the figures, described herein, are for illustration purposes only. It is to be understood that, in some instances, various aspects of the invention may be shown exaggerated or enlarged to facilitate an understanding of the invention. In the drawings, like reference characters generally refer to like features, functionally similar and/or structurally similar elements throughout the various figures. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the teachings. The drawings are

not intended to limit the scope of the present teachings in any way.

[0029] The features and advantages of the present invention will become more apparent from the detailed description set forth below when taken in conjunction with the drawings.

[0030] When describing embodiments in reference to the drawings, direction references (“above,” “below,” “top,” “bottom,” “left,” “right,” “horizontal,” “vertical,” etc.) may be used. Such references are intended merely as an aid to the reader viewing the drawings in a normal orientation. These directional references are not intended to describe a preferred or only orientation of an embodied device. A device may be embodied in other orientations.

[0031] As is apparent from the detailed description, the examples depicted in the figures and further described for the purpose of illustration throughout the application describe non-limiting embodiments, and in some cases may simplify certain processes or omit features or steps for the purpose of clearer illustration.

[0032] FIGS. 1A-1B show an example of polypeptide sequencing by detection (FIG. 1A) and analysis (FIG. 1B) of single molecule binding interactions.

[0033] FIGS. 1C-1E show various examples of labeled affinity reagents and methods of use in accordance with the application. FIG. 1C depicts example configurations of labeled affinity reagents, including labeled enzymes and labeled aptamers which selectively bind one or more types of terminal amino acids. FIG. 1D generically depicts a degradation-based process of polypeptide sequencing using labeled affinity reagents. FIG. 1E shows an example of polypeptide sequencing using labeled aptamers by repeated cycles of terminal amino acid detection, modification, and cleavage.

[0034] FIG. 2 shows an example of polypeptide sequencing in real-time using labeled exopeptidases that each selectively binds and cleaves a different type of terminal amino acid.

[0035] FIGS. 3A-3B show examples of polypeptide sequencing in real-time by evaluating binding interactions of terminal and/or internal amino acids with labeled affinity reagents and a labeled cleaving reagent (e.g., a labeled non-specific exopeptidase). FIG. 3A shows an example of real-time sequencing by detecting a series of pulses in a signal output. FIG. 3B schematically depicts a temperature-dependent sequencing process.

[0036] FIG. 4 shows an example of polypeptide sequencing in real-time by evaluating binding interactions of terminal and internal amino acids with labeled affinity reagents and a labeled non-specific exopeptidase.

[0037] FIGS. 5A-5E show non-limiting examples of affinity reagents labeled through a shielding element. FIG. 5A illustrates single-molecule peptide sequencing with an affinity reagent labeled through a conventional covalent linkage. FIG. 5B illustrates single-molecule peptide sequencing with an affinity reagent comprising a shielding element. FIGS. 5C-5E illustrate various examples of shielding elements in accordance with the application.

[0038] FIG. 6 shows an example of identifying polypeptides based on a unique combination of amino acids detected in a labeled polypeptide.

[0039] FIG. 7 shows an example of polypeptide sequencing by detecting luminescence of a labeled polypeptide which is subjected to repeated cycles of terminal amino acid modification and cleavage.

[0040] FIGS. 8A-8C show an example of polypeptide sequencing by processive enzymatic cleavage of a labeled polypeptide. FIG. 8A shows an example of sequencing by processive enzymatic cleavage of a labeled polypeptide by an immobilized terminal peptidase. FIG. 8B shows an example of sequencing by processive enzymatic cleavage of an immobilized labeled polypeptide by a terminal peptidase. FIG. 8C schematically illustrates an example of a real-time sequencing process performed in accordance with FIG. 8B.

[0041] FIG. 9 schematically illustrates an example of sequencing by cofactor-based FRET using an immobilized ATP-dependent protease, donor-labeled ATP, and acceptor-labeled amino acids of a polypeptide substrate.

[0042] FIGS. **10A-10C** show various examples of preparing samples and sample well surfaces for analysis of polypeptides and proteins in accordance with the application. FIG. **10A** generically depicts an example process of preparing terminally modified polypeptides from a protein sample. FIG. **10B** generically depicts an example process of conjugating a solubilizing linker to a polypeptide. FIG. **10C** shows an example schematic of a sample well having modified surfaces which may be used to promote single molecule immobilization to a bottom surface.

[0043] FIG. **11** is a diagram of an illustrative sequence data processing pipeline for analyzing data obtained during a polypeptide degradation process, in accordance with some embodiments of the technology described herein.

[0044] FIG. **12** is a flow chart of an illustrative process for determining an amino acid sequence of a polypeptide molecule, in accordance with some embodiments of the technology described herein.

[0045] FIG. **13** is a flow chart of an illustrative process for determining an amino acid sequence representative of a polypeptide, in accordance with some embodiments of the technology described herein.

[0046] FIG. **14** is a block diagram of an illustrative computer system that may be used in implementing some embodiments of the technology described herein.

[0047] FIGS. **15A-15C** show experimental data for select peptide-linker conjugates prepared and evaluated for enhanced solubility provided by different solubilizing linkers. FIG. **15A** shows example structures of peptide-linker conjugates that were synthesized and evaluated.

[0048] FIG. **15B** shows results from LCMS which demonstrate peptide cleavage at the N-terminus.

[0049] FIG. **15C** shows results from a loading experiment.

[0050] FIG. **16** shows a summary of amino acid cleavage activities for select exopeptidases based on experimental results.

[0051] FIGS. **17A-17C** show experimental data for a dye/peptide conjugate assay for detecting and cleaving terminal amino acids. FIG. **17A** shows example schemes and structures used for performing a dye/peptide conjugate assay. FIG. **17B** shows imaging results for peptide-linker conjugate loading into sample wells in an on-chip assay. FIG. **17C** shows example signal traces which detected peptide-conjugate loading and terminal amino acid cleavage.

[0052] FIGS. **18A-18F** show experimental data for a FRET dye/peptide conjugate assay for detecting and cleaving terminal amino acids. FIG. **18A** shows example schemes and structures used for performing a FRET dye/peptide conjugate assay. FIG. **18B** shows FRET imaging results for different time points. FIG. **18C** shows cutting efficiency at the different time points. FIG. **18D** shows cutting displayed at each of the different time points. FIG. **18E** shows additional FRET imaging results for different time points with a proline iminopeptidase from *Yersinia pestis* (yPIP). FIG. **18F** shows FRET imaging results for different time points with an aminopeptidase from *Vibrio proteolyticus* (VPr).

[0053] FIGS. **19A-19H** show experimental data for terminal amino acid discrimination by a labeled affinity reagent. FIG. **19A** shows a crystal structure of a ClpS2 protein that was labeled for these experiments. FIG. **19B** shows single molecule intensity traces which illustrate N-terminal amino acid discrimination by the labeled ClpS2 protein. FIG. **19C** is a plot showing mean pulse duration for different terminal amino acids. FIG. **19D** is a plot showing mean interpulse duration for different terminal amino acids. FIG. **19E** shows plots further illustrating discriminant pulse durations among the different terminal amino acids.

[0054] FIGS. **19F, 19G, and 19H** show example results from dwell time analysis demonstrating leucine recognition by a ClpS protein from *Thermosynochoccus elongatus* (teClpS). FIG. **19I** shows example results from dwell time analysis demonstrating differentiable recognition of phenylalanine, leucine, tryptophan, and tyrosine by *A. tumefaciens* ClpS1. FIG. **19J** shows example results from dwell time analysis demonstrating leucine recognition by *S. elongatus* ClpS2. FIGS. **19K-19L** show example results from dwell time analysis demonstrating proline recognition by GID4.

[0055] FIGS. 20A-20D show example results from polypeptide sequencing reactions conducted in real-time using a labeled ClpS2 recognition protein and an aminopeptidase cleaving reagent in the same reaction mixture. FIG. 20A shows signal trace data for a first sequencing reaction. FIG. 20B shows pulse duration statistics for the signal trace data shown in FIG. 20A. FIG. 20C shows signal trace data for a second sequencing reaction. FIG. 20D shows pulse duration statistics for the signal trace data shown in FIG. 20C.

[0056] FIGS. 21A-21F show experimental data for terminal amino acid identification and cleavage by a labeled exopeptidase. FIG. 21A shows a crystal structure of a proline iminopeptidase (yPIP) that was site-specifically labeled for these experiments. FIG. 21B shows the degree of labeling for the purified protein product. FIG. 21C is an image of SDS page confirming site-specific labeling of yPIP. FIG. 21D is an overexposed image of the SDS page gel confirming site-specific labeling. FIG. 21E is an image of a Coomassie stained gel confirming purity of labeled protein product. FIG. 21F is an HPLC trace demonstrating cleavage activity of the labeled exopeptidase. The sequence YPPYPK corresponds to SEQ ID NO: 82. The sequence PYPYPK corresponds to SEQ ID NO: 83.

[0057] FIGS. 22A-22F show data from experiments evaluating recognition of amino acids containing specific post-translational modifications. FIG. 22A shows representative traces which demonstrated phospho-tyrosine recognition by an SH2 domain-containing protein;

[0058] FIG. 22B shows pulse duration data corresponding to the traces of FIG. 22A; and FIG. 22C shows statistics determined for the traces. FIGS. 22D-22F show representative traces from negative control experiments.

[0059] FIG. 23 is a plot showing median pulse duration from experiments evaluating the effects of penultimate amino acids on pulse duration.

[0060] FIGS. 24A-24C show data from experiments evaluating simultaneous amino acid recognition by differentially labeled recognition molecules. FIG. 24A shows a representative trace. FIG. 24B is a plot comparing pulse duration data obtained during these experiments for each recognition molecule. FIG. 24C shows pulse duration statistics for these experiments.

[0061] FIGS. 25A-25C show data from experiments evaluating the photostability of peptides during single-molecule recognition. FIG. 25A shows a representative trace from recognition using at ClpS2-V1 labeled with a dye ~2 nm from the amino acid binding site. FIG. 25B shows a visualization of the structure of the ClpS2 protein used in these experiments. FIG. 25C shows a representative trace from recognition using ClpS2 labeled with a dye >10 nm from the amino acid binding site through a DNA/protein linker.

[0062] FIGS. 26A-26D show representative traces from polypeptide sequencing reactions conducted in real-time on a complementary metal-oxide-semiconductor (CMOS) chip using a ClpS2 recognition protein labeled through a DNA/streptavidin linker in the presence of an aminopeptidase cleaving reagent.

[0063] FIG. 27 shows representative traces from polypeptide sequencing reactions conducted in real-time using at ClpS2-V1 recognition protein labeled through a DNA/streptavidin linker in the presence of *Pyrococcus horikoshii* TET aminopeptidase cleaving reagent.

[0064] FIGS. 28A-28J show representative trace data from polypeptide sequencing reactions conducted in real-time using multiple types of exopeptidases with differential cleavage specificities. FIG. 28A shows a representative trace from a reaction performed with hTET exopeptidase, with expanded pulse pattern regions shown in FIG. 28B. The sequence YAAWAAFADDDWK in FIG. 28A corresponds to SEQ ID NO: 78. FIG. 28C shows a representative trace from a reaction performed with both hTET and yPIP exopeptidases, with expanded pulse pattern regions shown in FIG. 28D, and additional representative traces shown in FIG. 28E. The sequence FYPLPWPDDDYK in FIG. 28C corresponds to SEQ ID NO: 80. FIG. 28F shows a representative trace from a further reaction performed with both hTET and yPIP exopeptidases, with expanded pulse pattern regions shown in FIG. 28G, and additional

representative traces shown in FIG. 28H. FIG. 28I shows a representative trace from a reaction performed with both PfuTET and yPIP exopeptidases, with expanded pulse pattern regions shown in FIG. 28J. The sequence YPLPWPDDDYK in FIGS. 28F and 28I corresponds to SEQ ID NO: 81.

DETAILED DESCRIPTION

[0065] Aspects of the application relate to methods of protein sequencing and identification, methods of polypeptide sequencing and identification, methods of amino acid identification, and compositions for performing such methods.

[0066] In some aspects, the application relates to the discovery of polypeptide sequencing techniques which may be implemented using existing analytic instruments with few or no device modifications. For example, previous polypeptide sequencing strategies have involved iterative cycling of different reagent mixtures through a reaction vessel containing a polypeptide being analyzed. Such strategies may require modification of an existing analytic instrument, such as a nucleic acid sequencing instrument, which may not be equipped with a flow cell or similar apparatus capable of reagent cycling. The inventors have recognized and appreciated that certain polypeptide sequencing techniques of the application do not require iterative reagent cycling, thereby permitting the use of existing instruments without significant modifications which might increase instrument size. Accordingly, in some aspects, the application provides methods of polypeptide sequencing that permit the use of smaller sequencing instruments. In some aspects, the application relates to the discovery of polypeptide sequencing techniques that allow both genomic and proteomic analyses to be performed using the same sequencing instrument.

[0067] The inventors have further recognized and appreciated that differential binding interactions can provide an additional or alternative approach to conventional labeling strategies in polypeptide sequencing. Conventional polypeptide sequencing can involve labeling each type of amino acid with a uniquely identifiable label. This process can be laborious and prone to error, as there are at least twenty different types of naturally occurring amino acids in addition to numerous post-translational variations thereof. In some aspects, the application relates to the discovery of techniques involving the use of amino acid recognition molecules which differentially associate with different types of amino acids to produce detectable characteristic signatures indicative of an amino acid sequence of a polypeptide. Accordingly, aspects of the application provide techniques that do not require polypeptide labeling and/or harsh chemical reagents used in certain conventional polypeptide sequencing approaches, thereby increasing throughput and/or accuracy of sequence information obtained from a sample.

[0068] In some aspects, the application relates to the discovery that a polypeptide sequencing reaction can be monitored in real-time using only a single reaction mixture (e.g., without requiring iterative reagent cycling through a reaction vessel). As detailed above, conventional polypeptide sequencing reactions can involve exposing a polypeptide to different reagent mixtures to cycle between steps of amino acid detection and amino acid cleavage. Accordingly, in some aspects, the application relates to an advancement in next generation sequencing that allows for the analysis of polypeptides by amino acid detection throughout an ongoing degradation reaction in real-time. Approaches for such polypeptide analysis by dynamic sequencing are described below.

[0069] As described herein, in some aspects, the application provides methods of sequencing a polypeptide by obtaining data during a polypeptide degradation process, and analyzing the data to determine portions of the data corresponding to amino acids that are sequentially exposed at a terminus of the polypeptide during the degradation process. In some embodiments, the portions of the data comprise a series of signal pulses indicative of association of one or more amino acid recognition molecules with successive amino acids exposed at the terminus of the polypeptide (e.g., during a degradation). In some embodiments, the series of signal pulses corresponds to a series of reversible single molecule binding interactions at the terminus of the polypeptide during the degradation process.

[0070] A non-limiting example of polypeptide sequencing by detecting single molecule binding interactions during a polypeptide degradation process is schematically illustrated in FIG. 1A. An example signal trace (I) is shown with a series of panels (II) that depict different association events at times corresponding to changes in the signal. As shown, an association event between an amino acid recognition molecule (stippled shape) and an amino acid at the terminus of a polypeptide (shown as beads-on-a-string) produces a change in magnitude of the signal that persists for a duration of time.

[0071] Panels (A) and (B) depict different association events between an amino acid recognition molecule and a first amino acid exposed at the terminus of the polypeptide (e.g., a first terminal amino acid). Each association event produces a change in the signal trace (I) characterized by a change in magnitude of the signal that persists for the duration of the association event. Accordingly, the time duration between the association events of panels (A) and (B) may correspond to a duration of time within which the polypeptide is not detectably associated with an amino acid recognition molecule.

[0072] Panels (C) and (D) depict different association events between an amino acid recognition molecule and a second amino acid exposed at the terminus of the polypeptide (e.g., a second terminal amino acid). As described herein, an amino acid that is “exposed” at the terminus of a polypeptide is an amino acid that is still attached to the polypeptide and that becomes the terminal amino acid upon removal of the prior terminal amino acid during degradation (e.g., either alone or along with one or more additional amino acids). Accordingly, the first and second amino acids of the series of panels (II) provide an illustrative example of successive amino acids exposed at the terminus of the polypeptide, where the second amino acid became the terminal amino acid upon removal of the first amino acid.

[0073] As generically depicted, the association events of panels (C) and (D) produce changes in the signal trace (I) characterized by changes in magnitude that persist for time durations that are relatively shorter than that of panels (A) and (B), and the time duration between the association events of panels (C) and (D) is relatively shorter than that of panels (A) and (B). As described herein, in some embodiments, either one or both of these distinctive changes in signal may be used to determine characteristic patterns in the signal trace (I) which can discriminate between different types of amino acids. In some embodiments, a transition from one characteristic pattern to another is indicative of amino acid cleavage. As used herein, in some embodiments, amino acid cleavage refers to the removal of at least one amino acid from a terminus of a polypeptide (e.g., the removal of at least one terminal amino acid from the polypeptide). In some embodiments, amino acid cleavage is determined by inference based on a time duration between characteristic patterns. In some embodiments, amino acid cleavage is determined by detecting a change in signal produced by association of a labeled cleaving reagent with an amino acid at the terminus of the polypeptide. As amino acids are sequentially cleaved from the terminus of the polypeptide during degradation, a series of changes in magnitude, or a series of signal pulses, is detected. In some embodiments, signal pulse data can be analyzed as illustrated in FIG. 1B.

[0074] In some embodiments, signal data can be analyzed to extract signal pulse information by applying threshold levels to one or more parameters of the signal data. For example, panel (III) depicts a threshold magnitude level (“ML”) applied to the signal data of the example signal trace (I). In some embodiments, ML is a minimum difference between a signal detected at a point in time and a baseline determined for a given set of data. In some embodiments, a signal pulse (“sp”) is assigned to each portion of the data that is indicative of a change in magnitude exceeding ML and persisting for a duration of time. In some embodiments, a threshold time duration may be applied to a portion of the data that satisfies ML to determine whether a signal pulse is assigned to that portion. For example, experimental artifacts may give rise to a change in magnitude exceeding ML that does not persist for a duration of time sufficient to assign a signal pulse with a desired confidence (e.g., transient association events which could be non-discriminatory for amino acid

type, non-specific detection events such as diffusion into an observation region or reagent sticking within an observation region). Accordingly, in some embodiments, a signal pulse is extracted from signal data based on a threshold magnitude level and a threshold time duration.

[0075] Extracted signal pulse information is shown in panel (III) with the example signal trace (I) superimposed for illustrative purposes. In some embodiments, a peak in magnitude of a signal pulse is determined by averaging the magnitude detected over a duration of time that persists above ML. It should be appreciated that, in some embodiments, a “signal pulse” as used herein can refer to a change in signal data that persists for a duration of time above a baseline (e.g., raw signal data, as illustrated by the example signal trace (I)), or to signal pulse information extracted therefrom (e.g., processed signal data, as illustrated in panel (IV)).

[0076] Panel (IV) shows the signal pulse information extracted from the example signal trace (I). In some embodiments, signal pulse information can be analyzed to identify different types of amino acids in a sequence based on different characteristic patterns in a series of signal pulses. For example, as shown in panel (IV), the signal pulse information is indicative of a first type of amino acid based on a first characteristic pattern (“CP.sub.1”) and a second type of amino acid based on a second characteristic pattern (“CP.sub.2”). By way of example, the two signal pulses detected at earlier time points provide information indicative of the first amino acid at the terminus of the polypeptide based on CP.sub.1, and the two signal pulses detected at later time points provide information indicative of the second amino acid at the terminus of the polypeptide based on CP.sub.2.

[0077] Also as shown in panel (IV), each signal pulse comprises a pulse duration (“pd”) corresponding to an association event between the amino acid recognition molecule and the amino acid of the characteristic pattern. In some embodiments, the pulse duration is characteristic of a dissociation rate of binding. Also as shown, each signal pulse of a characteristic pattern is separated from another signal pulse of the characteristic pattern by an interpulse duration (“ipd”). In some embodiments, the interpulse duration is characteristic of an association rate of binding. In some embodiments, a change in magnitude (“AM”) can be determined for a signal pulse based on a difference between baseline and the peak of a signal pulse. In some embodiments, a characteristic pattern is determined based on pulse duration. In some embodiments, a characteristic pattern is determined based on pulse duration and interpulse duration. In some embodiments, a characteristic pattern is determined based on any one or more of pulse duration, interpulse duration, and change in magnitude.

[0078] Accordingly, as illustrated by FIGS. 1A-1B, in some embodiments, polypeptide sequencing is performed by detecting a series of signal pulses indicative of association of one or more amino acid recognition molecules with successive amino acids exposed at the terminus of a polypeptide in an ongoing degradation reaction. The series of signal pulses can be analyzed to determine characteristic patterns in the series of signal pulses, and the time course of characteristic patterns can be used to determine an amino acid sequence of the polypeptide.

[0079] In some embodiments, the series of signal pulses comprises a series of changes in magnitude of an optical signal over time. In some embodiments, the series of changes in the optical signal comprises a series of changes in luminescence produced during association events. In some embodiments, luminescence is produced by a detectable label associated with one or more reagents of a sequencing reaction. For example, in some embodiments, each of the one or more amino acid recognition molecules comprises a luminescent label. In some embodiments, a cleaving reagent comprises a luminescent label. Examples of luminescent labels and their use in accordance with the application are provided elsewhere herein.

[0080] In some embodiments, the series of signal pulses comprises a series of changes in magnitude of an electrical signal over time. In some embodiments, the series of changes in the electrical signal comprises a series of changes in conductance produced during association events. In some embodiments, conductivity is produced by a detectable label associated with one or more

reagents of a sequencing reaction. For example, in some embodiments, each of the one or more amino acid recognition molecules comprises a conductivity label. Examples of conductivity labels and their use in accordance with the application are provided elsewhere herein. Methods for identifying single molecules using conductivity labels have been described (see, e.g., U.S. Patent Publication No. 2017/0037462).

[0081] In some embodiments, the series of changes in conductance comprises a series of changes in conductance through a nanopore. For example, methods of evaluating receptor-ligand interactions using nanopores have been described (see, e.g., Thakur, A. K. & Movileanu, L. (2019) *Nature Biotechnology* 37(1)). The inventors have recognized and appreciated that such nanopores may be used to monitor polypeptide sequencing reactions in accordance with the application. Accordingly, in some embodiments, the application provides methods of polypeptide sequencing comprising contacting a single polypeptide molecule with one or more amino acid recognition molecules, where the single polypeptide molecule is immobilized to a nanopore. In some embodiments, the methods further comprise detecting a series of changes in conductance through the nanopore indicative of association of the one or more terminal amino acid recognition molecules with successive amino acids exposed at a terminus of the single polypeptide while the single polypeptide is being degraded, thereby sequencing the single polypeptide molecule.

[0082] In some aspects, the application provides methods of sequencing and/or identifying an individual protein in a complex mixture of proteins by identifying one or more types of amino acids of a polypeptide from the mixture. In some embodiments, one or more amino acids (e.g., terminal amino acids and/or internal amino acids) of the polypeptide are labeled (e.g., directly or indirectly, for example using a binding agent such as an amino acid recognition molecule) and the relative positions of the labeled amino acids in the polypeptide are determined. In some embodiments, the relative positions of amino acids in a polypeptide are determined using a series of amino acid labeling and cleavage steps. However, in some embodiments, the relative position of labeled amino acids in a polypeptide can be determined without removing amino acids from the polypeptide but by translocating a labeled polypeptide through a pore (e.g., a protein channel) and detecting a signal (e.g., a FRET signal) from the labeled amino acid(s) during translocation through the pore in order to determine the relative position of the labeled amino acids in the polypeptide molecule.

[0083] In some embodiments, the identity of a terminal amino acid (e.g., an N-terminal or a C-terminal amino acid) is assessed after which the terminal amino acid is removed and the identity of the next amino acid at the terminus is assessed, and this process is repeated until a plurality of successive amino acids in the polypeptide are assessed. In some embodiments, assessing the identity of an amino acid comprises determining the type of amino acid that is present. In some embodiments, determining the type of amino acid comprises determining the actual amino acid identity, for example by determining which of the naturally-occurring 20 amino acids is the terminal amino acid is (e.g., using a binding agent that is specific for an individual terminal amino acid). In some embodiments, the type of amino acid is selected from alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, proline, selenocysteine, serine, threonine, tryptophan, tyrosine, and valine.

[0084] However, in some embodiments assessing the identity of a terminal amino acid type can comprise determining a subset of potential amino acids that can be present at the terminus of the polypeptide. In some embodiments, this can be accomplished by determining that an amino acid is not one or more specific amino acids (and therefore could be any of the other amino acids). In some embodiments, this can be accomplished by determining which of a specified subset of amino acids (e.g., based on size, charge, hydrophobicity, post-translational modification, binding properties) could be at the terminus of the polypeptide (e.g., using a binding agent that binds to a specified subset of two or more terminal amino acids).

[0085] In some embodiments, assessing the identity of a terminal amino acid type comprises determining that an amino acid comprises a post-translational modification. Non-limiting examples of post-translational modifications include acetylation, ADP-ribosylation, caspase cleavage, citrullination, formylation, N-linked glycosylation, O-linked glycosylation, hydroxylation, methylation, myristoylation, neddylation, nitration, oxidation, palmitoylation, phosphorylation, prenylation, S-nitrosylation, sulfation, sumoylation, and ubiquitination.

[0086] In some embodiments, assessing the identity of a terminal amino acid type comprises determining that an amino acid comprises a side chain characterized by one or more biochemical properties. For example, an amino acid may comprise a nonpolar aliphatic side chain, a positively charged side chain, a negatively charged side chain, a nonpolar aromatic side chain, or a polar uncharged side chain. Non-limiting examples of an amino acid comprising a nonpolar aliphatic side chain include alanine, glycine, valine, leucine, methionine, and isoleucine. Non-limiting examples of an amino acid comprising a positively charged side chain includes lysine, arginine, and histidine. Non-limiting examples of an amino acid comprising a negatively charged side chain include aspartate and glutamate. Non-limiting examples of an amino acid comprising a nonpolar, aromatic side chain include phenylalanine, tyrosine, and tryptophan. Non-limiting examples of an amino acid comprising a polar uncharged side chain include serine, threonine, cysteine, proline, asparagine, and glutamine.

[0087] In some embodiments, a protein or polypeptide can be digested into a plurality of smaller polypeptides and sequence information can be obtained from one or more of these smaller polypeptides (e.g., using a method that involves sequentially assessing a terminal amino acid of a polypeptide and removing that amino acid to expose the next amino acid at the terminus).

[0088] In some embodiments, a polypeptide is sequenced from its amino (N) terminus. In some embodiments, a polypeptide is sequenced from its carboxy (C) terminus. In some embodiments, a first terminus (e.g., N or C terminus) of a polypeptide is immobilized and the other terminus (e.g., the C or N terminus) is sequenced as described herein.

[0089] As used herein, sequencing a polypeptide refers to determining sequence information for a polypeptide. In some embodiments, this can involve determining the identity of each sequential amino acid for a portion (or all) of the polypeptide. However, in some embodiments, this can involve assessing the identity of a subset of amino acids within the polypeptide (e.g., and determining the relative position of one or more amino acid types without determining the identity of each amino acid in the polypeptide). However, in some embodiments amino acid content information can be obtained from a polypeptide without directly determining the relative position of different types of amino acids in the polypeptide. The amino acid content alone may be used to infer the identity of the polypeptide that is present (e.g., by comparing the amino acid content to a database of polypeptide information and determining which polypeptide(s) have the same amino acid content).

[0090] In some embodiments, sequence information for a plurality of polypeptide products obtained from a longer polypeptide or protein (e.g., via enzymatic and/or chemical cleavage) can be analyzed to reconstruct or infer the sequence of the longer polypeptide or protein.

[0091] Accordingly, in some embodiments, the one or more types of amino acids are identified by detecting luminescence of one or more labeled affinity reagents that selectively bind the one or more types of amino acids. In some embodiments, the one or more types of amino acids are identified by detecting luminescence of a labeled polypeptide.

[0092] The inventors have further recognized and appreciated that the polypeptide sequencing techniques described herein may involve generating novel polypeptide sequencing data, particularly in contrast with conventional polypeptide sequencing techniques. Thus, conventional techniques for analyzing polypeptide sequencing data may not be sufficient when applied to the data generated using the polypeptide sequencing techniques described herein.

[0093] For example, conventional polypeptide sequencing techniques that involve iterative reagent

cycling may generate data associated with individual amino acids of a polypeptide being sequenced. In such instances, analyzing the data generated may simply involve determining which amino acid is being detected at a particular time because the data being detected corresponds to only one amino acid. In contrast, the polypeptide sequencing techniques described herein may generate data during a polypeptide degradation process while multiple amino acids of the polypeptide molecule are being detected, resulting in data where it may be difficult to discern between sections of the data corresponding to different amino acids of the polypeptide.

Accordingly, the inventors have developed new computational techniques for analyzing such data generated by the polypeptide sequencing techniques described herein that involve determining sections of the data that correspond to individual amino acids, such as by segmenting the data into portions that correspond to respective amino acid association events. Those sections may be then further analyzed to identify the amino acid being detected during those individual sections.

[0094] As another example, conventional sequencing techniques that involve using uniquely identifiable labels for each type of amino acid may involve simply analyzing which label is being detected at a particular time without taking into consideration any dynamics in how individual amino acids interact with other molecules. In contrast, the polypeptide sequencing techniques described herein generate data indicating how amino acids interact with recognition molecules. As discussed above, the data may include a series of characteristic patterns corresponding to association events between amino acids and their respective recognition molecules. Accordingly, the inventors have developed new computational techniques for analyzing the characteristic patterns to determine a type of amino acid corresponding to that portion of the data, allowing for an amino acid sequence of a polypeptide to be determined by analyzing a series of different characteristic patterns.

Labeled Affinity Reagents and Methods of Use

[0095] In some embodiments, methods provided herein comprise contacting a polypeptide with a labeled affinity reagent (also referred to herein as an amino acid recognition molecule, which may or may not comprise a label) that selectively binds one type of terminal amino acid. As used herein, in some embodiments, a terminal amino acid may refer to an amino-terminal amino acid of a polypeptide or a carboxy-terminal amino acid of a polypeptide. In some embodiments, a labeled affinity reagent selectively binds one type of terminal amino acid over other types of terminal amino acids. In some embodiments, a labeled affinity reagent selectively binds one type of terminal amino acid over an internal amino acid of the same type. In yet other embodiments, a labeled affinity reagent selectively binds one type of amino acid at any position of a polypeptide, e.g., the same type of amino acid as a terminal amino acid and an internal amino acid.

[0096] As used herein, in some embodiments, a type of amino acid refers to one of the twenty naturally occurring amino acids or a subset of types thereof. In some embodiments, a type of amino acid refers to a modified variant of one of the twenty naturally occurring amino acids or a subset of unmodified and/or modified variants thereof. Examples of modified amino acid variants include, without limitation, post-translationally-modified variants (e.g., acetylation, ADP-ribosylation, caspase cleavage, citrullination, formylation, N-linked glycosylation, O-linked glycosylation, hydroxylation, methylation, myristoylation, neddylation, nitration, oxidation, palmitoylation, phosphorylation, prenylation, S-nitrosylation, sulfation, sumoylation, and ubiquitination), chemically modified variants, unnatural amino acids, and proteinogenic amino acids such as selenocysteine and pyrrolysine. In some embodiments, a subset of types of amino acids includes more than one and fewer than twenty amino acids having one or more similar biochemical properties. For example, in some embodiments, a type of amino acid refers to one type selected from amino acids with charged side chains (e.g., positively and/or negatively charged side chains), amino acids with polar side chains (e.g., polar uncharged side chains), amino acids with nonpolar side chains (e.g., nonpolar aliphatic and/or aromatic side chains), and amino acids with hydrophobic side chains.

[0097] In some embodiments, methods provided herein comprise contacting a polypeptide with one or more labeled affinity reagents that selectively bind one or more types of terminal amino acids. As an illustrative and non-limiting example, where four labeled affinity reagents are used in a method of the application, any one reagent selectively binds one type of terminal amino acid that is different from another type of amino acid to which any of the other three selectively binds (e.g., a first reagent binds a first type, a second reagent binds a second type, a third reagent binds a third type, and a fourth reagent binds a fourth type of terminal amino acid). For the purposes of this discussion, one or more labeled affinity reagents in the context of a method described herein may be alternatively referred to as a set of labeled affinity reagents.

[0098] In some embodiments, a set of labeled affinity reagents comprises at least one and up to six labeled affinity reagents. For example, in some embodiments, a set of labeled affinity reagents comprises one, two, three, four, five, or six labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises ten or fewer labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises eight or fewer labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises six or fewer labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises four or fewer labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises three or fewer labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises two or fewer labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises four labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises at least two and up to twenty (e.g., at least two and up to ten, at least two and up to eight, at least four and up to twenty, at least four and up to ten) labeled affinity reagents. In some embodiments, a set of labeled affinity reagents comprises more than twenty (e.g., **20 to 25**, **20 to 30**) affinity reagents. It should be appreciated, however, that any number of affinity reagents may be used in accordance with a method of the application to accommodate a desired use.

[0099] In accordance with the application, in some embodiments, one or more types of amino acids are identified by detecting luminescence of a labeled affinity reagent (e.g., an amino acid recognition molecule comprising a luminescent label). In some embodiments, a labeled affinity reagent comprises an affinity reagent that selectively binds one type of amino acid and a luminescent label having a luminescence that is associated with the affinity reagent. In this way, the luminescence (e.g., luminescence lifetime, luminescence intensity, and other luminescence properties described elsewhere herein) may be associated with the selective binding of the affinity reagent to identify an amino acid of a polypeptide. In some embodiments, a plurality of types of labeled affinity reagents may be used in a method according to the application, wherein each type comprises a luminescent label having a luminescence that is uniquely identifiable from among the plurality. Suitable luminescent labels may include luminescent molecules, such as fluorophore dyes, and are described elsewhere herein.

[0100] In some embodiments, one or more types of amino acids are identified by detecting one or more electrical characteristics of a labeled affinity reagent. In some embodiments, a labeled affinity reagent comprises an affinity reagent that selectively binds one type of amino acid and a conductivity label that is associated with the affinity reagent. In this way, the one or more electrical characteristics (e.g., charge, current oscillation color, and other electrical characteristics) may be associated with the selective binding of the affinity reagent to identify an amino acid of a polypeptide. In some embodiments, a plurality of types of labeled affinity reagents may be used in a method according to the application, wherein each type comprises a conductivity label that produces a change in an electrical signal (e.g., a change in conductance, such as a change in amplitude of conductivity and conductivity transitions of a characteristic pattern) that is uniquely identifiable from among the plurality. In some embodiments, the plurality of types of labeled affinity reagents each comprises a conductivity label having a different number of charged groups (e.g., a different number of negatively and/or positively charged groups). Accordingly, in some

embodiments, a conductivity label is a charge label. Examples of charge labels include dendrimers, nanoparticles, nucleic acids and other polymers having multiple charged groups. In some embodiments, a conductivity label is uniquely identifiable by its net charge (e.g., a net positive charge or a net negative charge), by its charge density, and/or by its number of charged groups. [0101] In some embodiments, an affinity reagent (e.g., an amino acid recognition molecule) may be engineered by one skilled in the art using conventionally known techniques. In some embodiments, desirable properties may include an ability to bind selectively and with high affinity to one type of amino acid only when it is located at a terminus (e.g., an N-terminus or a C-terminus) of a polypeptide. In yet other embodiments, desirable properties may include an ability to bind selectively and with high affinity to one type of amino acid when it is located at a terminus (e.g., an N-terminus or a C-terminus) of a polypeptide and when it is located at an internal position of the polypeptide. In some embodiments, desirable properties include an ability to bind selectively and with low affinity (e.g., with a $K_{sub.D}$ of about 50 nM or higher, for example, between about 50 nM and about 50 μ M, between about 100 nM and about 10 μ M, between about 500 nM and about 50 μ M) to more than one type of amino acid. For example, in some aspects, the application provides methods of sequencing by detecting reversible binding interactions during a polypeptide degradation process. Advantageously, such methods may be performed using an affinity reagent that reversibly binds with low affinity to more than one type of amino acid (e.g., a subset of amino acid types).

[0102] As used herein, in some embodiments, the terms “selective” and “specific” (and variations thereof, e.g., selectively, specifically, selectivity, specificity) refer to a preferential binding interaction. For example, in some embodiments, a labeled affinity reagent that selectively binds one type of amino acid preferentially binds the one type over another type of amino acid. A selective binding interaction will discriminate between one type of amino acid (e.g., one type of terminal amino acid) and other types of amino acids (e.g., other types of terminal amino acids), typically more than about 10- to 100-fold or more (e.g., more than about 1,000- or 10,000-fold).

Accordingly, it should be appreciated that a selective binding interaction can refer to any binding interaction that is uniquely identifiable to one type of amino acid over other types of amino acids. For example, in some aspects, the application provides methods of polypeptide sequencing by obtaining data indicative of association of one or more amino acid recognition molecules with a polypeptide molecule. In some embodiments, the data comprises a series of signal pulses corresponding to a series of reversible amino acid recognition molecule binding interactions with an amino acid of the polypeptide molecule, and the data may be used to determine the identity of the amino acid. As such, in some embodiments, a “selective” or “specific” binding interaction refers to a detected binding interaction that discriminates between one type of amino acid and other types of amino acids.

[0103] In some embodiments, a labeled affinity reagent (e.g., an amino acid recognition molecule) selectively binds one type of amino acid with a dissociation constant ($K_{sub.D}$) of less than about $10^{sup.-6}$ M (e.g., less than about $10^{sup.-7}$ M, less than about $10^{sup.-8}$ M, less than about $10^{sup.-9}$ M, less than about $10^{sup.-10}$ M, less than about $10^{sup.-11}$ M, less than about $10^{sup.-12}$ M, to as low as $10^{sup.-16}$ M) without significantly binding to other types of amino acids. In some embodiments, a labeled affinity reagent selectively binds one type of amino acid (e.g., one type of terminal amino acid) with a $K_{sub.D}$ of less than about 100 nM, less than about 50 nM, less than about 25 nM, less than about 10 nM, or less than about 1 nM. In some embodiments, a labeled affinity reagent selectively binds one type of amino acid with a $K_{sub.D}$ of between about 50 nM and about 50 μ M (e.g., between about 50 nM and about 500 nM, between about 50 nM and about 5 μ M, between about 500 nM and about 50 μ M, between about 5 μ M and about 50 μ M, or between about 10 μ M and about 50 μ M). In some embodiments, a labeled affinity reagent selectively binds one type of amino acid with a $K_{sub.D}$ of about 50 nM.

[0104] In some embodiments, a labeled affinity reagent (e.g., an amino acid recognition molecule)

selectively binds two or more types of amino acids with a dissociation constant ($K_{sub.D}$) of less than about $10^{sup.-6}$ M (e.g., less than about $10^{sup.-7}$ M, less than about $10^{sup.-8}$ M, less than about $10^{sup.-9}$ M, less than about $10^{sup.-10}$ M, less than about $10^{sup.-11}$ M, less than about $10^{sup.-12}$ M, to as low as $10^{sup.-16}$ M). In some embodiments, a labeled affinity reagent selectively binds two or more types of amino acids with a $K_{sub.D}$ of less than about 100 nM, less than about 50 nM, less than about 25 nM, less than about 10 nM, or less than about 1 nM. In some embodiments, a labeled affinity reagent selectively binds two or more types of amino acids with a $K_{sub.D}$ of between about 50 nM and about 50 μ M (e.g., between about 50 nM and about 500 nM, between about 50 nM and about 5 μ M, between about 500 nM and about 50 μ M, between about 5 μ M and about 50 μ M, or between about 10 μ M and about 50 μ M). In some embodiments, a labeled affinity reagent selectively binds two or more types of amino acids with a $K_{sub.D}$ of about 50 nM. [0105] In accordance with the methods and compositions provided herein, FIG. 1C shows various example configurations and uses of labeled affinity reagents. In some embodiments, a labeled affinity reagent **100** comprises a luminescent label **110** (e.g., a label) and an affinity reagent (shown as stippled shapes) that selectively binds one or more types of terminal amino acids of a polypeptide **120**. In some embodiments, an affinity reagent is selective for one type of amino acid or a subset (e.g., fewer than the twenty common types of amino acids) of types of amino acids at a terminal position or at both terminal and internal positions.

[0106] As described herein, an affinity reagent (also known as a “recognition molecule”) may be any biomolecule capable of selectively or specifically binding one molecule over another molecule (e.g., one type of amino acid over another type of amino acid, as with an “amino acid recognition molecule” referred to herein). In some embodiments, an affinity reagent is not a peptidase or does not have peptidase activity. For example, in some embodiments, methods of polypeptide sequencing of the application involve contacting a polypeptide molecule with one or more affinity reagents and a cleaving reagent. In such embodiments, the one or more affinity reagents do not have peptidase activity, and removal of one or more amino acids from the polypeptide molecule (e.g., amino acid removal from a terminus of the polypeptide molecule) is performed by the cleaving reagent.

[0107] Affinity reagents (e.g., recognition molecules) include, for example, proteins and nucleic acids, which may be synthetic or recombinant. In some embodiments, an affinity reagent or recognition molecule may be an antibody or an antigen-binding portion of an antibody, an SH2 domain-containing protein or fragment thereof, or an enzymatic biomolecule, such as a peptidase, an aminotransferase, a ribozyme, an aptazyme, or a tRNA synthetase, including aminoacyl-tRNA synthetases and related molecules described in U.S. patent application Ser. No. 15/255,433, filed Sep. 2, 2016, titled “MOLECULES AND METHODS FOR ITERATIVE POLYPEPTIDE ANALYSIS AND PROCESSING.”

[0108] In some embodiments, an affinity reagent or recognition molecule of the application is a degradation pathway protein. Examples of degradation pathway proteins suitable for use as recognition molecules include, without limitation, N-end rule pathway proteins, such as Arg/N-end rule pathway proteins, Ac/N-end rule pathway proteins, and Pro/N-end rule pathway proteins. In some embodiments, a recognition molecule is an N-end rule pathway protein selected from a Gid protein (e.g., Gid4 or Gid10 protein), a UBR box protein (e.g., UBR1, UBR2) or UBR box domain-containing protein fragment thereof, a p62 protein or ZZ domain-containing fragment thereof, and a ClpS protein (e.g., ClpS1, ClpS2).

[0109] In some embodiments, an affinity reagent or recognition molecule of the application is a ClpS protein, such as *Agrobacterium tumefaciens* ClpS1, *Agrobacterium tumefaciens* ClpS2, *Synechococcus elongatus* ClpS1, *Synechococcus elongatus* ClpS2, *Thermosynechococcus elongatus* ClpS, *Escherichia coli* ClpS, or *Plasmodium falciparum* ClpS. In some embodiments, the recognition molecule is an L/F transferase, such as *Escherichia coli* leucyl/phenylalanyl-tRNA-protein transferase. In some embodiments, the recognition molecule is a D/E leucyltransferase,

such as *Vibrio vulnificus* Aspartate/glutamate leucyltransferase Bpt. In some embodiments, the recognition molecule is a UBR protein or UBR-box domain, such as the UBR protein or UBR-box domain of human UBR1 and UBR2 or *Saccharomyces cerevisiae* UBR1. In some embodiments, the recognition molecule is a p62 protein, such as *H. sapiens* p62 protein or *Rattus norvegicus* p62 protein, or truncation variants thereof that minimally include a ZZ domain. In some embodiments, the recognition molecule is a Gid4 protein, such as *H. sapiens* GID4 or *Saccharomyces cerevisiae* GID4. In some embodiments, the recognition molecule is a Gid10 protein, such as *Saccharomyces cerevisiae* GID10. In some embodiments, the recognition molecule is an N-meristoyltransferase, such as *Leishmania major* N-meristoyltransferase or *H. sapiens* N-meristoyltransferase NMT1. In some embodiments, the recognition molecule is a BIR2 protein, such as *Drosophila melanogaster* BIR2. In some embodiments, the recognition molecule is a tyrosine kinase or SH2 domain of a tyrosine kinase, such as *H. sapiens* Fyn SH2 domain, *H. sapiens* Src tyrosine kinase SH2 domain, or variants thereof, such as *H. sapiens* Fyn SH2 domain triple mutant superbinder. In some embodiments, the recognition molecule is an antibody or antibody fragment, such as a single-chain antibody variable fragment (scFv) against phosphotyrosine or another post-translationally modified amino acid variant described herein.

[0110] Table 1 provides a list of example sequences of amino acid recognition molecules. Also shown are the amino acid binding preferences of each molecule with respect to amino acid identity at a terminal position of a polypeptide unless otherwise specified in Table 1. It should be appreciated that these sequences and other examples described herein are meant to be non-limiting, and recognition molecules in accordance with the application can include any homologs, variants thereof, or fragments thereof minimally containing domains or subdomains responsible for peptide recognition.

TABLE-US-00001 TABLE 1 Non-limiting examples of amino acid recognition proteins.				
SEQ ID	Binding	Name	Pref.*	NO:
MSDSPVDLKP	KPKVKPKLERPKLYKVMLLNDDYTPMSFV	<i>Agrobacterium tumifaciens</i>	F, W, Y	1
TVVLKAVFRMSEDTGRRVMMTAHRFGSAVVVVCERDIAE		variant		1
TKAKEATDLGKEAGFPLMFTTEPEE		<i>Agrobacterium tumifaciens</i>	F, W, Y	2
MSDSPVDLKP	KPKVKPKLERPKLYKVMLLNDDYTPREFV	<i>Agrobacterium tumifaciens</i>	F, W, Y	3
TVVLKAVFRMSEDTGRRVMMTAHRFGSAVVVVCERDIAE				
TKAKEATDLGKEAGFPLMFTTEPEE		<i>Agrobacterium tumifaciens</i>	F, W, Y	3
MSDSPVDLKP	KPKVKPKLERPKLYKVMLLNDDYTPREFV	<i>Agrobacterium tumifaciens</i>	F, W, Y	3
TVVLKAVFRMSEDTGRRVMMTAHRFGSAVVVVSERDIAE		C71S		
TKAKEATDLGKEAGFPLMFTTEPEE		<i>Agrobacterium tumifaciens</i>	F, W, Y, L	4
MIAEPICMQGEDGEDGGTNRGTSVITRVKPKTKRPPLY		<i>Agrobacterium tumifaciens</i>	F, W, Y, L	4
RVLLLNDDYTPMEFVIHILERFFQKDREAATRIMLVHQ				
HGVGECGVFTYEVAETKVSQVMDFARQHQHPLQCVMEKK		<i>Agrobacterium tumifaciens</i>	F, W, Y	5
MSDSPVDLKP	KPKVKPKLERPKLYKVMLLNDDYTPMSFV	<i>Agrobacterium tumifaciens</i>	F, W, Y	5
TVVLKAVFRMSEDTGRRVMMTAHRFGSAVVVVSERDIAE		variant		1
TKAKEATDLGKEAGFPLMFTTEPEE		<i>Agrobacterium tumifaciens</i>	F, W, Y, L	6
MIAEPISMQGEDGEDGGTNRGTSVITRVKPKTKRPPLY		<i>Agrobacterium tumifaciens</i>	F, W, Y, L	6
RVLLLNDDYTPMEFVIHILERFFQKDREAATRIMLVHQ		C7S		
HGVGECGVFTYEVAETKVSQVMDFARQHQHPLQCVMEKK		<i>Agrobacterium tumifaciens</i>	F, W, Y, L	7
MIAEPISMQGEDGEDGGTNRGTSVITRVKPKTKRPPLY		<i>Agrobacterium tumifaciens</i>	F, W, Y, L	7
RVLLLNDDYTPMEFVIHILERFFQKDREAATRIMLVHQ		C7S		
HGVGESGVFTYEVAETKVSQVMDFARQHQHPLQSVMEKK		<i>Agrobacterium tumifaciens</i>	F, W, Y	8
MSDSPVDLKP	KPKVKPKLERPKLYKVILLNDDYTPMEFV	<i>Agrobacterium tumifaciens</i>	F, W, Y	8
VEVLKRVFNMSEEQARRVMMTAHKKGKAVVGVCPRDIAE		thermostable		
TKAKQATDLAREAGFPLMFTTEPEE		variant		
MSDSPVDLKP	KPKVKPKLERPKLYKVILLNDDYTPMEFV	<i>Agrobacterium tumifaciens</i>	F, W, Y	9
MSDSPVDLKP	KPKVKPKLERPKLYKVILLNDDYTPMEFV	<i>Agrobacterium tumifaciens</i>	F, W, Y	9

VEVLKRVFNMSQARVMTAHKKAVVGVSPPRDIAE thermostable
TKAKQATDLAREAGFPLMFTTEPEE variant C72S *Synechococcus* F, W, Y 10
MAVETIQKPETTTKRKIAPRYRVLLHNDDFNPM EYVVMV *elongatus* ClpS1
LMQTVPSLTQPQAVDIMMEAHTNGTGLVITCDIEPAEFY CEQLKSHGLSSSIEPDD
Synechococcus F, W, Y, L, 11 MSPQPDESVL SILGVPRPCVKKRSRND A FVLT VLTCSLQ
elongatus ClpS2 V, I AIAAPATAPGTTTTRVRQPYPHERVIVLDDD VNTFQHVA
ECLLK YIPGMTGDRAWDLTNQVHYEGAATVWSGPQEQA E LYHEQLRREGLTMAPLEAA
Thermosynechococcus F, W, Y, L 12
MPQERQQVTRKHYPNYKVIVLNDDFNTEFQHVAACLMKYI *elongatus* ClpS
PNMTSDRAWELTNQVHYEGQAIVWVG PQEQAELYHEQLL RAGLTMAPLEPE
Escherichia coli F, W, Y, L 13
MGKTNDWLDFDQLAEEKVRDAL KPPSMYKVILVNDDYTP ClpS
MEFVIDVLQKFFSYDVERATQLMLAVHYQGKAICGVFTA
EVAETKVAMVNKYARENEHPLLCTLEKA *Escherichia coli* F, W, Y, L 14
MGKTNDWLDFDQLAEEKVRDAL KPPSMYKVILVNDDYTP ClpS M40A
AEFVIDVLQKFFSYDVERATQLMLAVHYQGKAICGVFTA
EVAETKVAMVNKYARENEHPLLCTLEKA *Plasmodium* F, W, Y, L, 15
MFKDLKPFFLCIILLLLIIYKCTHSYNIKNKNCPLNFMN *falciparum* ClpS I
SCVRINNVNKNTNISFPKELQKRPSLVYSQKNFNLEKIK
KLRNVIKEIKKDNIKEADEHEKKEREKETSAWKVILYND
DIHNFTYVTDVIVKVVGQISKAKAHTITVEAHSTGQALI
LSTWKS KAEKYCQELQQNGLT VSIHESQLKDKQKK *Escherichia coli* K, R 16
MRLVQLSRHSIAFPSPEGALREPNGLLALGGDLSPARLL leucyl/phenylalanyl-
MAYQRGIFPWFSPGDPILWWSPDPRAVLWPESLHISRSM tRNA-protein
KRFHKRSPYRVTMNYAFGQVIEGCASDREEGTWITRGVV transferase
EAYHRLHELGHHSIEVWREDELVGGMYGVAQGT LFCGE
SMFSRMENASKTALLVFCEEFIGHGGKLIDCQVLNDHTA
SLGACEIPRRDYLNYLNQMRLGRLPNNFWVPRCLFSPQE LE *Vibrio vulnificus* D, E 17
MSSDIHQIKIGLTDNHPCSYLPERKERVAVALEADMHTA Aspartate/glutamate
DNYEVLLANGFRRSGNTIYKPHCDSCHSCQPIRISVPDI leucyltransferase
ELSRSQKRL LAKARSLSWSMKRNMDENWFDLYSRYIVAR Bpt
HRNGTMYPPKKDDFAHFSRNQWLTTQFLHIYEGQRLIAV
AVTDIMDHCASAFYTF FEPEHEL SLGTLAVLFQLEFCQE
EKKQWLYLGYQIDEC PAMNYKVRFH RHQKLVNQRWQ *Saccharomyces* K, R, H 18
MGSVHKHTGRNCGRKFKIGEPLYRCHECGDDTCVLCIH *cerevisiae* UBR1
CFNPKDHVNHVCTDICTEFTSGICDCGDEEAWNSPLHC KAE EQ *H. sapiens* GID4 P 19
MSGSKFRGHQKSKGNSYDVEVVLQHVD TGNSYLCGYLKI
KGLTEEYPTLTTFEGEIISKHPFLTRKWD ADEDVDRK
HWGKFLAFYQYAKSFNSDDFDYEELKNGDYVFMRWKEQF
LVPDHTIKDISGASFAGFY YICFQKSAASIEGYYYHRSS EWYQSLNLTHV *Saccharomyces* P
20 MINNPKVDSVAEKPKAVTSKQSEQAASPEPTAPPVSRN *cerevisiae* GID4
QYPITFNL TSTAPFHLHDRHRYLQE QDLYKCASRDSLSS
LQQLAHTPNGSTRKKYIVEDQSPYSEN PVIVTSSYNHT
VCTNYLRPRMQFTGYQISGYKRYQVTVNLKTVDLPKKDC
TSLSPHL SGFLSIRGLTNQHPEISTYFEAYAVNHKELGF
LSSSWKDEPVLNEFKATDQTDLEHWINFPSFRQLFLMSQ
KNGLNSTDDNGTTNA AKKLPPQQLPTTPSADAGNISRI F
SQEKQEDNYLNERFIFMKWKEKFLVPDALLMEGV D GASY
DGFYYIVHDQVTGNIQGFYYHQDAEKFQQL ELVPSLKNK VESSDCSFEFA Single-chain
phospho-Y 21 MMEVQLQQSGPELVKPGASVMISCRTSAYTFTENTVHWV antibody variable

KQSHGESLWYGGISFSPKFKATLTVDKSS fragment (scFv)
STAYMELRSLTSEDSAVYYCARRAGAYYFDYWGQGTTLT against
VSSGGGSGGGSGGGSENVLTQSPAIMASAPGEKVTMTCR phosphotyrosine**
ASSSVSSSYLHWYRQKSGASPKLWIYSTSNLASGVPARF
SGSGSGTSYSLTISSVEAEDAATYYCQQYSGYRTFGGGT KLEIKR *H. sapiens* Fyn
phospho-Y 22 MGAMDSIQAEWYEGKLGRKDAERQLLSFGNPRGTFLIR SH2 domain**
ESETTKGAYSLSIRDWDDMKGDHVKHYKIRKLDNGGYI
TTRAQFETLQQLVQHYSERAAGLSSRLVVP SHK *H. sapiens* Fyn phospho-Y 23
MGAMDSIQAEWYEGKLGRKDAERQLLSFGNPRGTFLIR SH2 domain triple
ESETVKGAYALSIRDWDDMKGDHVKHYLIRKLDNGGYI mutant
TTRAQFETLQQLVQHYSERAAGLSSRLVVP SHK superbinder** *H. sapiens* Src phospho-Y
24 MGAMDSIQAEWYEGKITRRESERLLLNAENPRGTFLVR tyrosine kinase
ESETTKGAYSLSVSDFDNAKGLNVKHYKIRKLDSSGGFYI SH2 domain**
TSRTQFNSLQQLVAYYSKHADGLCHRLTTVCPTSK *H. sapiens* Src phospho-Y 25
MGAMDSIQAEWYEGKITRRESERLLLNAENPRGTFLVR tyrosine kinase
ESEVTKGAYALSVDSDFDNAKGLNVKHYLIRKLDSSGGFYI SH2 domain triple
TSRTQFNSLQQLVAYYSKHADGLCHRLTTVCPTSK mutant** *H. sapiens* p62 K, R, H,
26 MASLTVKAYLLGKEDAAREIRRF SFCCSPEPEAEAEAAA fragment 1-310 W, F, Y
GPGPCERLLSRVAALFPALRPGGFQAHYRDEDGDLVAFS
SDEELTMAMSYVKDDIFRIYIKEKKECRRDHRPPCAQEA
PRNMVHPNVICDGCNGPVVGTRYKCSVCPDYDLCSVCEG
KGLHRGHTKLAFSPFGHLSEGFSSRWLRKVKHGHFGW
PGWEMGPPGNWSPRPPRAGEARPGPTAESASGPSEDPSV
NFLKNVGESVAAALSPLGIEVDIDVEHGGKRSRLTPVSP
ESSSTEEKSSSQPSSCCSDPSKPGGNVEGATQSLAEQ *H. sapiens* p62 K, R, H, 27
MASLTVKAYLLGKEDAAREIRRF SFCCSPEPEAEAEAAA fragment 1-180 W, F, Y
GPGPCERLLSRVAALFPALRPGGFQAHYRDEDGDLVAFS
SDEELTMAMSYVKDDIFRIYIKEKKECRRDHRPPCAQEA
PRNMVHPNVICDGCNGPVVGTRYKCSVCPDYDLCSVCEG
KGLHRGHTKLAFSPFGHLSEGFSSRWLRKVKHGHFGW
PGWEMGPPGNWSPRPPRAGEARPGPTAESASGPSEDPSV
NFLKNVGESVAAALSPLGIEVDIDVEHGGKRSRLTPVSP
ESSSTEEKSSSQPSSCCSDPSKPGGNVEGATQSLAEQ *H. sapiens* p62 K, R, H, 28
MASLTVKAYLLGKEDAAREIRRF SFCCSPEPEAEAEAAA fragment 126-180 W, F, Y
GPGPCERLLSRVAALFPALRPGGFQAHYRDEDGDLVAFS
SDEELTMAMSYVKDDIFRIYIKEKKECRRDHRPPCAQEA
PRNMVHPNVICDGCNGPVVGTRYKCSVCPDYDLCSVCEG
KGLHRGHTKLAFSPFGHLSEGFSSRWLRKVKHGHFGW
PGWEMGPPGNWSPRPPRAGEARPGPTAESASGPSEDPSV
NFLKNVGESVAAALSPLGIEVDIDVEHGGKRSRLTPVSP
ESSSTEEKSSSQPSSCCSDPSKPGGNVEGATQSLAEQ *H. sapiens* p62 K, R, H, 29
MASLTVKAYLLGKEDAAREIRRF SFCCSPEPEAEAEAAA protein W, F, Y
GPGPCERLLSRVAALFPALRPGGFQAHYRDEDGDLVAFS
SDEELTMAMSYVKDDIFRIYIKEKKECRRDHRPPCAQEA
PRNMVHPNVICDGCNGPVVGTRYKCSVCPDYDLCSVCEG
KGLHRGHTKLAFSPFGHLSEGFSSRWLRKVKHGHFGW
PGWEMGPPGNWSPRPPRAGEARPGPTAESASGPSEDPSV
NFLKNVGESVAAALSPLGIEVDIDVEHGGKRSRLTPVSP
ESSSTEEKSSSQPSSCCSDPSKPGGNVEGATQSLAEQMR
KIALESEGRPEEQMESDNCSGGDDDWTHLSSKEVDPSTG

ELQSLQMPESQPEGLPESQEGPTGLKEAALYPHLPPEA
DPRLESLSQMLSMGFSDEGGWLTRLLQTKNYDIGAALD TIQYSKHPPPL *Rattus*
norvegicus K, R, H, 30 MASLTVKAYLLGKEEAAREIRRFSCFSPEPEAEAAAGP p62
protein W, F, Y GPCERLLSRVAVLFPALRPGGFQAHYRDEDGDLVAFSSD
EELTMAMSYVKDDIFRIYIKEKKECRREHRPPCAQEAR
MVHPNVICDGCNGPVVGTRYKCSVCPDYDLCSVCEGKGL
HREHSLKIFPNPFGHLSDSFHSRWLRKLKHGHFGWPGW
EMGPPGNWSRPPRAGDGRPCPTAESASAPSEDPNVNFI
KNVGESVAAALSPLGIEVDIDVEHGGKRSRLTPTSAESS
STGTEDKSGTQPSSCSSEVSKPDGAGEGPAQSLTEQMKK
IALESVGQPEELMESDNCSGGDDDWTHLSSKEVDPSTGE
LQSLQMPESQPEGLPESQEGPTGLKEAALYPHLPPEA
PRLIESLSQMLSMGFSDEGGWLTRLLQTKNYDIGAALDT IQYSKHPPPL *Saccharomyces*
P, M, V 31 MTSLNIMGRKFILERAKRNDNIEEIYTSAYVSLPSSTD *cerevisiae* GID10
RLPHFKAKEEDCDVYEEGTNLVGKNAKYTYRSLGRHLDF
LRPGLRFGGSQSSKYTYTYTVEVKIDTVNLPLYKDSRSLD
PHVTGTFTIKNLTPVLDKVVTLFEGYVINYNQFPLCSLH
WPAEETLDPYMAQRESDCSHWKRFHFGSDNWSLTERNF
GQYNHESAEFMNQRYTYLKWKERFLLDDEEQENQMLDDN
HHLEGASFEGFYVCLDQLTGSVEGYYYHPACELFQKLE LVPTNCDALNTYSSGFEIA
UBR-box domain K, R, H 32
MGPLGSLCGRVEKSGETTYSCRDCAIDPTCVLCMDCFQD from *Homo sapiens*
SVHKNHRYKMHTSTGGGFCDCGDTEAWKTGPFCVNHEP UBR1 UBR-box domain K,
R, H 33 MGPLGSLCGRVFKVGEPTYSCRDCAVDPTCVLCMECFLG from *Homo sapiens*
SIHRDHRYRMTTSGGGGFCDCGDTEAWKEGPYCQKHE UBR2 *Leishmania major* G 34
MSRNPNSDAAHAFWSTQPVPQTEDETEKIVFAGPMDEP N-
KTVADIPEEPYPIASTFEWWTPNMEAADDIHAIYELLRD meristoyltrans-
NYVEDDDSMERFNYSEEFLQWALCPPNYIPDWHVAVRRK ferase
ADKKLLAFIAGVPVTLRMGTPKYMKVKAQEKGEGEAAK
YDEPRHICEINFLCVHKQLREKRLAPILIKEATRNVNRT
NVWQAVYTAGVLLPTPYASGQYFHRSLNPEKLVEIRFSG
IPAQYQKFQNPAMMLKRNYQLPSAPKNSGLREMKPSDVP
QVRRILMNYLDSFDVGPVFSDAEISHYLLPRDGVVFTYV
VENDKKVTDFFSFYRIPSTVIGNSNYNLLNAAYVHYAA
TSIPLHQLILDLLIVAHSRGEDVCNMVEILDNRSFVEQL
KFGAGDGHRLRYFYFNWAYPKIKPSQVALVML *H. sapiens* N- G 35
MADESETAVKPPAPPLPQMMEGNGNGHEHCSDCENEEDN meristoyltrans-
SYNRGGLSPANDTGAKKKKKKQKKKKEKGSETDSAQDQP ferase NMT1
VKMNSLPAERIQEIQAIELFSVGQGPAKTMEEASKRSY
QFWDTPQVPKLGEVVNTHGPVEPDKNIRQEPYTL PQGF
TWDALDLGDRGVLELYTLLNENYVEDDDNMFRFDYSPE
FLLWALRPPGWL PQWHCGVRVVSRLVGFISAIPANIH
IYDTEKKMVEINFLCVHKKLRSKRVAPVLIREITRRVHL
EGIFQAVYTAGVVL PKPVGTCRYWHRSLNPRKLIEVKFS
HLSRNMTMQRTMKLYRLPETPKTAGLRPMETKDIPVVHQ
LLTRYLKQFHLTPVMSQEEVEHWFYPQENIIDTFVVENA
NGEVTDFLSFYTL PSTIMNHPHSLKAAYSFYNVHTQT
PLLDLMSDALVLAKMKGFDVFNALDLMENKTFLEKLFKFG
IGDGNLQYYLYNWKCPSMGAEKVGLVLQ *Drosophila* A 36
MGDVQPETCRPSAASGNYFPQYPEYAIETARLRTFEAWP *melanogaster* BIR2

RNLKQKQHALWLSQCRFVKLMKGQLYIDTVAAKPVLAAEEK EESTSIGGDT *Binding

preferences are inferred from published scientific literature and/or further demonstrated by the inventors in single-molecule experiments, as described herein. **Binding to phosphotyrosine may occur at a peptide terminus or at an internal position.

[0111] Accordingly, in some embodiments, the application provides an amino acid recognition molecule having an amino acid sequence selected from Table 1 (or having an amino acid sequence that has at least 50%, at least 60%, at least 70%, at least 80%, 80-90%, 90-95%, 95-99%, or higher, amino acid sequence identity to an amino acid sequence selected from Table 1). In some embodiments, an amino acid recognition molecule has 25-50%, 50-60%, 60-70%, 70-80%, 80-90%, 90-95%, or 95-99%, or higher, amino acid sequence identity to an amino acid recognition molecule listed in Table 1. In some embodiments, an amino acid recognition molecule is a modified amino acid recognition molecule and includes one or more amino acid mutations relative to a sequence set forth in Table 1.

[0112] In some embodiments, an amino acid recognition molecule comprises a tag sequence that provides one or more functions other than amino acid binding. For example, in some embodiments, a tag sequence comprises a biotin ligase recognition sequence that permits biotinylation of the recognition molecule (e.g., incorporation of one or more biotin molecules, including biotin and bis-biotin moieties). Additional examples of functional sequences in a tag sequence include purification tags, cleavage sites, and other moieties useful for purification and/or modification of recognition molecules. Table 2 provides a list of non-limiting sequences of terminal tag sequences, any one or more of which may be used in combination with any one of the amino acid recognition molecules of the application (e.g., in combination with a sequence set forth in Table 1). It should be appreciated that the tag sequences shown in Table 2 are meant to be non-limiting, and recognition molecules in accordance with the application can include any one or more of the tag sequences (e.g., His-tags and/or biotinylation tags) at the N- or C-terminus of a recognition molecule polypeptide, split between the N- and C-terminus, or otherwise rearranged as practiced in the art.

TABLE-US-00002 TABLE 2 Non-limiting examples of terminal tag sequences. SEQ ID

Name NO: Sequence Biotinylation 37 GGGSGGGSGGGSGLNDFFEAQKIEWHE tag Bis- 38

GGGSGGGSGGGSGLNDFFEAQKIEWHE biotinylation

GGGSGGGSGGGSGLNDFFEAQKIEWHE tag Bis- 39

GSGGGSGGGSGGGSGLNDFFEAQKIEW biotinylation

HEGGSGGGSGGGSGLNDFFEAQKIEW tag HE His/ 40

GHHHHHHHHHHHGGGSGGGSGGGSGLNDFFEAQKIEWHE tag His/bis- 41

GHHHHHHHHHHHGGGSGGGSGGGSGLNDFFEAQKIEWHE biotinylation

FFEAQKIEWHEGGGSGGGSGGGSGLNDFFEAQKIEWHE His/bis- 42

GGSHHHHHHHHHHGGGSGGGSGGGSGLNDFFEAQKIEWHE biotinylation

NDFFEAQKIEWHEGGGSGGGSGGGSGLNDFFEAQKIEWHE His/bis- 43

GSHHHHHHHHHHGGGSGGGSGGGSGLNDFFEAQKIEWHE biotinylation

DFFEAQKIEWHEGGGSGGGSGGGSGLNDFFEAQKIEWHE Bis- 44

GGGSGGGSGGGSGLNDFFEAQKIEWHE biotinylation/

GGGSGGGSGGGSGLNDFFEAQKIEWHE His tag GHHHHHHH

[0113] In some embodiments, a recognition molecule or affinity reagent of the application is a peptidase. A peptidase, also referred to as a protease or proteinase, is an enzyme that catalyzes the hydrolysis of a peptide bond. Peptidases digest polypeptides into shorter fragments and may be generally classified into endopeptidases and exopeptidases, which cleave a polypeptide chain internally and terminally, respectively. In some embodiments, labeled affinity reagent **100** comprises a peptidase that has been modified to inactivate exopeptidase or endopeptidase activity. In this way, labeled affinity reagent **100** selectively binds without also cleaving the amino acid from a polypeptide. In yet other embodiments, a peptidase that has not been modified to inactivate

exopeptidase or endopeptidase activity may be used. For example, in some embodiments, a labeled affinity reagent comprises a labeled exopeptidase **102**.

[0114] In accordance with certain embodiments of the application, protein sequencing methods may comprise iterative detection and cleavage at a terminal end of a polypeptide. In some embodiments, labeled exopeptidase **102** may be used as a single reagent that performs both steps of detection and cleavage of an amino acid. As generically depicted, in some embodiments, labeled exopeptidase **102** has aminopeptidase or carboxypeptidase activity such that it selectively binds and cleaves an N-terminal or C-terminal amino acid, respectively, from a polypeptide. It should be appreciated that, in certain embodiments, labeled exopeptidase **102** may be catalytically inactivated by one skilled in the art such that labeled exopeptidase **102** retains selective binding properties for use as a non-cleaving labeled affinity reagent **100**, as described herein.

[0115] An exopeptidase generally requires a polypeptide substrate to comprise at least one of a free amino group at its amino-terminus or a free carboxyl group at its carboxy-terminus. In some embodiments, an exopeptidase in accordance with the application hydrolyses a bond at or near a terminus of a polypeptide. In some embodiments, an exopeptidase hydrolyses a bond not more than three residues from a polypeptide terminus. For example, in some embodiments, a single hydrolysis reaction catalyzed by an exopeptidase cleaves a single amino acid, a dipeptide, or a tripeptide from a polypeptide terminal end.

[0116] In some embodiments, an exopeptidase in accordance with the application is an aminopeptidase or a carboxypeptidase, which cleaves a single amino acid from an amino- or a carboxy-terminus, respectively. In some embodiments, an exopeptidase in accordance with the application is a dipeptidyl-peptidase or a peptidyl-dipeptidase, which cleave a dipeptide from an amino- or a carboxy-terminus, respectively. In yet other embodiments, an exopeptidase in accordance with the application is a tripeptidyl-peptidase, which cleaves a tripeptide from an amino-terminus. Peptidase classification and activities of each class or subclass thereof is well known and described in the literature (see, e.g., Gurupriya, V. S. & Roy, S. C. *Proteases and Protease Inhibitors in Male Reproduction. Proteases in Physiology and Pathology* 195-216 (2017); and Brix, K. & Stöcker, W. *Proteases: Structure and Function. Chapter 1*). In some embodiments, a peptidase in accordance with the application removes more than three amino acids from a polypeptide terminus. Accordingly, in some embodiments, the peptidase is an endopeptidase, e.g., that cleaves preferentially at particular positions (e.g., before or after a particular amino acid). In some embodiments, the size of a polypeptide cleavage product of endopeptidase activity will depend on the distribution of cleavage sites (e.g., amino acids) within the polypeptide being analyzed.

[0117] An exopeptidase in accordance with the application may be selected or engineered based on the directionality of a sequencing reaction. For example, in embodiments of sequencing from an amino-terminus to a carboxy-terminus of a polypeptide, an exopeptidase comprises aminopeptidase activity. Conversely, in embodiments of sequencing from a carboxy-terminus to an amino-terminus of a polypeptide, an exopeptidase comprises carboxypeptidase activity. Examples of carboxypeptidases that recognize specific carboxy-terminal amino acids, which may be used as labeled exopeptidases or inactivated to be used as non-cleaving labeled affinity reagents described herein, have been described in the literature (see, e.g., Garcia-Guerrero, M. C., et al. (2018) *PNAS* 115 (17)).

[0118] Suitable peptidases for use as cleaving reagents and/or affinity reagents (e.g., recognition molecules) include aminopeptidases that selectively bind one or more types of amino acids. In some embodiments, an aminopeptidase recognition molecule is modified to inactivate aminopeptidase activity. In some embodiments, an aminopeptidase cleaving reagent is non-specific such that it cleaves most or all types of amino acids from a terminal end of a polypeptide. In some embodiments, an aminopeptidase cleaving reagent is more efficient at cleaving one or more types of amino acids from a terminal end of a polypeptide as compared to other types of amino acids at

the terminal end of the polypeptide. For example, an aminopeptidase in accordance with the application specifically cleaves alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, proline, selenocysteine, serine, threonine, tryptophan, tyrosine, and/or valine. In some embodiments, an aminopeptidase is a proline aminopeptidase. In some embodiments, an aminopeptidase is a proline iminopeptidase. In some embodiments, an aminopeptidase is a glutamate/aspartate-specific aminopeptidase. In some embodiments, an aminopeptidase is a methionine-specific aminopeptidase. In some embodiments, an aminopeptidase is an aminopeptidase set forth in Table 3. In some embodiments, an aminopeptidase cleaving reagent cleaves a peptide substrate as set forth in Table 3.

[0119] In some embodiments, an aminopeptidase is a non-specific aminopeptidase. In some embodiments, a non-specific aminopeptidase is a zinc metalloprotease. In some embodiments, a non-specific aminopeptidase is an aminopeptidase set forth in Table 4. In some embodiments, a non-specific aminopeptidase cleaves a peptide substrate as set forth in Table 4.

[0120] Accordingly, in some embodiments, the application provides an aminopeptidase (e.g., an aminopeptidase recognition molecule, an aminopeptidase cleaving reagent) having an amino acid sequence selected from Table 3 or Table 4 (or having an amino acid sequence that has at least 50%, at least 60%, at least 70%, at least 80%, 80-90%, 90-95%, 95-99%, or higher, amino acid sequence identity to an amino acid sequence selected from Table 3 or Table 4). In some embodiments, an aminopeptidase has 25-50%, 50-60%, 60-70%, 70-80%, 80-90%, 90-95%, or 95-99%, or higher, amino acid sequence identity to an aminopeptidase listed in Table 3 or Table 4. In some embodiments, an aminopeptidase is a modified aminopeptidase and includes one or more amino acid mutations relative to a sequence set forth in Table 3 or Table 4.

TABLE-US-00003 TABLE 3 Non-limiting examples of aminopeptidases. SEQ ID Name NO: Sequence *L. pneumophila* M1 45

MMVKQGVFMKTDQSKVKKLSDYKSLDYFVIHVDLQIDLSKKPVESK Aminopeptidase
ARLTVVPNLNVDSHSNDLVLDGENMTLVSLQMNDNLLKENEYELTK (Glu/Asp Specific)
DSLIIKNIPQNTPTFTIEMTSLLGENTDLFGLYETEGVALVKAASEG
LRRVFYLPDRPDNLATYKTTIIANQEDYPVLLSNGVLIEKKELPLG
LHSVTLWDDVPKPSYLFALVAGNLQRSVTYYQTKSGRELPIEFYVP
PSATSKCDFAKEVLKEAMAWDERTFNLECALRQHMVAGVDKYASGA
SEPTGLNLFNTENLFASPETKTDLGILRVLEVVAHEFFHYWSGDRV
TIRDWFNLPLKEGLTTFRAAMFREELFGTDLIRLLDGKNLDERAPR
QSAYTAVRSLYTAAAYEKSADIFRMMMLFIGKEPFIEAVAKFFKDN
DGGAVTLEDIESISNSSGKDLRSFSLSWFTESGIPELIVTDELNPD
TKQYFLKIKTVNGRNRPIPILMGLLDSSGAEIVADKLLIVDQEEIE
FQFENIQTRPIPSLLRSFSAPVHMKYEYSYQDLLLLMQFDTNLYNR
CEAAKQLISALINDFCIGKKIELSPQFFAVYKALLSDNSLNEWMLA
ELITLPSLEELIENQDKPDFEKLNEGRQLIQNALANELKTDFYNLL
FRIQISGDDDKQKLKGFDLKQAGLRRLKSVCFSYLLNVDFEKTKEK
LILQFEDALGKNMTETALASMLCEINCEEADVALEDYYHYWKNNDP
GAVNNWFSIQALAHSPDVIERVKKLMRHGDFDLNPNKVYALLGSF
IKNPFGFHSVTGEGYQLVADAIFDLDKINPTLAANLTEKFTYWDKY
DVNRQAMMISTLKIIYSNATSSDVRTMAKKGLDKVKEDLPLPIHLT
FHGGSTMQDRTAQLIADGNKENAYQLH *E. coli* methionine 46
MGTAISIKTPEDIEKMRVAGRLAAEVLEMIEPYVKPGVSTGELDRI aminopeptidase
CNDYIVNEQHAVSACLGYPKSVCSINEVVCHGIPDDAKLLKD (Met specific)
GDIVNIDVTVIKDGFGDTSKMFIVGKPTIMGERLCRITQESLYLA
LRMVKPGINLREIGAAIQKFVEAEGFSVVREYCGHGIGRGEHEEPQ
VLHYDSRETNVVLKPGMTFTIEPMVNAGKKEIRTMKDGWTVKTKDR

SLSAQYEYTIIVTDNGCEITLRLKIDTIPAIISHD *M. smegmatis* Proline 47
MGTLEANTNGPGSMLSRMPVSSRTVPFGDHETWVQVTTPENAQPHA iminopeptidase
LPLIVLHGGPGMAHNYVANIALADETGRTVIHYDQVGCGNSTHLP (Pro specific)
DAPADEWTPQLFVDEFHAVCTALGIERYHVLGQSWGGMLGAEIAVR
QPSGLVSLAICNSPASMRLWSEAAGDLRAQLPAETRAALDRHEAAG
TITHPDYLQAAAEFYRRHVCRVVPTPQDFADSVAQMEAEPTVYHTM
NGPNEFHVVGTLDGWSVIDRLPDVTAPVLVIAGEHDEATPKTWQPF
VDHIPDVRSHVFPGTSHCTHLEKPEEFRAVVAQFLHQHDLAADARV *Y. pestis* Proline 48
MTQQEYQNRRLQALLAKMAPGSAIIFAAPEATRSDSEYPYRQNSD iminopeptidase
FSYLTGFNEPEAVLILVKSEETHNHSVLFNRIRDLTAEIWFGRRLG (Pro Specific)
QEAAPTKLAVDRALPFDEINEQLYLLLNRDLVVIYHAQQGYAYADNI
VFAALEKLRHGERKNLRAPATLIDWRPWLHEMRLEKSAAEIIAVLRR
AGEISALAHTRAMEKCRPGMFEYQLEGEILHEFTRHGARYPAYNTI
VGGGENGCILHYTENECELRDGDLVLIDAGCEYRGYAGDITRTFPV
NGKFTPAQRAVYDIVLAAINKSLTLFRPGTSIREVTEEVVRIMVVG
LVELGILKGDIEQLIAEQAHRPFFMHGLSHWLGMVDVHDVGDYGDSSD
RGRILEPGMVLTVEPGLYIAPDADVPPQYRGIGIRIEDDIVITATG
NENLTASVVKDPDDIEALMALNHAGENLYFQE *P. furiosus* 49
MDTEKLMKAGEIAKKVREKAIKLARPGMLLLELAESIEKMIMELGG methionine
KPAFPVNLISINEIAAHYTPYKGDTTVLKEGDYLIKIDVGVHIDGFIA aminopeptidase
DTAVTVRVGMEEDELMEAAKEALNAAISVARAGVEIKELGKAIENE
IRKRGFKPIVNLSGHKIERYLHAGISIPNIYRPHDNYVLKEGDVF
AIEPFATIGAGQVIEVPPTLIYMYVRDVPVRVAQARFLLAKIKREY
GTLPFAYRWLQNDMPEGQLKLALKTLEKAGAIYGYPVVLKEIRNGIV
AQFEHTIIVEKDSVIVTQDMINKSTLE *Aeromonas sobria* 50
HMSSPLHYVLDGIHCEPHFFTVPDLHQQPDDDEETITLFGRTLCKRD Proline
RLDDELPWLLYLQGGPGFGAPRPSANGGWIKRALQEFRVLLLDQRG aminopeptidase
TGHSTPIHAELLAHLNPRQQADYLSHFRADSIVRDAELIREQLSPD
HPWSLLGQSFGGFCSLTYLSLFPDSLHEVYLTGGVAPIGRSADEVY
RATYQRVADKNRAFFARFPHAQAIANRLATHLQRHDTVRLPNGQRLT
VEQLQQQGLDLGASGAFEELYLLEDAFIGEKLNPFLYQVQAMQP
FNTNPVFAILHELIYCEGAASHWAAERVRGEFPALAWAQGKDFAT
GEMIFPWMFEQFRELIPKEAAHLLAEKADWGPLYDPVQLARNKVP
VACAVYAEDMYVEFDYSRETLKGLSNSRAWITNEYEHNGLRVDGEQ ILDRILRLNRDCLE
Pyrococcus furiosus 51
MKERLEKLVKFMDENSIDRVFIAKPVNVYYYFSGTSPLGGGYIIVDG Proline
DEATLYVPELEYEMAKEESKLPVVKFKKFDEIYEILKNTETLGIEG Aminopeptidase (X/-
TLSYSMVENFKEKSNVKEFKKIDDVIKDLRIIKTKEEIEIEKACE Pro)
IADKAVMAAIEEITEGKREREVA AKVEYLMKMNGAEKPAFDTHAS
GHRALPHGVASDKRIERGDLVVIDLGALYNHYNDSITRTIVVGSP
NEKQREIYEIVLEAQKRAVEAAKPGMTAKELDSIAREIIKEYGYGD
YFIHSLGHGVGLEIHEWPRISQYDETIVLKEGMVITIEPGIYIPKLG
GVRIEDTVLITENGAKRLTKTERELL *Elizabethkingia* 52
MIPITTPVGNEKVWTKREGTNP KIKVLLLHGGPAMTHEYMECFETF *meningoseptica*
FQREGFEEYEYDQLGSYYSDQPIDEKLNWIDRFVDEVEQVRKAIHA Proline
DKENFYVLGNSWGGILAMEYALKYQQNLKGLIVANMMASAPEYVKY aminopeptidase
AEVLSKQMKPEVLAEVRAIEAKKDYANPRYTELLFPNYAQHICRL
KEWPDALNRSLKHVNSTVYTLMQGPSELGMSSDARLAKWDIKNRLH
EIAPTLMIGARYDTMDPKAMEEQSKLVQKGRYLYCPNGSHLAMWD
DQKVFMDGVIFIKDVDTKSFN *N. gonorrhoeae* 53

MYEIKQPFHSGVQVSEIYHQIYQWESHPVIFLHGGPGAGAS Proline
PECRGFFNPDPVFRIVIIDQRGCGRSHPHYACAEDNTTWDLVADIEKV Iminopeptidase
REMLGIGKWLTVFGGSWGSTLSLAYAQTHPERVKGLVLRGIFLCRPS
ETAWLNEAGGVSRIYPEQWQKFVAPIAENRRNRRLIEAYHGLLFHQD
EEVCLSAKAWADWESYLIRFEPEGVDEDAYASLAIAARLENHYFVN
GGWLQGDKAILNNIGKIRHIPTVIVQGRYDLCTPMQSAWELSKAFP
EAELRVVQAGHCAFDPLADALVQAVEDILPRL
TABLE-US-00004 TABLE 4 Non-limiting examples of non-specific aminopeptidases.
SEQ ID Name NO: Sequence *E. coli* 54
MTQQPQAKYRHDYRAPDYQITDIDLTFDLDAQKTVVTAVSQAVRHG Aminopeptidase N*
ASDAPLRLNGEDLKLVSVINDEPWTAWKEEEGALVISNLPERFTL (Zinc
KIINEISPAANTALEGLYQSGDALCTQCEAEGFRHITYYLDRLPDVL Metalloprotease)
ARFTTKIADKIKYPFLLSNGNRVAQGELENGRHVWVQWQDPFPKPC
YLFALVAGDFDVLRLDTFTTRSGREVALELYVDRGNLDRAPWAMTSL
KNSMKWDEERFGLDYDLIYMIVAVDFFNMGAMENKGLNIFNSKYV
LARTDTATDKDYLDIERVIGHEYFHNWTGNRVTCRDWFQLSLKEGL
TVFRDQEFSSDLGSRVNRINNVRTMRGLQFAEDASPMAPHPRPDM
VIEMNNFYTLTVYEKGAEVIRMIHTLLGEENFQKGMQLYFERHDGS
AATCDDFVQAMEDASNVDLSHFRRWYSQSGTPIVTVKDDYNPETEQ
YTLTISQRTPATPDQAEKQPLHIPFAIELYDNEGKVIPLQKGGHPV
NSVLNVTQAEQTFVFDNVYFQVPALLCEFSAPVKLEYKWSQQQLT
FLMRHARNDFSRWDAQAQLLATYIKLNVARHQQGQPLSLPVHVADA
FRAVLLDEKIDPALAAEILTLPVSNEMAELFDIIDPTAIAEVREAL
TRTLATELADELLAIYNANYQSEYRVEHEDIAKRTLRLNACLRFLAF
GETHLADVLSKQFHEANNMTDALAALSAVAQAQLPCRDALMQEYD
DKWHQNGLVMDKWFIQATSPAANVLETVRGLLQHRSTMSNPNR
RSLIGAFAGSNPAAFAEDGSGYLFLVEMLTDLNSRNPQVASRLIE
PLIRLKRYDAKRQEKMRALAEQLKGLNLSGDLYEKITKALA *P. falciparum* M1 55
PKIHYRKDYKPSGFIINQVTLNINIHDQETIVRSVLDMDISKHNVG aminopeptidase**
EDLVFDGVGLKINEISINNKKLVEGEEYTYDNEFLTIFSKFVPSK
FAFSSEVIIHPETNYALTGLYKSKNIIVSQCEATGFRRITFFIDRP
DMMAKYDVTVTADKEKYPVLLSNGDKVNEFEIPGGRHGARFNDPPL
KPCYLFAVVAGDLKHLSATYITKYTKKKVELYVFSEEKYVSKLQWA
LECLKKSMAFDEDYFGLDYDL SRLNLVAVSDFNVGAMENKGLNIFN
ANSLASKKNSIDFSYARILTVVGHEYFHQYTGNRVTLRDWFQTL
KEGLTVHRENLFSEEMTKTVTTRL SHVDLLRSVQFLEDSSPLSHPI
RPESYVSMENFYTTTVYDKGSEVMRMYLTILGEEYKKGFDIYIKK
NDGNTATCEDFNYAMEQAYKMKKADNSANLNQYLLWFSQSGTTPHVS
FKYNYDAEKKQYSIHVNQYTKPDENQKEKKPLFIPISVGLINPENG
KEMISQTTLELTKESDTFVFNNIAVKPIPSLFRGFSAPVYIEDQLT
DEERILLKYDSDAFVRYNSCTNIYMKQILMNYNEFLKAKNEKLES
FQLTPVNAQFIDAIKYLLDPHADAGFKSYIVSLPQDRYIINFVSN
LDTDVLADTKEYIYKQIGDKLNDVYYKMFKSLEAKADDLTIFYNDES
HVDQDQMNMRTRLRNTLLSLLSKAQYPNILEHHSKSPYPSNWL
SLSVSAYFDKYFELYDKTYKLSKDDELLEQLWLVSRSDRKDIYE
ILKKLENEVLKDSKNPNDIRAVYLPFTNNLRRFHDISGKGYKLIAE
VITKTDKFNPMVATQLCEPFLWNLKLDTKRQELMLNEMNTMLQEPQ
ISNNLKEYLLRLTNK Puromycin-sensitive 56
MWLAAAAPSLARRLLFLGPPPPPLLLLVSRSRRRLHSLGLAAMP aminopeptidase
EKRPFERLPADVSPINYSCLKPDLLDFTFEGKLEAAAQVRQATNQ (“NPEPPS”)

IVMNCADIDIITASYAPEGDEEIHATGFNYQNEDEKVTL SFPSTLQ
TGTGTLKIDFVGELNDKMKGFYRSKYTTSPSGEVRYA AVTQFEATDA
RRAFPCWDEPAIKATFDISLVVPKDRVALSNMNVIDRKPYPPDENL
VEVKFARTPVMSTYLVAFVVG EYDFVETR SKDGV CVRVYTPVGKAE
QGKFALEVA AKTLPFYKDYFNVPYPLPKIDLIAIADFAAGAMENWG
LVTYRETALLIDPKNSC SSSRQWVALVVGHEL AHQWFGNLVTMEWW
THLWLNEG FASWIEYLCVDHCFPEYDIWTQFVSADYTRAQELDALD
NSHPIEVS VGH PSEVDEIFDAISYSK GASVIRMLHDYIGDKDFKKG
MNMYLTKFQQKNAATEDLWESLENASGKP IAAVMNTWTKQMGFPLI
YVEAEQVEDDRLLRLS QKKFCAGGSYVGEDCPQWMVPITISTSEDP
NQAKLKILMDKPEMNVVLKNVKPDQWVKLNLGTVG FYRTQYSSAML
ESLLPGIRDLSLPPVDRLGLQNDLFS LARAGIISTVEVLKVMEAFV
NEPNYTVWSDLSCNLGILSTLLSHTDEYEEIQEFVKD VESPIGERL
GWDPKPGEGHLDALLRGLVLGKLGKAGHKATLEEARRRFKDHVEGK
QILSADLRSPVYLTVLKHGDGTTLDIMLKLHKQADMQEEKNR IERV
LGATLLPDLIQKVLTFALSEEVRPQDTVSVIGGVAGGSKHGRKAAW
KFIKDNWEELYNRYQGGFLISRLIKLSVEGFAVDKMAGEVK AFFES
HPAPSAERTIQCCENILLNAAWLKRDAESIHQYLLQRKASPPTV NPEPPS E366V 57
MWLAAAAPSLARLLFLGPPPPPLLLL VFSRSSRRRLHSLGLAAMP
EKRPFERLPADVSPINYS LCLKPDLLDFTFEGKLEAAAQVRQATNQ
IVMNCADIDIITASYAPEGDEEIHATGFNYQNEDEKVTL SFPSTLQ
TGTGTLKIDFVGELNDKMKGFYRSKYTTSPSGEVRYA AVTQFEATDA
RRAFPCWDEPAIKATFDISLVVPKDRVALSNMNVIDRKPYPPDENL
VEVKFARTPVMSTYLVAFVVG EYDFVETR SKDGV CVRVYTPVGKAE
QGKFALEVA AKTLPFYKDYFNVPYPLPKIDLIAIADFAAGAMENWG
LVTYRETALLIDPKNSC SSSRQWVALVVGHVLAHQWFGNLVTMEWW
THLWLNEG FASWIEYLCVDHCFPEYDIWTQFVSADYTRAQELDALD
NSHPIEVS VGH PSEVDEIFDAISYSK GASVIRMLHDYIGDKDFKKG
MNMYLTKFQQKNAATEDLWESLENASGKP IAAVMNTWTKQMGFPLI
YVEAEQVEDDRLLRLS QKKFCAGGSYVGEDCPQWMVPITISTSEDP
NQAKLKILMDKPEMNVVLKNVKPDQWVKLNLGTVG FYRTQYSSAML
ESLLPGIRDLSLPPVDRLGLQNDLFS LARAGIISTVEVLKVMEAFV
NEPNYTVWSDLSCNLGILSTLLSHTDFYEEIQEFVKD VFSP IGERL
GWDPKPGEGHLDALLRGLVLGKLGKAGHKATLEEARRRFKDHVEGK
QILSADLRSPVYLTVLKHGDGTTLDIMLKLHKQADMQEEKNR IERV
LGATLLPDLIQKVLTFALSEEVRPQDTVSVIGGVAGGSKHGRKAAW
KFIKDNWEELYNRYQGGFLISRLIKLSVEGFAVDKMAGEVK AFFES
HPAPSAERTIQCCENILLNAAWLKRDAESIHQYLLQRKASPPTV *Francisella tularensis* 58
MIYEFVMTDPKIKY LKDYKPSNYLIDETHLIFELDESKTRVTANLY Aminopeptidase N
IVANRENRENNTLVLDGVELKLLSIKLNNKHLSPAEF AVNENQLII
NNVPEKFVLQTVVEINPSANTSLEGLYKSGDV FSTQCEATGFRKIT
YYLDRPDVMAAFTVKI IADKKKYPIILSNGDKIDSGDISDNQHFAV
WKDPFKKPCYLFALVAGDLASIKDTYITKSQRKVSLEIYAFKQDID
KCHYAMQAVKDSMKWDEDRFGLEYDLDTFMIVAVPDFNAGAMENKG
LNIFNTKYIMASNKTATDKDFELVQSVVGHEYFHNWTGDRVICRDW
FQLSLKEGLTVFRDQEFTSDLNSRDVKRIDDVRIIRSAQFAEDASP
MSHPIRPESYIEMNNEYTVTVYNKGAEIIRMIHTLLGEEGFQKGMK
LYFERHDGQAVTCDDFVNAMADANNRDFSLFKRWYAQSGTPNIKVS
ENYDASSQTYSLTLEQTTLPTADQKEKQALHIPVKMGLINPEGKNI
AEQVIELKEQKQTYTFENIAAKPVASLFRDFSAPVKVEH KRSEKDL

[illegible]

HKAGFSAAMPFEAKFNDYNPNIHTPNDTLQNSDPTASHAVKFTKLG
 LAYAIEMASTTGGTPPTGNVLKDGVPVNGLSGATGSQVHYSFELP
 AQKNLQISTAGGSGDVDLYVSFGSEATKQNWDCRPYRNGNNEVCTF
 AGATPGTYSIMLDGYRQFSGVTLKASTQ *Yersinia pestis* 64
 MTQQPQAKYRHDYRAPDYTTITDIDLDFALDAQKTTVTAVSKVKRQG AminopeptidaseN
 TDVTPLILNGEDLTLSISVSDGQAWPHYRQQDNTLVIEQLPADFTL
 TIVNDIHPATNSALEGLYLSGEALCTQCEAEGFRHITYYLDPRDVL
 ARFTTRIVADKSRYPYLLSNGNRVGGQELDDGRHWVKWEDPFPKPS
 YLFALVAGDFDVLQDKFITRSGREVALEIFVDRGNLDRADWAMTSL
 KNSMKWDETRFGLLEYDLDIYMIVAVDFFNMGAMENKGLNVFNSKYV
 LAKAETATDKDYLNIEAVIGHEYFHNWTGNRVTCRDWFQLSLKEGL
 TVFRDQEFSSDLGSRSVNRIENVVRVMRAAQFAEDASPMHAIRPDK
 VIEMNNFYTLTVYEKGSEVIRMMHTLLGEQQFQAGMRLYFERHDGS
 AATCDDFVQAMEDVSNVDLSLFRRWYSQSGTPLLTVHDDYDVEKQQ
 YHLFVSQKTLPTADQPEKLPLHIPLDIELYDSKGNVIPLQHNGLPV
 HHVLNVTEAEQTFTFDNVAQKPIPSLLREFSAPVKLDYPYSDQQLT
 FLMQHARNEFSRWDAQAQSLLATYIKLNVAKYQQQQPLSLPAHVADA
 FRAILLDEHLDPALAAQILTLPSENEMAELFTTIDPQAISTVHEAI
 TRCLAQELSDELLAVYVANMTPVYRIEHGDIKRALRNTCLNYLAF
 GDEEFANKLVSLQYHQADNMTDSLAAALAAVAAQLPCRDELLAAFD
 VRWNHDGLVMDKWFALQATSPAANVLVQVRTLLKHPAFSLSNPNT
 RSLIGSFASGNPAAFHAADGSGYQFLVEILSDLNTRNPQVAARLIE
 PLIRLKRYDAGRQALMRKALEQLKTLNLSGDLYEKITKALAA *Vibrio anguillarum* 65
 MEEKVWISIGGDATQTALRSGAQSLLENLINQTSVWVGQVPVSEL Aminopeptidase
 ATLSHEMHENHQRCGGYMVHPSAQSAMSVSAMPLNLNAFSAPEITQ
 QTTVNAWLPSVSAQQITSTITTLTQFKNRFYTTSTGAQASNWIADH
 WRSLSASLPASKVEQITHSGYNQKSVMLTITGSEKPDEWVVGHL
 DSTLGSRTNESSIAPGADDDASGIAGVTEIIRLLSEQNFRPKRSIA
 FMAYAAEEVGLRGSQDLANRFKAEGKKVMSVMQLDMTNYQGSREDI
 VFITDYTDSNFTQYLTQLLDEYLPSTLYGFDTCGYACSDHASWHAV
 GYPAAMPFESKFNDYNPNIHSPQDTLQNSDPTGFHAVKFTKLGLAY
 VVEMGNASTPPTPSNQLKNGVPVNGLSASRNSKTWYQFELQEAGNL
 SIVLSGGSGDADLYVKYQTDADLQQYDCRPYRSGNNETCQFSNAQP
 GRYSILLHGYNNYSNASLVANAQ *Salinivibrio* spYCSC6 66
 MEDKKVWISIGADAQQTALSSGAQPLLAQSVAHNGQAWIGEVSESE Aminopeptidase
 LAALSHEMHENHHRCGGYIVHSSAQSAMAAASN MPLSRASFIAPAI
 QQALVTPWISQIDSALIVNTIDRLTDFPNREYTTTSGAQASDWIKQ
 RWQSLSAGLAGASVTQISHSGYNQASVMLTIEGSESPDEWVVVGHL
 LDSTIGSRTNEQSIAPGADDDASGIAAVTEVIRVLAQNNFQPKRSI
 AFVAYAAEEVGLRGSQDVANQFKQAGKDV RGV LQLDMTNYQGS AED
 IVFITDYTDNQLTQYLTQLLDEYLPSTLYGFDTCGYACSDHASWHQ
 VGYPAMPFEAKFNDYNPNIHTPQDTLANSDSEGAAHAKFTKLGLA
 YTVELANADSSPNPGNELKLGEPIGLSGARGNEKYFNYRLDQSGE
 LVIRTYGGSGDVDLYVKANGDVSTGNWDCRPYRSGNDEVCRFDNAT
 PGNYAVMLRGYRTYDNVSLIVE *Vibrio proteolyticus* 67
 MPPITQQATVTAWLPQVDASQITGTISSLESFTNRFYTTTSGAQAS Aminopeptidase I
 DWIASEWQALSASLPNASVKQVSHSGYNQKSVVMTITGSEAPDEWI
 VIGGHLDSTIGSHTNEQSVAPGADDDASGIAAVTEVIRVLSENN FQ
 PKRSIAFMAYAAEEVGLRGSQDLANQYKSEGKNVVSALQLDMTNYK
 GSAQDVVFITDYTDSNFTQYLTQLMDEYLPSTLYGFDTCGYACSDH

ASWHNAGYPAAMPFESKFNDYNPRIHTTQDTLANSDPTGSHAKKFT
 QLGLAYAIEMGSATGDTPTPGNQL *Vibrio proteolyticus* 68
 MPPITQQATVTAWLPQVDASQITGTISSLESFTNRFYTTTSGAQAS Aminopeptidase I
 DWIASEWQFLSASLPNASVKQVSHSGYNQKSVVMTITGSEAPDEWI (A55F)
 VIGGHLDSTIGSHTNEQSVAPGADDDASGIAAVTEVIRVLSENNFQ
 PKRSIAFMAYAAEEVGLRGSQDLANQYKSEGKNVVSALQLDMTNYK
 GSAQDVVFITDYTDSNFTQYLTQLMDEYLPSLTYGFDTCGYACSDH
 ASWHNAGYPAAMPFESKFNDYNPRIHTTQDTLANSDPTGSHAKKFT
 QLGLAYAIEMGSATGDTPTPGNQL *P. furiosus* 69
 MVDWELMKKIHESPGVSGYEH LGIRDLVVDILKDVADEVKIDKLG N Aminopeptidase I
 VIAHFKGSAPKVMVA AHMDKIGLMVNHIDKDG YLRVVP IGGVLPET
 LIAQKIRFFTEKGERYG VVGVLPPHLRREAKDQGGKIDWDSIIVDV
 GASSREEAEEMGFRIGTIGEFAPNFTRLSEHRFATPYLDDRICLYA
 MIEAARQLGEHEAD IYIVASVQEEIGLRGARVASFAIDPEVGIAMD
 VTFAKQPNDKGKIVPELGKGPVMDVGP NINPKLRQFADEVAKKYEI
 PLQVEPSRPTGTDANVMQINREGVATAVLSIPIRYMHSQVELADA
 RDVDNTIKLAKALLEELKPMDFTPLE *Cleavage efficiency (from most to least): arginine >
 lysine > hydrophobic residues (including alanine, leucine, methionine, and phenylalanine) >
 proline (see, e.g., Matthews Biochemistry 47, 2008, 5303-5311). **Cleavage efficiency (from most
 to least): leucine > alanine > arginine > phenylalanine > proline; does not cleave after
 glutamate and aspartate.

[0121] For the purposes of comparing two or more amino acid sequences, the percentage of
 “sequence identity” between a first amino acid sequence and a second amino acid sequence (also
 referred to herein as “amino acid identity”) may be calculated by dividing [the number of amino
 acid residues in the first amino acid sequence that are identical to the amino acid residues at the
 corresponding positions in the second amino acid sequence] by [the total number of amino acid
 residues in the first amino acid sequence] and multiplying by [100], in which each deletion,
 insertion, substitution or addition of an amino acid residue in the second amino acid sequence
 compared to the first amino acid sequence is considered as a difference at a single amino acid
 residue (position). Alternatively, the degree of sequence identity between two amino acid sequences
 may be calculated using a known computer algorithm (e.g., by the local homology algorithm of
 Smith and Waterman (1970) Adv. Appl. Math. 2:482c, by the homology alignment algorithm of
 Needleman and Wunsch, J. Mol. Biol. (1970) 48:443, by the search for similarity method of
 Pearson and Lipman. Proc. Natl. Acad. Sci. USA (1998) 85:2444, or by computerized
 implementations of algorithms available as Blast, Clustal Omega, or other sequence alignment
 algorithms) and, for example, using standard settings. Usually, for the purpose of determining the
 percentage of “sequence identity” between two amino acid sequences in accordance with the
 calculation method outlined hereinabove, the amino acid sequence with the greatest number of
 amino acid residues will be taken as the “first” amino acid sequence, and the other amino acid
 sequence will be taken as the “second” amino acid sequence.

[0122] Additionally, or alternatively, two or more sequences may be assessed for the identity
 between the sequences. The terms “identical” or percent “identity” in the context of two or more
 nucleic acids or amino acid sequences, refer to two or more sequences or subsequences that are the
 same. Two sequences are “substantially identical” if two sequences have a specified percentage of
 amino acid residues or nucleotides that are the same (e.g., at least 80%, 85%, 90%, 95%, 96%,
 97%, 98%, 99%, 99.5%, 99.6%, 99.7%, 99.8%, or 99.9% identical) over a specified region or over
 the entire sequence, when compared and aligned for maximum correspondence over a comparison
 window, or designated region as measured using one of the above sequence comparison algorithms
 or by manual alignment and visual inspection. Optionally, the identity exists over a region that is at
 least about 25, 50, 75, or 100 amino acids in length, or over a region that is 100 to 150, 150 to 200,

100 to 200, or 200 or more, amino acids in length.

[0123] Additionally, or alternatively, two or more sequences may be assessed for the alignment between the sequences. The terms “alignment” or percent “alignment” in the context of two or more nucleic acids or amino acid sequences, refer to two or more sequences or subsequences that are the same. Two sequences are “substantially aligned” if two sequences have a specified percentage of amino acid residues or nucleotides that are the same (e.g., at least 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.6%, 99.7%, 99.8% or 99.9% identical) over a specified region or over the entire sequence, when compared and aligned for maximum correspondence over a comparison window, or designated region as measured using one of the above sequence comparison algorithms or by manual alignment and visual inspection. Optionally, the alignment exists over a region that is at least about 25, 50, 75, or 100 amino acids in length, or over a region that is 100 to 150, 150 to 200, 100 to 200, or 200 or more amino acids in length.

[0124] In addition to protein molecules, nucleic acid molecules possess a variety of advantageous properties for use as affinity reagents (e.g., amino acid recognition molecules) in accordance with the application.

[0125] Nucleic acid aptamers are nucleic acid molecules that have been engineered to bind desired targets with high affinity and selectivity. Accordingly, nucleic acid aptamers may be engineered to selectively bind a desired type of amino acid using selection and/or enrichment techniques known in the art. Thus, in some embodiments, an affinity reagent comprises a nucleic acid aptamer (e.g., a DNA aptamer, an RNA aptamer). As shown in FIG. 1C, in some embodiments, a labeled affinity reagent is a labeled aptamer **104** that selectively binds one type of terminal amino acid. For example, in some embodiments, labeled aptamer **104** selectively binds one type of amino acid (e.g., a single type of amino acid or a subset of types of amino acids) at a terminus of a polypeptide, as described herein. Although not shown, it should be appreciated that labeled aptamer **104** may be engineered to selectively bind one type of amino acid at any position of a polypeptide (e.g., at a terminal position or at terminal and internal positions of a polypeptide) in accordance with a method of the application.

[0126] In some embodiments, a labeled affinity reagent comprises a label having binding-induced luminescence. For example, in some embodiments, a labeled aptamer **106** comprises a donor label **112** and an acceptor label **114** and functions as illustrated in panels (I) and (II) of FIG. 1C. As depicted in panel (I), labeled aptamer **106** as a free molecule adopts a conformation in which donor label **112** and acceptor label **114** are separated by a distance that limits detectable FRET between the labels (e.g., about 10 nm or more). As depicted in panel (II), labeled aptamer **106** as a selectively bound molecule adopts a conformation in which donor label **112** and acceptor label **114** are within a distance that promotes detectable FRET between the labels (e.g., about 10 nm or less). In yet other embodiments, labeled aptamer **106** comprises a quenching moiety and functions analogously to a molecular beacon, wherein luminescence of labeled aptamer **106** is internally quenched as a free molecule and restored as a selectively bound molecule (see, e.g., Hamaguchi, et al. (2001) *Analytical Biochemistry* 294, 126-131). Without wishing to be bound by theory, it is thought that these and other types of mechanisms for binding-induced luminescence may advantageously reduce or eliminate background luminescence to increase overall sensitivity and accuracy of the methods described herein.

[0127] In addition to methods of identifying a terminal amino acid of a polypeptide, the application provides methods of sequencing polypeptides using labeled affinity reagents. In some embodiments, methods of sequencing may involve subjecting a polypeptide terminus to repeated cycles of terminal amino acid detection and terminal amino acid cleavage. For example, in some embodiments, the application provides a method of determining an amino acid sequence of a polypeptide comprising contacting a polypeptide with one or more labeled affinity reagents described herein and subjecting the polypeptide to Edman degradation.

[0128] Conventional Edman degradation involves repeated cycles of modifying and cleaving the

terminal amino acid of a polypeptide, wherein each successively cleaved amino acid is identified to determine an amino acid sequence of the polypeptide. As an illustrative example of a conventional Edman degradation, the N-terminal amino acid of a polypeptide is modified using phenyl isothiocyanate (PITC) to form a PITC-derivatized N-terminal amino acid. The PITC-derivatized N-terminal amino acid is then cleaved using acidic conditions, basic conditions, and/or elevated temperatures. It has also been shown that the step of cleaving the PITC-derivatized N-terminal amino acid may be accomplished enzymatically using a modified cysteine protease from the protozoa *Trypanosoma cruzi*, which involves relatively milder cleavage conditions at a neutral or near-neutral pH. Non-limiting examples of useful enzymes are described in U.S. patent application Ser. No. 15/255,433, filed Sep. 2, 2016, titled “MOLECULES AND METHODS FOR ITERATIVE POLYPEPTIDE ANALYSIS AND PROCESSING.”

[0129] An example of sequencing by Edman degradation using labeled affinity reagents in accordance with the application is depicted in FIG. 1D. In some embodiments, sequencing by Edman degradation comprises providing a polypeptide **122** that is immobilized to a surface **130** of a solid support (e.g., immobilized to a bottom or sidewall surface of a sample well) through a linker **124**. In some embodiments, as described herein, polypeptide **122** is immobilized at one terminus (e.g., an amino-terminal amino acid or a carboxy-terminal amino acid) such that the other terminus is free for detecting and cleaving of a terminal amino acid. Accordingly, in some embodiments, the reagents used in Edman degradation methods described herein preferentially interact with terminal amino acids at the non-immobilized (e.g., free) terminus of polypeptide **122**. In this way, polypeptide **122** remains immobilized over repeated cycles of detecting and cleaving. To this end, in some embodiments, linker **124** may be designed according to a desired set of conditions used for detecting and cleaving, e.g., to limit detachment of polypeptide **122** from surface **130** under chemical cleavage conditions. Suitable linker compositions and techniques for immobilizing a polypeptide to a surface are described in detail elsewhere herein.

[0130] In accordance with the application, in some embodiments, a method of sequencing by Edman degradation comprises a step (1) of contacting polypeptide **122** with one or more labeled affinity reagents that selectively bind one or more types of terminal amino acids. As shown, in some embodiments, a labeled affinity reagent **108** interacts with polypeptide **122** by selectively binding the terminal amino acid. In some embodiments, step (1) further comprises removing any of the one or more labeled affinity reagents that do not selectively bind the terminal amino acid (e.g., the free terminal amino acid) of polypeptide **122**.

[0131] In some embodiments, the method further comprises identifying the terminal amino acid of polypeptide **122** by detecting labeled affinity reagent **108**. In some embodiments, detecting comprises detecting a luminescence from labeled affinity reagent **108**. As described herein, in some embodiments, the luminescence is uniquely associated with labeled affinity reagent **108**, and the luminescence is thereby associated with the type of amino acid to which labeled affinity reagent **108** selectively binds. As such, in some embodiments, the type of amino acid is identified by determining one or more luminescence properties of labeled affinity reagent **108**.

[0132] In some embodiments, a method of sequencing by Edman degradation comprises a step (2) of removing the terminal amino acid of polypeptide **122**. In some embodiments, step (2) comprises removing labeled affinity reagent **108** (e.g., any of the one or more labeled affinity reagents that selectively bind the terminal amino acid) from polypeptide **122**. In some embodiments, step (2) comprises modifying the terminal amino acid (e.g., the free terminal amino acid) of polypeptide **122** by contacting the terminal amino acid with an isothiocyanate (e.g., PITC) to form an isothiocyanate-modified terminal amino acid. In some embodiments, an isothiocyanate-modified terminal amino acid is more susceptible to removal by a cleaving reagent (e.g., a chemical or enzymatic cleaving reagent) than an unmodified terminal amino acid.

[0133] In some embodiments, step (2) comprises removing the terminal amino acid by contacting polypeptide **122** with a protease **140** that specifically binds and cleaves the isothiocyanate-modified

terminal amino acid. In some embodiments, protease **140** comprises a modified cysteine protease. In some embodiments, protease **140** comprises a modified cysteine protease, such as a cysteine protease from *Trypanosoma cruzi* (see, e.g., Borgo, et al. (2015) Protein Science 24:571-579). In yet other embodiments, step (2) comprises removing the terminal amino acid by subjecting polypeptide **122** to chemical (e.g., acidic, basic) conditions sufficient to cleave the isothiocyanate-modified terminal amino acid.

[0134] In some embodiments, a method of sequencing by Edman degradation comprises a step (3) of washing polypeptide **122** following terminal amino acid cleavage. In some embodiments, washing comprises removing protease **140**. In some embodiments, washing comprises restoring polypeptide **122** to neutral pH conditions (e.g., following chemical cleavage by acidic or basic conditions). In some embodiments, a method of sequencing by Edman degradation comprises repeating steps (1) through (3) for a plurality of cycles.

[0135] An example method of sequencing by Edman degradation is shown in FIG. **1E**. In some embodiments, a sample containing a complex mixture of polypeptides (e.g., a mixture of proteins) can be degraded using common enzymes into short polypeptide fragments of approximately 6 to 40 amino acids. In some embodiments, sequencing of this polypeptide library in accordance with methods of the application would reveal the identity and abundance of each of the polypeptides present in the original complex mixture. As described herein and in the literature, most polypeptides in the size range of 6 to 40 amino acids can be uniquely identified by determining the number and location of just four amino acids within a polypeptide chain.

[0136] Accordingly, in some embodiments, a method of sequencing by Edman degradation may be performed using a set of labeled aptamers **150** comprising four DNA aptamer types, each type recognizing a different N-terminal amino acid. Each aptamer type may be labeled with a different luminescent label, such that the different aptamer types can be distinguished based on one or more luminescence properties. For illustrative purposes, the example set of labeled aptamers **150** includes: a cysteine-specific aptamer labeled with a first luminescent label (“dye 1”); a lysine-specific aptamer labeled with a second luminescent label (“dye 2”); a tryptophan-specific aptamer labeled with a third luminescent label (“dye 3”); and a glutamate-specific aptamer labeled with a fourth luminescent label (“dye 4”).

[0137] In some embodiments, a method of sequencing by Edman degradation in accordance with the application proceeds according to a process **152** shown in FIG. **1E**. In some embodiments, prior to step (1), single polypeptide molecules from a polypeptide library are immobilized to a surface of a solid support, e.g., at a bottom or sidewall surface of a sample well of an array of sample wells. In some embodiments, as described elsewhere herein, moieties that enable surface immobilization (e.g., biotin) or improve solubility (e.g., oligonucleotides) may be chemically or enzymatically attached to the C-terminus of the polypeptides. To determine the sequence of each polypeptide, in some embodiments, immobilized polypeptides are subjected to repeated cycles of N-terminal amino acid detection and N-terminal amino acid cleavage, as illustrated by process **152**. In some embodiments, process **152** comprises reagent addition and wash steps which are performed by injection into a flowcell above the detection surface using an automated fluidic system. In some embodiments, steps (1) through (4) illustrate one cycle of detection and cleavage using labeled aptamers **150**.

[0138] In some embodiments, a method of sequencing by Edman degradation according to process **152** comprises a step (1) of flowing in a mixture of four orthogonally labeled DNA aptamers and incubating to allow the aptamers to bind to any immobilized polypeptides (e.g., polypeptides immobilized within a sample well of an array) that contain one of the four correct amino acids at the N-terminus. In some embodiments, the method further comprises washing the immobilized polypeptides to remove unbound aptamers. In some embodiments, the method further comprises imaging the immobilized polypeptides (“Imaging step 1”). In some embodiments, the acquired images contain enough information to determine the location of aptamer-bound polypeptides (e.g.,

location within an array of sample wells) and which of the four aptamers is bound at each location. In some embodiments, the method further comprises washing the immobilized polypeptides using an appropriate buffer to remove the aptamers from the immobilized polypeptides.

[0139] In some embodiments, a method of sequencing according to process **152** comprises a step (2) of flowing in a solution containing a reactive molecule (e.g., PITC, as shown) that specifically modifies the N-terminal amine group. An isothiocyanate molecule such as PITC, in some embodiments, modifies the N-terminal amino acid into a substrate for cleavage by a modified protease such as the cysteine protease cruzain from *Trypanosoma Cruzi*.

[0140] In some embodiments, a method of sequencing according to process **152** comprises a step (3) of washing the immobilized polypeptides before flowing in a suitable modified protease that recognizes and cleaves the modified N-terminal amino acid from the immobilized polypeptide. In some embodiments, the method comprises a step (4) of washing the immobilized polypeptides after enzymatic cleavage. In some embodiments, steps (1) through (4) depict one cycle of Edman degradation. Accordingly, step (1') as shown is the start of the next reaction cycle which proceeds as steps (1') through (4') performed as described above for steps (1) through (4). In some embodiments, steps (1) through (4) are repeated for approximately 20-40 cycles.

[0141] In some embodiments, a labeled isothiocyanate (e.g., a dye-labeled PITC) may be used to monitor sample loading. For example, in some embodiments, prior to subjecting a polypeptide sample to a method of sequencing as shown in process **152**, the polypeptide sample is pre-conjugated with a luminescent label at a terminal end by modification of the terminal end using a dye-labeled PITC. In this way, loading of the polypeptide sample into an array of sample wells may be monitored by detecting luminescence from the labels prior to initiating process **152**. In some embodiments, the luminescence is used to determine single occupancy of sample wells in the array (e.g., a fraction of sample wells containing a single polypeptide molecule), which may advantageously increase the amount of information reliably obtained for a given sample. Once a desired sample loading status is determined by luminescence, process **152** may be initiated by chemical or enzymatic cleavage, as described, before proceeding with step (1).

[0142] In some embodiments, a labeled isothiocyanate (e.g., a dye-labeled PITC) may be used to monitor reaction progress for a polypeptide sample in an array. For example, in some embodiments, step (2) comprises flowing in a solution containing a dye-labeled PITC that specifically modifies and labels N-terminal amine groups of polypeptides in the sample. In some embodiments, luminescence from the labels may be detected during or after step (2) to evaluate N-terminal PITC modification of polypeptides in the sample. Accordingly, in some embodiments, luminescence is used to determine whether or when to proceed from step (2) to step (3). In some embodiments, luminescence from the labels may be detected during or after step (3) to evaluate N-terminal amino acid cleavage of polypeptides in the sample—e.g., to determine whether or when to proceed from step (3) to step (4).

[0143] A method of sequencing according to process **152** utilizes separate reagents for detecting and cleaving a terminal amino acid of a polypeptide. Nonetheless, in some aspects, the application provides a method of sequencing in which a single reagent comprising a peptidase may be used for detecting and cleaving a terminal amino acid of a polypeptide. FIG. 2 shows an example of polypeptide sequencing using a set of labeled exopeptidases **200**, wherein each labeled exopeptidase selectively binds and cleaves a different type of terminal amino acid.

[0144] As generically illustrated in the example of FIG. 2, labeled exopeptidases **200** include a lysine-specific exopeptidase comprising a first luminescent label, a glycine-specific exopeptidase comprising a second luminescent label, an aspartate-specific exopeptidase comprising a third luminescent label, and a leucine-specific exopeptidase comprising a fourth luminescent label. In accordance with certain embodiments described herein, each of labeled exopeptidases **200** selectively binds and cleaves its respective amino acid only when that amino acid is at an amino- or carboxy-terminus of a polypeptide. Accordingly, as sequencing by this approach proceeds from one

terminus of a peptide toward the other, labeled exopeptidases **200** are engineered or selected such that all reagents of the set will possess either aminopeptidase or carboxypeptidase activity.

[0145] As further shown in FIG. 2, process **202** schematically illustrates a real-time sequencing reaction using labeled exopeptidases **200**. Panels (I) through (IX) illustrate a progression of events involving iterative detection and cleavage at a terminal end of a polypeptide in relation to a signal output shown below, and corresponding to, the event depicted in each panel. For illustrative purposes, a polypeptide is shown having an arbitrarily selected amino acid sequence of “KLDG . . .” (proceeding from one terminus toward the other).

[0146] Panel (I) depicts the start of a sequencing reaction, wherein a polypeptide is immobilized to a surface of a solid support, such as a bottom or sidewall surface of a sample well. In some embodiments, sequencing methods in accordance with the application comprise single molecule sequencing in real-time. In some embodiments, a plurality of single molecule sequencing reactions are performed simultaneously in an array of sample wells. In such embodiments, polypeptide immobilization prevents diffusion of a polypeptide out of a sample well by anchoring the polypeptide within the sample well for single molecule analysis.

[0147] Panel (II) depicts a detection event, wherein the lysine-specific exopeptidase from the set of labeled affinity reagents **200** selectively binds the terminal lysine residue of the polypeptide. As shown in the signal trace below panels (I) and (II), signal output reports on this binding event by displaying an increase in signal intensity, which may be used to identify the luminescent label of the lysine-specific exopeptidase to thereby identify the terminal amino acid. Panel (III) illustrates that, after selectively binding a terminal amino acid, a labeled peptidase cleaves the terminal amino acid. As a result, these components are free to diffuse away from an observation region for luminescence detection, which is reported in the signal output by a drop in signal intensity, as shown in the trace below panel (III). Panels (IV) through (IX) proceed analogously to the process as described for panels (I) through (III). That is, a labeled exopeptidase binds and cleaves a corresponding terminal amino acid to produce a corresponding increase and decrease, respectively, in signal output.

[0148] In some aspects, the application provides methods of polypeptide sequencing in real-time by evaluating binding interactions of terminal amino acids with labeled amino acid recognition molecules (e.g., labeled affinity reagents) and a labeled cleaving reagent (e.g., a labeled non-specific exopeptidase). FIG. 3A shows an example of a method of sequencing in which discrete binding events give rise to signal pulses of a signal output **300**. The inset panel of FIG. 3A illustrates a general scheme of real-time sequencing by this approach. As shown, a labeled affinity reagent **310** selectively associates with (e.g., binds to) and dissociates from a terminal amino acid (shown here as lysine), which gives rise to a series of pulses in signal output **300** which may be used to identify the terminal amino acid. In some embodiments, the series of pulses provide a pulsing pattern (e.g., a characteristic pattern) which may be diagnostic of the identity of the corresponding terminal amino acid.

[0149] Without wishing to be bound by theory, labeled affinity reagent **310** selectively binds according to a binding affinity ($K_{sub.D}$) defined by an association rate, or an “on” rate, of binding (k_{on}) and a dissociation rate, or an “off” rate, of binding (k_{off}). The rate constants k_{off} and k_{on} are the critical determinants of pulse duration (e.g., the time corresponding to a detectable binding event) and interpulse duration (e.g., the time between detectable binding events), respectively. In some embodiments, these rates can be engineered to achieve pulse durations and pulse rates (e.g., the frequency of signal pulses) that give the best sequencing accuracy.

[0150] As shown in the inset panel, a sequencing reaction mixture further comprises a labeled non-specific exopeptidase **320** comprising a luminescent label that is different than that of labeled affinity reagent **310**. In some embodiments, labeled non-specific exopeptidase **320** is present in the mixture at a concentration that is less than that of labeled affinity reagent **310**. In some embodiments, labeled non-specific exopeptidase **320** displays broad specificity such that it cleaves

most or all types of terminal amino acids. Accordingly, a dynamic sequencing approach can involve monitoring affinity reagent binding at a terminus of a polypeptide over the course of a degradation reaction catalyzed by exopeptidase cleavage activity.

[0151] As illustrated by the progress of signal output **300**, in some embodiments, terminal amino acid cleavage by labeled non-specific exopeptidase **320** gives rise to a signal pulse, and these events occur with lower frequency than the binding pulses of a labeled affinity reagent **310**. In this way, amino acids of a polypeptide may be counted and/or identified in a real-time sequencing process. As further illustrated in signal output **300**, in some embodiments, a plurality of labeled affinity reagents may be used, each with a diagnostic pulsing pattern (e.g., characteristic pattern) which may be used to identify a corresponding terminal amino acid. For example, in some embodiments, different characteristic patterns (as illustrated by each of lysine, phenylalanine, and glutamine in signal output **300**) correspond to the association of more than one labeled affinity reagent with different types of terminal amino acids. As described herein, it should be appreciated that a single affinity reagent that associates with more than one type of amino acid may be used in accordance with the application. Accordingly, in some embodiments, different characteristic patterns correspond to the association of one labeled affinity reagent with different types of terminal amino acids.

[0152] As described herein, signal pulse information may be used to identify an amino acid based on a characteristic pattern in a series of signal pulses. In some embodiments, a characteristic pattern comprises a plurality of signal pulses, each signal pulse comprising a pulse duration. In some embodiments, the plurality of signal pulses may be characterized by a summary statistic (e.g., mean, median, time decay constant) of the distribution of pulse durations in a characteristic pattern. In some embodiments, the mean pulse duration of a characteristic pattern is between about 1 millisecond and about 10 seconds (e.g., between about 1 ms and about 1 s, between about 1 ms and about 100 ms, between about 1 ms and about 10 ms, between about 10 ms and about 10 s, between about 100 ms and about 10 s, between about 1 s and about 10 s, between about 10 ms and about 100 ms, or between about 100 ms and about 500 ms). In some embodiments, different characteristic patterns corresponding to different types of amino acids in a single polypeptide may be distinguished from one another based on a statistically significant difference in the summary statistic. For example, in some embodiments, one characteristic pattern may be distinguishable from another characteristic pattern based on a difference in mean pulse duration of at least 10 milliseconds (e.g., between about 10 ms and about 10 s, between about 10 ms and about 1 s, between about 10 ms and about 100 ms, between about 100 ms and about 10 s, between about 1 s and about 10 s, or between about 100 ms and about 1 s). It should be appreciated that, in some embodiments, smaller differences in mean pulse duration between different characteristic patterns may require a greater number of pulse durations within each characteristic pattern to distinguish one from another with statistical confidence.

[0153] As detailed above, a real-time sequencing process as illustrated by FIG. 3A can generally involve cycles of terminal amino acid recognition and terminal amino acid cleavage, where the relative occurrence of recognition and cleavage can be controlled by a concentration differential between a labeled affinity reagent **310** and a labeled non-specific exopeptidase **320**. In some embodiments, the concentration differential can be optimized such that the number of signal pulses detected during recognition of an individual amino acid provides a desired confidence interval for identification. For example, if an initial sequencing reaction provides signal data with too few signal pulses between cleavage events to permit determination of characteristic patterns with a desired confidence interval, the sequencing reaction can be repeated using a decreased concentration of non-specific exopeptidase relative to affinity reagent.

[0154] In some embodiments, polypeptide sequencing in accordance with the application may be carried out by contacting a polypeptide with a sequencing reaction mixture comprising one or more amino acid recognition molecules (e.g., affinity reagents) and/or one or more cleaving reagents

(e.g., exopeptidases). In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule at a concentration of between about 10 nM and about 10 μ M. In some embodiments, a sequencing reaction mixture comprises a cleaving reagent at a concentration of between about 500 nM and about 500 μ M.

[0155] In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule at a concentration of between about 100 nM and about 10 μ M, between about 250 nM and about 10 μ M, between about 100 nM and about 1 μ M, between about 250 nM and about 1 μ M, between about 250 nM and about 750 nM, or between about 500 nM and about 1 μ M. In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule at a concentration of about 100 nM, about 250 nM, about 500 nM, about 750 nM, or about 1 μ M.

[0156] In some embodiments, a sequencing reaction mixture comprises a cleaving reagent at a concentration of between about 500 nM and about 250 μ M, between about 500 nM and about 100 μ M, between about 1 μ M and about 100 μ M, between about 500 nM and about 50 μ M, between about 1 μ M and about 100 μ M, between about 10 μ M and about 200 μ M, or between about 10 μ M and about 100 μ M. In some embodiments, a sequencing reaction mixture comprises a cleaving reagent at a concentration of about 1 μ M, about 5 μ M, about 10 μ M, about 30 μ M, about 50 μ M, about 70 μ M, or about 100 μ M.

[0157] In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule at a concentration of between about 10 nM and about 10 μ M, and a cleaving reagent at a concentration of between about 500 nM and about 500 μ M. In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule at a concentration of between about 100 nM and about 1 μ M, and a cleaving reagent at a concentration of between about 1 μ M and about 100 μ M. In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule at a concentration of between about 250 nM and about 1 μ M, and a cleaving reagent at a concentration of between about 10 μ M and about 100 μ M. In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule at a concentration of about 500 nM, and a cleaving reagent at a concentration of between about 25 μ M and about 75 μ M.

[0158] In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule and a cleaving reagent in a ratio of about 500:1, about 400:1, about 300:1, about 200:1, about 100:1, about 75:1, about 50:1, about 25:1, about 10:1, about 5:1, about 2:1, or about 1:1. In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule and a cleaving reagent in a ratio of between about 10:1 and about 200:1. In some embodiments, a sequencing reaction mixture comprises an amino acid recognition molecule and a cleaving reagent in a ratio of between about 50:1 and about 150:1.

[0159] While the example illustrated by FIG. 3A relates to a sequencing process using a labeled cleaving reagent, the sequencing process is not intended to be limited in this respect. As described elsewhere herein, the inventors have demonstrated single-molecule sequencing using an unlabeled cleaving reagent. In some embodiments, the approximate frequency with which a cleaving reagent removes successive terminal amino acids is known, e.g., based on a known activity and/or concentration of the enzyme being used. In some embodiments, terminal amino acid cleavage by the reagent is inferred, e.g., based on signal detected for amino acid recognition or a lack of signal detected. The inventors have recognized further techniques for controlling real-time sequencing reactions, which may be used in combination with, or alternatively to, the concentration differential approach as described.

[0160] An example of a temperature-dependent real-time sequencing process is shown in FIG. 3B. Panels (I) through (III) illustrate a sequencing reaction involving cycles of temperature-dependent terminal amino acid recognition and terminal amino acid cleavage. Each cycle of the sequencing reaction is carried out over two temperature ranges: a first temperature range ("T.sub.1") that is optimal for affinity reagent activity over exopeptidase activity (e.g., to promote terminal amino acid recognition), and a second temperature range ("T.sub.2") that is optimal for exopeptidase

activity over affinity reagent activity (e.g., to promote terminal amino acid cleavage). The sequencing reaction progresses by alternating the reaction mixture temperature between the first temperature range T.sub.1 (to initiate amino acid recognition) and the second temperature range T.sub.2 (to initiate amino acid cleavage). Accordingly, progression of a temperature-dependent sequencing process is controllable by temperature, and alternating between different temperature ranges (e.g., between T.sub.1 and T.sub.2) may be carried through manual or automated processes. In some embodiments, affinity reagent activity (e.g., binding affinity (K.sub.D) for an amino acid) within the first temperature range T.sub.1 as compared to the second temperature range T.sub.2 is increased by at least 10-fold, at least 100-fold, at least 1,000-fold, at least 10,000-fold, at least 100,000-fold, or more. In some embodiments, exopeptidase activity (e.g., rate of substrate conversion to cleavage product) within the second temperature range T.sub.2 as compared to the first temperature range T.sub.1 is increased by at least 2-fold, 10-fold, at least 25-fold, at least 50-fold, at least 100-fold, at least 1,000-fold, or more.

[0161] In some embodiments, the first temperature range T.sub.1 is lower than the second temperature range T.sub.2. In some embodiments, the first temperature range T.sub.1 is between about 15° C. and about 40° C. (e.g., between about 25° C. and about 35° C., between about 15° C. and about 30° C., between about 20° C. and about 30° C.). In some embodiments, the second temperature range T.sub.2 is between about 40° C. and about 100° C. (e.g., between about 50° C. and about 90° C., between about 60° C. and about 90° C., between about 70° C. and about 90° C.). In some embodiments, the first temperature range T.sub.1 is between about 20° C. and about 40° C. (e.g., approximately 30° C.), and the second temperature range T.sub.2 is between about 60° C. and about 100° C. (e.g., approximately 80° C.).

[0162] In some embodiments, the first temperature range T.sub.1 is higher than the second temperature range T.sub.2. In some embodiments, the first temperature range T.sub.1 is between about 40° C. and about 100° C. (e.g., between about 50° C. and about 90° C., between about 60° C. and about 90° C., between about 70° C. and about 90° C.). In some embodiments, the second temperature range T.sub.2 is between about 15° C. and about 40° C. (e.g., between about 25° C. and about 35° C., between about 15° C. and about 30° C., between about 20° C. and about 30° C.). In some embodiments, the first temperature range T.sub.1 is between about 60° C. and about 100° C. (e.g., approximately 80° C.), and the second temperature range T.sub.2 is between about 20° C. and about 40° C. (e.g., approximately 30° C.).

[0163] Panel (I) depicts a sequencing reaction mixture at a temperature that is within a first temperature range T.sub.1 which is optimal for affinity reagent activity over exopeptidase activity. For illustrative purposes, a polypeptide of amino acid sequence “KFVAG . . .” is shown. When the reaction mixture temperature is within the first temperature range T.sub.1, labeled affinity reagents in the mixture are activated (e.g., renatured) to initiate amino acid recognition by associating with the polypeptide terminus. Also within the first temperature range T.sub.1, labeled exopeptidases in the mixture are inactivated (e.g., denatured) to prevent amino acid cleavage during recognition. In panel (I), a first affinity reagent is shown reversibly associating with lysine at the polypeptide terminus, while a labeled exopeptidase (e.g., Pfu aminopeptidase I (Pfu API)) is shown denatured. In some embodiments, amino acid recognition occurs for a predetermined duration of time before initiating cleavage of the amino acid. In some embodiments, amino acid recognition occurs for a duration of time required to reach a desired confidence interval for identification before initiating cleavage of the amino acid. Following amino acid recognition, the reaction proceeds by changing the temperature of the mixture to within a second temperature range T.sub.2.

[0164] Panel (II) depicts the sequencing reaction mixture at a temperature that is within a second temperature range T.sub.2 which is optimal for exopeptidase activity over affinity reagent activity. For illustrative purposes of this example, the second temperature range T.sub.2 is higher than the first temperature range T.sub.1, although it should be appreciated that reagent activity may be optimized for any desired temperature range. Accordingly, progression from panel (I) to panel (II)

is carried out by raising the reaction mixture temperature using a suitable source of heat. When the reaction mixture reaches a temperature that is within the second temperature range $T_{\text{sub.2}}$, labeled exopeptidases in the mixture are activated (e.g., renatured) to initiate terminal amino acid cleavage by exopeptidase activity. Also within the second temperature range $T_{\text{sub.2}}$, labeled affinity reagents in the mixture are inactivated (e.g., denatured) to prevent amino acid recognition during cleavage. In panel (II), a labeled exopeptidase is shown cleaving the terminal lysine residue, while labeled affinity reagents are denatured. In some embodiments, amino acid cleavage occurs for a predetermined duration of time before initiating recognition of a successive amino acid at the polypeptide terminus. In some embodiments, amino acid cleavage occurs for a duration of time required to detect cleavage before initiating recognition of a successive amino acid. Following amino acid cleavage, the reaction proceeds by changing the temperature of the mixture to within the first temperature range $T_{\text{sub.1}}$.

[0165] Panel (III) depicts the beginning of the next cycle in the sequencing reaction, wherein the reaction mixture temperature has been reduced back to within the first temperature range $T_{\text{sub.1}}$. Accordingly, in this example, progression from panel (II) to panel (III) can be carried out by removing the reaction mixture from the source of heat or otherwise cooling the reaction mixture (e.g., actively or passively) to within the first temperature range $T_{\text{sub.1}}$. As shown, labeled affinity reagents are renatured, including a second affinity reagent that reversibly associates with phenylalanine at the polypeptide terminus, while the labeled exopeptidase is shown denatured. The sequencing reaction continues by further cycling between amino acid recognition and amino acid cleavage in a temperature-dependent fashion as illustrated by this example.

[0166] Accordingly, a dynamic sequencing approach can involve reaction cycling that is controlled at the level of protein activity or function of one or more proteins within a reaction mixture. It should be appreciated that the temperature-dependent polypeptide sequencing process depicted in FIG. 3B and described above may be illustrative of a general approach to polypeptide sequencing by controllable cycling of condition-dependent recognition and cleavage. For example, in some embodiments, the application provides a luminescence-dependent sequencing process using luminescence-activated reagents. In some embodiments, a luminescence-dependent sequencing process involves cycles of luminescence-dependent amino acid recognition and cleavage. Each cycle of the sequencing reaction may be carried out by exposing a sequencing reaction mixture to two different luminescent conditions: a first luminescent condition that is optimal for affinity reagent activity over exopeptidase activity (e.g., to promote amino acid recognition), and a second luminescent condition that is optimal for exopeptidase activity over affinity reagent activity (e.g., to promote amino acid cleavage). The sequencing reaction progresses by alternating between exposing the reaction mixture to the first luminescent condition (to initiate amino acid recognition) and exposing the reaction mixture to the second luminescent condition (to initiate amino acid cleavage). By way of example and not limitation, in some embodiments, the two different luminescent conditions comprise a first wavelength and a second wavelength.

[0167] In some aspects, the application provides methods of polypeptide sequencing in real-time by evaluating binding interactions of one or more labeled affinity reagents with terminal and internal amino acids and binding interactions of a labeled non-specific exopeptidase with terminal amino acids. FIG. 4 shows an example of a method of sequencing in which the method described and illustrated for the approach in FIGS. 3A-3B is modified by using a labeled affinity reagent **410** that selectively binds to and dissociates from one type of amino acid (shown here as lysine) at both terminal and internal positions (FIG. 4, inset panel). As described in the previous approach, the selective binding gives rise to a series of pulses in signal output **400**. In this approach, however, the series of pulses occur at a rate that is determined by the number of the type of amino acid throughout the polypeptide. Accordingly, in some embodiments, the rate of pulsing corresponding to binding events would be diagnostic of the number of cognate amino acids currently present in the polypeptide.

[0168] As in the previous approach, a labeled non-specific peptidase **420** would be present at a relatively lower concentration than labeled affinity reagent **410**, e.g., to give optimal time windows in between cleavage events (FIG. 4, inset panel). Additionally, in certain embodiments, uniquely identifiable luminescent label of labeled non-specific peptidase **420** would indicate when cleavage events have occurred. As the polypeptide undergoes iterative cleavage, the rate of pulsing corresponding to binding by labeled affinity reagent **410** would drop in a step-wise manner whenever a terminal amino acid is cleaved by labeled non-specific peptidase **420**. This concept is illustrated by plot **402**, which generally depicts pulse rate as a function of time, with cleavage events in time denoted by arrows. Thus, in some embodiments, amino acids may be identified-and polypeptides thereby sequenced—in this approach based on a pulsing pattern and/or on the rate of pulsing that occurs within a pattern detected between cleavage events.

[0169] In some embodiments, terminal polypeptide sequence information (e.g., determined as described herein) can be combined with polypeptide sequence information obtained from one or more other sources. For example, terminal polypeptide sequence information could be combined with internal polypeptide sequence information. In some embodiments, internal polypeptide sequence information can be obtained using one or more amino acid recognition molecules that associate with internal amino acids, as described herein. Internal or other polypeptide sequence information can be obtained before or during a polypeptide degradation process. In some embodiments, sequence information obtained from these methods can be combined with polypeptide sequence information using other techniques, e.g., sequence information obtained using one or more internal amino acid recognition molecules.

Shielded Recognition Molecules

[0170] In accordance with embodiments described herein, single-molecule polypeptide sequencing methods can be carried out by illuminating a surface-immobilized polypeptide with excitation light, and detecting luminescence produced by a label attached to an amino acid recognition molecule (e.g., a labeled affinity reagent). In some cases, radiative and/or non-radiative decay produced by the label can result in photodamage to the polypeptide. For example, FIG. 5A illustrates an example sequencing reaction in which a recognition molecule is shown associated with a polypeptide immobilized to a surface.

[0171] In the presence of excitation illumination, the label can produce fluorescence through radiative decay which results in a detectable association event. However, in some cases, the label produces non-radiative decay which can result in the formation of reactive oxygen species **500**. The reactive oxygen species **500** can eventually damage the immobilized peptide, such that the reaction ends before obtaining complete sequence information for the polypeptide. This photodamage can occur, for example, at the exposed polypeptide terminus (top open arrow), at an internal position (middle open arrow), or at the surface linker attaching the polypeptide to the surface (bottom open arrow).

[0172] The inventors have found that photodamage can be mitigated and recognition times extended by incorporation of a shielding element into an amino acid recognition molecule. FIG. 5B illustrates an example sequencing reaction using a shielded recognition molecule that includes a shield **502**. Shield **502** forms a covalent or non-covalent linkage group that provides increased distance between the label and polypeptide, such that damaging effects from reactive oxygen species **500** can be reduced due to free radical decay over the label-polypeptide separation distance. Shield **502** can also provide a steric barrier that shields the polypeptide from the label by absorbing damage from reactive oxygen species **500** and radiative and/or non-radiative decay.

[0173] Without wishing to be bound by theory, it is thought that a shield, positioned between a recognition component and a label component, can absorb, deflect, or otherwise block radiative and/or non-radiative decay emitted by the label component. In some embodiments, the shield prevents or limits the extent to which one or more labels (e.g., luminescent labels) interact with one or more amino acid recognition molecules. In some embodiments, the shield prevents or limits the

extent to which one or more labels interact with one or more molecules associated with an amino acid recognition molecule (e.g., a polypeptide associated with the recognition molecule). Accordingly, in some embodiments, the term shield can generally refer to a protective or shielding effect that is provided by some portion of a linkage group formed between a recognition component and a label component.

[0174] In some embodiments, a shield is attached to one or more amino acid recognition molecules (e.g., a recognition component) and to one or more labels (e.g., a label component). In some embodiments, the recognition and label components are attached at non-adjacent sites on the shield. For example, one or more amino acid recognition molecules can be attached to a first side of the shield, and one or more labels can be attached to a second side of the shield, where the first and second sides of the shield are distant from each other. In some embodiments, the attachment sites are on approximately opposite sides of the shield.

[0175] The distance between the site at which a shield is attached to a recognition molecule and the site at which the shield is attached to a label can be a linear measurement through space or a non-linear measurement across the surface of the shield. The distance between the recognition molecule and label attachment sites on a shield can be measured by modeling the three-dimensional structure of the shield. In some embodiments, this distance can be at least 2 nm, at least 4 nm, at least 6 nm, at least 8 nm, at least 10 nm, at least 12 nm, at least 15 nm, at least 20 nm, at least 30 nm, at least 40 nm, or more. Alternatively, the relative positions of the recognition molecule and label on a shield can be described by treating the structure of the shield as a quadratic surface (e.g., ellipsoid, elliptic cylinder). In some embodiments, the recognition molecule and label attachment sites are separated by a distance that is at least one eighth of the distance around an ellipsoidal shape representing the shield. In some embodiments, the recognition molecule and label are separated by a distance that is at least one quarter of the distance around an ellipsoidal shape representing the shield. In some embodiments, the recognition molecule and label are separated by a distance that is at least one third of the distance around an ellipsoidal shape representing the shield. In some embodiments, the recognition molecule and label are separated by a distance that is one half of the distance around an ellipsoidal shape representing the shield.

[0176] The size of a shield should be such that a label is unable or unlikely to directly contact the polypeptide when the amino acid recognition molecule is associated with the polypeptide. The size of a shield should also be such that an attached label is detectable when the amino acid recognition molecule is associated with the polypeptide. For example, the size should be such that an attached luminescent label is within an illumination volume to be excited.

[0177] It should be appreciated that there are a variety of parameters by which a practitioner could evaluate shielding effects. Generally, the effects of a shielding element can be evaluated by conducting a comparative assessment between a composition having the shielding element and a composition lacking the shielding element. For example, a shielding element can increase recognition time of an amino acid recognition molecule. In some embodiments, recognition time refers to the length of time in which association events between the recognition molecule and a polypeptide are observable in a polypeptide sequencing reaction as described herein. In some embodiments, recognition time is increased by about 10%, 25%, 50%, 75%, or more than 100%, for example by about 2-fold, 3-fold, 4-fold, 5-fold, or more, relative to a polypeptide sequencing reaction performed under the same conditions, with the exception that the amino acid recognition molecule lacks the shielding element but is otherwise similar or identical. In some embodiments, a shielding element can increase sequencing accuracy and/or sequence read length (e.g., by at least 5%, at least 10%, at least 15%, at least 25% or more, relative to a sequencing reaction performed under comparative conditions as described above).

[0178] Accordingly, in some aspects, the application provides shielded recognition molecules comprising at least one amino acid recognition molecule, at least one detectable label, and a shielding element (e.g., a “shield”) that forms a covalent or non-covalent linkage group between

the recognition molecule and label. In some embodiments, a shielding element is at least 2 nm, at least 5 nm, at least 10 nm, at least 12 nm, at least 15 nm, at least 20 nm, or more, in length (e.g., in an aqueous solution). In some embodiments, a shielding element is between about 2 nm and about 100 nm in length (e.g., between about 2 nm and about 50 nm, between about 10 nm and about 50 nm, between about 20 nm and about 100 nm).

[0179] In some embodiments, a shield (e.g., shielding element) forms a covalent or non-covalent linkage group between one or more amino acid recognition molecules (e.g., a recognition component) and one or more labels (e.g., a label component). As used herein, in some embodiments, covalent and non-covalent linkages or linkage groups refer to the nature of the attachments of the recognition and label components to the shield.

[0180] In some embodiments, a covalent linkage, or a covalent linkage group, refers to a shield that is attached to each of the recognition and label components through a covalent bond or a series of contiguous covalent bonds. Covalent attachment one or both components can be achieved by covalent conjugation methods known in the art. For example, in some embodiments, click chemistry techniques (e.g., copper-catalyzed, strain-promoted, copper-free click chemistry, etc.) can be used to attach one or both components to the shield. Such methods generally involve conjugating one reactive moiety to another reactive moiety to form one or more covalent bonds between the reactive moieties. Accordingly, in some embodiments, a first reactive moiety of a shield can be contacted with a second reactive moiety of a recognition or label component to form a covalent attachment. Examples of reactive moieties include, without limitation, reactive amines, azides, alkynes, nitrones, alkenes (e.g., cycloalkenes), tetrazines, tetrazoles, and other reactive moieties suitable for click reactions and similar coupling techniques.

[0181] In some embodiments, a non-covalent linkage, or a non-covalent linkage group, refers to a shield that is attached to one or both of the recognition and label components through one or more non-covalent coupling means, including but not limited to receptor-ligand interactions and oligonucleotide strand hybridization. Examples of receptor-ligand interactions are provided herein and include, without limitation, protein-protein complexes, protein-ligand complexes, protein-aptamer complexes, and aptamer-nucleic acid complexes. Various configurations and strategies for oligonucleotide strand hybridization are described herein and are known in the art (see, e.g., U.S. Patent Publication No. 2019/0024168).

[0182] In some embodiments, shield **502** comprises a polymer, such as a biomolecule or a dendritic polymer. FIG. 5C depicts examples of polymer shields and configurations of shielded recognition molecules of the application. A first shielded construct **504** shows an example of a protein shield **530**. In some embodiments, protein shield **530** forms a covalent linkage group between a recognition molecule and a label. For example, in some embodiments, protein shield **530** is attached to each of the recognition molecule and label through one or more covalent bonds, e.g., by covalent attachment through a side-chain of a natural or unnatural amino acid of protein shield **530**. In some embodiments, protein shield **530** forms a non-covalent linkage group between a recognition molecule and a label. For example, in some embodiments, protein shield **530** is a monomeric or multimeric protein comprising one or more ligand-binding sites. In some embodiments, a non-covalent linkage group is formed through one or more ligand moieties bound to the one or more ligand-binding sites. Additional examples of non-covalent linkages formed by protein shields are described elsewhere herein.

[0183] A second shielded construct **506** shows an example of a double-stranded nucleic acid shield comprising a first oligonucleotide strand **532** hybridized with a second oligonucleotide strand **534**. As shown, in some embodiments, the double-stranded nucleic acid shield can comprise a recognition molecule attached to first oligonucleotide strand **532**, and a label attached to second oligonucleotide strand **534**. In this way, the double-stranded nucleic acid shield forms a non-covalent linkage group between the recognition molecule and the label through oligonucleotide strand hybridization. In some embodiments, a recognition molecule and a label can be attached to

the same oligonucleotide strand, which can provide a single-stranded nucleic acid shield or a double-stranded nucleic acid shield through hybridization with another oligonucleotide strand. In some embodiments, strand hybridization can provide increased rigidity within a linkage group to further enhance separation between the recognition molecule and the label.

[0184] Where shielding element **502** comprises a nucleic acid, the separation distance between a label and a recognition molecule can be measured by the distance between attachment sites on the nucleic acid (e.g., direct attachment or indirect attachment, such as through one or more additional shield polymers). In some embodiments, the distance between attachment sites on a nucleic acid can be measured by the number of nucleotides within the nucleic acid that occur between the label and the recognition molecule. It should be understood that the number of nucleotides can refer to either the number of nucleotide bases in a single-stranded nucleic acid or the number of nucleotide base pairs in a double-stranded nucleic acid.

[0185] Accordingly, in some embodiments, the attachment site of a recognition molecule and the attachment site of a label can be separated by between 5 and 200 nucleotides (e.g., between 5 and 150 nucleotides, between 5 and 100 nucleotides, between 5 and 50 nucleotides, between 10 and 100 nucleotides). It should be appreciated that any position in a nucleic acid can serve as an attachment site for a recognition molecule, a label, or one or more additional polymer shields. In some embodiments, an attachment site can be at or approximately at 5' or 3' end, or at an internal position along a strand of the nucleic acid.

[0186] The non-limiting configuration of second shielded construct **506** illustrates an example of a shield that forms a non-covalent linkage through strand hybridization. A further example of non-covalent linkage is illustrated by a third shielded construct **508** comprising an oligonucleotide shield **536**. In some embodiments, oligonucleotide shield **536** is a nucleic acid aptamer that binds a recognition molecule to form a non-covalent linkage. In some embodiments, the recognition molecule is a nucleic acid aptamer, and oligonucleotide shield **536** comprises an oligonucleotide strand that hybridizes with the aptamer to form a non-covalent linkage.

[0187] A fourth shielded construct **510** shows an example of a dendritic polymer shield **538**. As used herein, in some embodiments, a dendritic polymer refers generally to a polyol or a dendrimer. Polyols and dendrimers have been described in the art, and may include branched dendritic structures optimized for a particular configuration. In some embodiments, dendritic polymer shield **538** comprises polyethylene glycol, tetraethylene glycol, poly(amidoamine), poly(propyleneimine), poly(propyleneamine), carbosilane, poly(L-lysine), or a combination of one or more thereof.

[0188] A dendrimer, or dendron, is a repetitively branched molecule that is typically symmetric around the core and that may adopt a spherical three-dimensional morphology. See, e.g., Astruc et al. (2010) Chem. Rev. 110:1857. Incorporation of such structures into a shield of the application can provide for a protective effect through the steric inhibition of contacts between a label and one or more biomolecules associated therewith (e.g., a recognition molecule and/or a polypeptide associated with the recognition molecule). Refinement of the chemical and physical properties of the dendrimer through variation in primary structure of the molecule, including potential functionalization of the dendrimer surface, allows the shielding effects to be adjusted as desired. Dendrimers may be synthesized by a variety of techniques using a wide range of materials and branching reactions, as is known in the art. Such synthetic variation allows the properties of the dendrimer to be customized as necessary. Examples of polyol and dendrimer compounds which can be used in accordance with shields of the application include, without limitation, compounds described in U.S. Patent Publication No. 20180346507.

[0189] FIG. 5D depicts further example configurations of shielded recognition molecules of the application. A protein-nucleic acid construct **512** shows an example of a shield comprising more than one polymer in the form of a protein and a double-stranded nucleic acid. In some embodiments, the protein portion of the shield is attached to the nucleic acid portion of the shield through a covalent linkage. In some embodiments, the attachment is through a non-covalent

linkage. For example, in some embodiments, the protein portion of the shield is a monovalent or multivalent protein that forms at least one non-covalent linkage through a ligand moiety attached to a ligand-binding site of the monovalent or multivalent protein. In some embodiments, the protein portion of the shield comprises an avidin protein.

[0190] In some embodiments, a shielded recognition molecule of the application is an avidin-nucleic acid construct **514**. In some embodiments, avidin-nucleic acid construct **514** includes a shield comprising an avidin protein **540** and a double-stranded nucleic acid. As described herein, avidin protein **540** may be used to form a non-covalent linkage between one or more amino acid recognition molecules and one or more labels, either directly or indirectly, such as through one or more additional shield polymers described herein.

[0191] Avidin proteins are biotin-binding proteins, generally having a biotin binding site at each of four subunits of the avidin protein. Avidin proteins include, for example, avidin, streptavidin, traptavidin, tamavidin, bradavidin, xenavidin, and homologs and variants thereof. In some cases, the monomeric, dimeric, or tetrameric form of the avidin protein can be used. In some embodiments, the avidin protein of an avidin protein complex is streptavidin in a tetrameric form (e.g., a homotetramer). In some embodiments, the biotin binding sites of an avidin protein provide attachment sites for one or more amino acid recognition molecules, one or more labels, and/or one or more additional shield polymers described herein.

[0192] An illustrative diagram of an avidin protein complex is shown in the inset panel of FIG. 5D. As shown in the inset panel, avidin protein **540** can include a binding site **542** at each of four subunits of the protein which can be bound to a biotin moiety (shown as white circles). The multivalency of avidin protein **540** can allow for various linkage configurations, which are generally shown for illustrative purposes. For example, in some embodiments, a biotin linkage moiety **544** can be used to provide a single point of attachment to avidin protein **540**. In some embodiments, a bis-biotin linkage moiety **546** can be used to provide two points of attachment to avidin protein **540**. As illustrated by avidin-nucleic acid construct **514**, an avidin protein complex may be formed by two bis-biotin linkage moieties, which form a trans-configuration to provide an increased separation distance between a recognition molecule and a label.

[0193] Various further examples of avidin protein shield configurations are shown. A first avidin construct **516** shows an example of an avidin shield attached to a recognition molecule through a bis-biotin linkage moiety and to two labels through separate biotin linkage moieties. A second avidin construct **518** shows an example of an avidin shield attached to two recognition molecules through separate biotin linkage moieties and to a label through a bis-biotin linkage moiety. A third avidin construct **520** shows an example of an avidin shield attached to two recognition molecules through separate biotin linkage moieties and to a labeled nucleic acid through a biotin linkage moiety of each strand of the nucleic acid. A fourth avidin construct **522** shows an example of an avidin shield attached to a recognition molecule and to a labeled nucleic acid through separate bis-biotin linkage moieties. As shown, the label is further shielded from the recognition molecule by a dendritic polymer between the label and nucleic acid.

[0194] It should be appreciated that the example configurations of shielded recognition molecules shown in FIGS. 5A-5D are provided for illustrative purposes. The inventors have conceived of various other shield configurations using one or more different polymers that form a covalent or non-covalent linkage between recognition and label components of a shielded recognition molecule. By way of example, FIG. 5E illustrates the modularity of shield configuration in accordance with the application.

[0195] As shown at the top of FIG. 5E, a shielded recognition molecule generally comprises a recognition component **550**, a shielding element **552**, and a label component **554**. For ease of illustration, recognition component **550** is depicted as one amino acid recognition molecule, and label component **554** is depicted as one label. It should be appreciated that shielded recognition molecules of the application can comprise shielding element **552** attached to one or more amino

acid recognition molecules and one or more labels. Where recognition component **550** comprises more than one recognition molecule, each recognition molecule can be attached to shielding element **552** at one or more attachment sites on shielding element **552**. Where label component **554** comprises more than one label, each label can be attached to shielding element **552** at one or more attachment sites on shielding element **552**.

[0196] In some embodiments, shielding element **552** comprises a protein **560**. In some embodiments, protein **560** is a monovalent or multivalent protein. In some embodiments, protein **560** is a monomeric or multimeric protein, such as a protein homodimer, protein heterodimer, protein oligomer, or other proteinaceous molecule. In some embodiments, shielding element **552** comprises a protein complex formed by a protein non-covalently bound to at least one other molecule. For example, in some embodiments, shielding element **552** comprises a protein-protein complex **562**. In some embodiments, protein-protein complex **562** comprises one proteinaceous molecule specifically bound to another proteinaceous molecule. In some embodiments, protein-protein complex **562** comprises an antibody or antibody fragment (e.g., scFv) bound to an antigen. In some embodiments, protein-protein complex **562** comprises a receptor bound to a protein ligand. Additional examples of protein-protein complexes include, without limitation, trypsin-aprotinin, barnase-barstar, and colicin E9-Im9 immunity protein.

[0197] In some embodiments, shielding element **552** comprises a protein-ligand complex **564**. In some embodiments, protein-ligand complex **564** comprises a monovalent protein and a non-proteinaceous ligand moiety. For example, in some embodiments, protein-ligand complex **564** comprises an enzyme bound to a small-molecule inhibitor moiety. In some embodiments, protein-ligand complex **564** comprises a receptor bound to a non-proteinaceous ligand moiety.

[0198] In some embodiments, shielding element **552** comprises a multivalent protein complex formed by a multivalent protein non-covalently bound to one or more ligand moieties. In some embodiments, shielding element **552** comprises an avidin protein complex formed by an avidin protein non-covalently bound to one or more biotin linkage moieties. Constructs **566**, **568**, **570**, and **572** provide illustrative examples of avidin protein complexes, any one or more of which may be incorporated into shielding element **552**.

[0199] In some embodiments, shielding element **552** comprises a two-way avidin complex **566** comprising an avidin protein bound to two bis-biotin linkage moieties. In some embodiments, shielding element **552** comprises a three-way avidin complex **568** comprising an avidin protein bound to two biotin linkage moieties and a bis-biotin linkage moiety. In some embodiments, shielding element **552** comprises a four-way avidin complex **570** comprising an avidin protein bound to four biotin linkage moieties.

[0200] In some embodiments, shielding element **552** comprises an avidin protein comprising one or two non-functional binding sites engineered into the avidin protein. For example, in some embodiments, shielding element **552** comprises a divalent avidin complex **572** comprising an avidin protein bound to a biotin linkage moiety at each of two subunits, where the avidin protein comprises a non-functional ligand-binding site **548** at each of two other subunits. As shown, in some embodiments, divalent avidin complex **572** comprises a trans-divalent avidin protein, although a cis-divalent avidin protein may be used depending on a desired implementation. In some embodiments, the avidin protein is a trivalent avidin protein. In some embodiments, the trivalent avidin protein comprises non-functional ligand-binding site **548** at one subunit and is bound to three biotin linkage moieties, or one biotin linkage moiety and one bis-biotin linkage moiety, at the other subunits.

[0201] In some embodiments, shielding element **552** comprises a dendritic polymer **574**. In some embodiments, dendritic polymer **574** is a polyol or a dendrimer, as described elsewhere herein. In some embodiments, dendritic polymer **574** is a branched polyol or a branched dendrimer. In some embodiments, dendritic polymer **574** comprises a monosaccharide-TEG, a disaccharide, an N-acetyl monosaccharide, a TEMPO-TEG, a trolox-TEG, or a glycerol dendrimer. Examples of

polyols useful in accordance with shielded recognition molecules of the application include polyether polyols and polyester polyols, e.g., polyethylene glycol, polypropylene glycol, and similar such polymers well known in the art. In some embodiments, dendritic polymer **574** comprises a compound of the following formula: $\text{---}(\text{CH}_2\text{CH}_2\text{O})_n\text{---}$, where n is an integer from 1 to 500, inclusive. In some embodiments, dendritic polymer **574** comprises a compound of the following formula: $\text{---}(\text{CH}_2\text{CH}_2\text{O})_n\text{---}$, wherein n is an integer from 1 to 100, inclusive.

[0202] In some embodiments, shielding element **552** comprises a nucleic acid. In some embodiments, the nucleic acid is single-stranded. In some embodiments, label component **554** is attached directly or indirectly to one end of the single-stranded nucleic acid (e.g., 5' end or 3' end) and recognition component **550** is attached directly or indirectly to the other end of the single-stranded nucleic acid (e.g., 3' end or 5' end). For example, the single-stranded nucleic acid can comprise a label attached to the 5' end of the nucleic acid and an amino acid recognition molecule attached to the 3' end of the nucleic acid.

[0203] In some embodiments, shielding element **552** comprises a double-stranded nucleic acid **576**. As shown, in some embodiments, double-stranded nucleic acid **576** can form a non-covalent linkage between recognition component **550** and label component **554** through strand hybridization. However, in some embodiments, double-stranded nucleic acid **576** can form a covalent linkage between recognition component **550** and label component **554** through attachment to the same oligonucleotide strand. In some embodiments, label component **554** is attached directly or indirectly to one end of the double-stranded nucleic acid and recognition component **550** is attached directly or indirectly to the other end of the double-stranded nucleic acid. For example, the double-stranded nucleic acid can comprise a label attached to the 5' end of one strand and an amino acid recognition molecule attached to the 5' end of the other strand.

[0204] In some embodiments, shielding element **552** comprises a nucleic acid that forms one or more structural motifs which can be useful for increasing steric bulk of the shield.

[0205] Examples of nucleic acid structural motifs include, without limitation, stem-loops, three-way junctions (e.g., formed by two or more stem-loop motifs), four-way junctions (e.g., Holliday junctions), and bulge loops.

[0206] In some embodiments, shielding element **552** comprises a nucleic acid that forms a stem-loop **578**. A stem-loop, or hairpin loop, is an unpaired loop of nucleotides on an oligonucleotide strand that is formed when the oligonucleotide strand folds and forms base pairs with another section of the same strand. In some embodiments, the unpaired loop of stem-loop **578** comprises three to ten nucleotides. Accordingly, stem-loop **578** can be formed by two regions of an oligonucleotide strand having inverted complementary sequences that hybridize to form a stem, where the two regions are separated by the three to ten nucleotides that form the unpaired loop. In some embodiments, the stem of stem-loop **578** can be designed to have one or more G/C nucleotides, which can provide added stability with the addition hydrogen bonding interaction that forms compared to A/T nucleotides. In some embodiments, the stem of stem-loop **578** comprises G/C nucleotides immediately adjacent to an unpaired loop sequence. In some embodiments, the stem of stem-loop **578** comprises G/C nucleotides within the first 2, 3, 4, or 5 nucleotides adjacent to an unpaired loop sequence. In some embodiments, an unpaired loop of stem-loop **578** comprises one or more attachment sites. In some embodiments, an attachment site occurs at an abasic site in the unpaired loop. In some embodiments, an attachment site occurs at a base of the unpaired loop.

[0207] In some embodiments, stem-loop **578** is formed by a double-stranded nucleic acid. As described herein, in some embodiments, the double-stranded nucleic acid can form a non-covalent linkage group through strand hybridization of first and second oligonucleotide strands. However, in some embodiments, shielding element **552** comprises a single-stranded nucleic acid that forms a stem-loop motif, e.g., to provide a covalent linkage group. In some embodiments, shielding element **552** comprises a nucleic acid that forms two or more stem-loop motifs. For example, in

some embodiments, the nucleic acid comprises two stem-loop motifs. In some embodiments, a stem of one stem-loop motif is adjacent to the stem of the other such that the motifs together form a three-way junction. In some embodiments, shielding element 552 comprises a nucleic acid that forms a four-way junction 578. In some embodiments, four-way junction 578 is formed through hybridization of two or more oligonucleotide strands (e.g., 2, 3, or 4 oligonucleotide strands). [0208] In some embodiments, shielding element 552 comprises one or more polymers selected from 560, 562, 564, 566, 568, 570, 572, 574, 576, 578, and 580 of FIG. 5E. It should be appreciated that the linkage moieties and attachment sites shown on each of 560, 562, 564, 566, 568, 570, 572, 574, 576, 578, and 580 are shown for illustrative purposes and are not intended to depict a preferred linkage or attachment site configuration. [0209] In some aspects, the application provides an amino acid recognition molecule of Formula (I):

$A-(Y)_{\text{sub}.n}-D$ (I),

wherein: A is an amino acid binding component comprising at least one amino acid recognition molecule; each instance of Y is a polymer that forms a covalent or non-covalent linkage group; n is an integer from 1 to 10, inclusive; and D is a label component comprising at least one detectable label. In some embodiments, the application provides a composition comprising a soluble amino acid recognition molecule of Formula (I).

[0210] In some embodiments, A comprises a plurality of amino acid recognition molecules. In some embodiments, each amino acid recognition molecule of the plurality is attached to a different attachment site on Y. In some embodiments, at least two amino acid recognition molecules of the plurality are attached to a single attachment site on Y. In some embodiments, the amino acid recognition molecule is a recognition protein or a nucleic acid aptamer, e.g., as described elsewhere herein.

[0211] In some embodiments, the detectable label is a luminescent label or a conductivity label. In some embodiments, the luminescent label comprises at least one fluorophore dye molecule. In some embodiments, D comprises 20 or fewer fluorophore dye molecules. In some embodiments, the ratio of the number of fluorophore dye molecules to the number of amino acid recognition molecules is between 1:1 and 20:1. In some embodiments, the luminescent label comprises at least one FRET pair comprising a donor label and an acceptor label. In some embodiments, the ratio of the donor label to the acceptor label is 1:1, 2:1, 3:1, 4:1, or 5:1. In some embodiments, the ratio of the acceptor label to the donor label is 1:1, 2:1, 3:1, 4:1, or 5:1.

[0212] In some embodiments, D is less than 200 Å in diameter. In some embodiments, $-(Y)_{\text{sub}.n}$ is at least 2 nm in length. In some embodiments, $-(Y)_{\text{sub}.n}$ is at least 5 nm in length. In some embodiments, $-(Y)_{\text{sub}.n}$ is at least 10 nm in length. In some embodiments, each instance of Y is independently a biomolecule, a polyol, or a dendrimer. In some embodiments, the biomolecule is a nucleic acid, a polypeptide, or a polysaccharide.

[0213] In some embodiments, the amino acid recognition molecule is of one of the following formulae:

$A-Y_{\text{sup}.1}-(Y)_{\text{sub}.m}-D$ or $A-(Y)_{\text{sub}.m}-Y_{\text{sup}.1}-D$,

wherein: $Y_{\text{sup}.1}$ is a nucleic acid or a polypeptide; and m is an integer from 0 to 10, inclusive.

[0214] In some embodiments, the nucleic acid comprises a first oligonucleotide strand. In some embodiments, the nucleic acid comprises a second oligonucleotide strand hybridized with the first oligonucleotide strand. In some embodiments, the nucleic acid forms a covalent linkage through the first oligonucleotide strand. In some embodiments, the nucleic acid forms a non-covalent linkage through the hybridized first and second oligonucleotide strands.

[0215] In some embodiments, the polypeptide is a monovalent or multivalent protein. In some embodiments, the monovalent or multivalent protein forms at least one non-covalent linkage

through a ligand moiety attached to a ligand-binding site of the monovalent or multivalent protein. In some embodiments, A, Y, or D comprises the ligand moiety.

[0216] In some embodiments, the amino acid recognition molecule is of one of the following formulae:

$A-(Y)_{\text{sub.m}}-Y_{\text{sup.2}}-D$ or $A-Y_{\text{sup.2}}-(Y)_{\text{sub.m}}-D$,

wherein: $Y_{\text{sup.2}}$ is a polyol or dendrimer; and m is an integer from 0 to 10, inclusive. In some embodiments, the polyol or dendrimer comprises polyethylene glycol, tetraethylene glycol, poly(amidoamine), poly(propyleneimine), poly(propyleneamine), carbosilane, poly(L-lysine), or a combination of one or more thereof.

[0217] In some aspects, the application provides an amino acid recognition molecule of Formula (II):

$A-Y_{\text{sup.1}}-D$ (II),

wherein: A is an amino acid binding component comprising at least one amino acid recognition molecule; $Y_{\text{sup.1}}$ is a nucleic acid or a polypeptide; D is a label component comprising at least one detectable label. In some embodiments, when $Y_{\text{sup.1}}$ is a nucleic acid, the nucleic acid forms a covalent or non-covalent linkage group. In some embodiments, when $Y_{\text{sup.1}}$ is a polypeptide, the polypeptide forms a non-covalent linkage group characterized by a dissociation constant ($K_{\text{sub.D}}$) of less than 50×10^{-9} M.

[0218] In some embodiments, $Y_{\text{sup.1}}$ is a nucleic acid comprising a first oligonucleotide strand. In some embodiments, the nucleic acid comprises a second oligonucleotide strand hybridized with the first oligonucleotide strand. In some embodiments, A is attached to the first oligonucleotide strand, and wherein D is attached to the second oligonucleotide strand. In some embodiments, A is attached to a first attachment site on the first oligonucleotide strand, and wherein D is attached to a second attachment site on the first oligonucleotide strand. In some embodiments, each oligonucleotide strand of the nucleic acid comprises fewer than 150, fewer than 100, or fewer than 50 nucleotides.

[0219] In some embodiments, $Y_{\text{sup.1}}$ is a monovalent or multivalent protein. In some embodiments, the monovalent or multivalent protein forms at least one non-covalent linkage through a ligand moiety attached to a ligand-binding site of the monovalent or multivalent protein. In some embodiments, at least one of A and D comprises the ligand moiety. In some embodiments, the polypeptide is an avidin protein (e.g., avidin, streptavidin, traptavidin, tamavidin, bradavidin, xenavidin, or a homolog or variant thereof). In some embodiments, the ligand moiety is a biotin moiety.

[0220] In some embodiments, the amino acid recognition molecule is of one of the following formulae:

$A-Y_{\text{sup.1}}-(Y)_{\text{sub.n}}-D$ or $A-(Y)_{\text{sub.n}}-Y_{\text{sup.1}}-D$,

wherein: each instance of Y is a polymer that forms a covalent or non-covalent linkage group; and n is an integer from 1 to 10, inclusive. In some embodiments, each instance of Y is independently a biomolecule, a polyol, or a dendrimer.

[0221] In other aspects, the application provides an amino acid recognition molecule comprising: a nucleic acid; at least one amino acid recognition molecule attached to a first attachment site on the nucleic acid; and at least one detectable label attached to a second attachment site on the nucleic acid. In some embodiments, the nucleic acid forms a covalent or non-covalent linkage group between the at least one amino acid recognition molecule and the at least one detectable label.

[0222] In some embodiments, the nucleic acid is a double-stranded nucleic acid comprising a first oligonucleotide strand hybridized with a second oligonucleotide strand. In some embodiments, the first attachment site is on the first oligonucleotide strand, and wherein the second attachment site is

on the second oligonucleotide strand. In some embodiments, the at least one amino acid recognition molecule is attached to the first attachment site through a protein that forms a covalent or non-covalent linkage group between the at least one amino acid recognition molecule and the nucleic acid. In some embodiments, the at least one detectable label is attached to the second attachment site through a protein that forms a covalent or non-covalent linkage group between the at least one detectable label and the nucleic acid. In some embodiments, the first and second attachment sites are separated by between 5 and 100 nucleotide bases or nucleotide base pairs on the nucleic acid.

[0223] In yet other aspects, the application provides an amino acid recognition molecule comprising: a multivalent protein comprising at least two ligand-binding sites; at least one amino acid recognition molecule attached to the protein through a first ligand moiety bound to a first ligand-binding site on the protein; and at least one detectable label attached to the protein through a second ligand moiety bound to a second ligand-binding site on the protein.

[0224] In some embodiments, the multivalent protein is an avidin protein comprising four ligand-binding sites. In some embodiments, the ligand-binding sites are biotin binding sites, and wherein the ligand moieties are biotin moieties. In some embodiments, at least one of the biotin moieties is a bis-biotin moiety, and wherein the bis-biotin moiety is bound to two biotin binding sites on the avidin protein. In some embodiments, the at least one amino acid recognition molecule is attached to the protein through a nucleic acid comprising the first ligand moiety. In some embodiments, the at least one detectable label is attached to the protein through a nucleic acid comprising the second ligand moiety.

[0225] As described elsewhere herein, shielded recognition molecules of the application may be used in a polypeptide sequencing method in accordance with the application, or any method known in the art. For example, in some embodiments, a shielded recognition molecule provided herein may be used in an Edman-type degradation reaction provided herein, or conventionally known in the art, which can involve iterative cycling of multiple reaction mixtures in a polypeptide sequencing reaction. In some embodiments, a shielded recognition molecule provided herein may be used in a dynamic sequencing reaction of the application, which involves amino acid recognition and degradation in a single reaction mixture.

Sequencing by Degradation of Labeled Polypeptides

[0226] In some aspects, the application provides a method of sequencing a polypeptide by identifying a unique combination of amino acids corresponding to a known polypeptide sequence. For example, FIG. 6 shows a method of sequencing by detecting selectively labeled amino acids of a labeled polypeptide **600**. In some embodiments, labeled polypeptide **600** comprises selectively modified amino acids such that different amino acid types comprise different luminescent labels. As used herein, unless otherwise indicated, a labeled polypeptide refers to a polypeptide comprising one or more selectively labeled amino acid sidechains. Methods of selective labeling and details relating to the preparation and analysis of labeled polypeptides are known in the art (see, e.g., Swaminathan, et al. PLOS Comput Biol. 2015, 11(2): e1004080).

[0227] As shown, in some embodiments, labeled polypeptide **600** is immobilized and exposed to an excitation source. An aggregate luminescence from labeled polypeptide **600** is detected and, in some embodiments, exposure to luminescence over time results in a loss in detected signal due to luminescent label degradation (e.g., degradation due to photobleaching). In some embodiments, labeled polypeptide **600** comprises a unique combination of selectively labeled amino acids that give rise to an initial detected signal. As generically illustrated, degradation of luminescent labels over time results in a corresponding decrease in a detected signal for the photobleached labeled polypeptide **602**. In some embodiments, the signal can be deconvoluted by analysis of one or more luminescence properties (e.g., signal deconvolution by luminescence lifetime analysis). In some embodiments, the unique combination of selectively labeled amino acids of labeled polypeptide **600** have been computationally precomputed and empirically verified—e.g., based on known polypeptide sequences of a proteome. In some embodiments, the combination of detected amino

acid labels are compared against a database of known sequences of a proteome of an organism to identify a particular polypeptide of the database corresponding to labeled polypeptide **600**.

[0228] In some embodiments, the approach illustrated in FIG. **6** may be modified by determining an optimal sample concentration for performing a sequencing reaction that maximizes sampling in massively parallel analysis. In some embodiments, the concentration is selected so that a desired fraction of the sample wells of an array (e.g., 30%) are occupied at any given time. Without wishing to be bound by theory, it is thought that while a polypeptide is bleached over a period of time, the same well continues to be available for further analysis. Through diffusion, approximately 30% of the sample wells of an array can be used for analysis every 3 minutes. As an illustrative example, in a million sample well chip, 6,000,000 polypeptides per hour may be sampled, or 24,000,000 over a 4 hour period.

[0229] In some aspects, the application provides a method of sequencing a polypeptide by detecting luminescence of a labeled polypeptide which is subjected to repeated cycles of terminal amino acid modification and cleavage. For example, FIG. **7** shows a method of sequencing a labeled polypeptide by Edman degradation in accordance with the application. In some embodiments, the method generally proceeds as described herein for other methods of sequencing by Edman degradation. For example, in some embodiments, steps (1) and (2) shown in FIG. **7** may be performed as described elsewhere herein for terminal amino acid modification and terminal amino acid cleavage, respectively, in an Edman degradation reaction.

[0230] As shown in the example depicted in FIG. **7**, in some embodiments, the method comprises a step of (1) modifying the terminal amino acid of a labeled polypeptide. As described elsewhere herein, in some embodiments, modifying comprises contacting the terminal amino acid with an isothiocyanate (e.g., PITC) to form an isothiocyanate-modified terminal amino acid. In some embodiments, an isothiocyanate modification **710** converts the terminal amino acid to a form that is more susceptible to removal by a cleaving reagent (e.g., a chemical or enzymatic cleaving reagent, as described herein). Accordingly, in some embodiments, the method comprises a step of (2) removing the modified terminal amino acid using chemical or enzymatic means detailed elsewhere herein for Edman degradation.

[0231] In some embodiments, the method comprises repeating steps (1) through (2) for a plurality of cycles, during which luminescence of the labeled polypeptide is detected, and cleavage events corresponding to the removal of a labeled amino acid from the terminus may be detected as a decrease in detected signal. In some embodiments, no change in signal following step (2) as shown in FIG. **7** identifies an amino acid of unknown type. Accordingly, in some embodiments, partial sequence information may be determined by evaluating a signal detected following step (2) during each sequential round by assigning an amino acid type by a determined identity based on a change in detected signal or identifying an amino acid type as unknown based on no change in a detected signal.

[0232] In some aspects, a method of sequencing a polypeptide in accordance with the application comprises sequencing by processive enzymatic cleavage of a labeled polypeptide, as generally illustrated in FIGS. **8A-8C**. As shown, in some embodiments, a labeled polypeptide is subjected to degradation using a modified processive exopeptidase that continuously cleaves a terminal amino acid from one terminus to another terminus. Exopeptidases are described in detail elsewhere herein. FIG. **8A** depicts an example in which a labeled polypeptide **800** is subjected to degradation by an immobilized processive exopeptidase **810**. FIG. **8B** depicts an example in which an immobilized labeled polypeptide **820** is subjected to degradation by a processive exopeptidase **830**.

[0233] FIG. **8C** schematically illustrates an example of a real-time sequencing process performed in accordance with the method depicted in FIG. **8B**. As shown, panels (I) through (IV) show a progression of labeled polypeptide degradation, with a corresponding signal trace over time shown below each panel. As shown, each cleavage event corresponding to a labeled amino acid gives rise to a concomitant drop in signal. In some embodiments, the rate of processivity of processive

exopeptidase **830** is known, such that the timing between a detected decrease in signal may be used to calculate the number of unlabeled amino acids between each detection event. For example, if a polypeptide of **40** amino acids was cleaved in such a way that an amino acid was removed every second, a labeled polypeptide having 3 signals would show all 3 initially (panel (I)), then 2 (panel (II)), then 1 (panel (III)), and finally no signal. In this way, the order of the labeled amino acids can be determined. Accordingly, these methods may be used to determine partial sequence information, e.g., for proteomic analysis based on polypeptide fragment sequencing.

[0234] In some embodiments, single molecule protein sequencing can be achieved using an ATP-based Förster resonance energy transfer (FRET) scheme (e.g., with one or more labeled cofactors), for example as illustrated in FIG. **9**. In some embodiments, sequencing by cofactor-based FRET can be performed using an immobilized ATP-dependent protease, donor-labeled ATP, and acceptor-labeled amino acids of a polypeptide substrate. In some embodiments, amino acids can be labeled with acceptors, and the one or more cofactors can be labeled with donors.

[0235] For example, in some embodiments, extracted proteins are denatured, and cysteines and lysines are labeled with fluorescent dyes. In some embodiments, an engineered version of a protein translocase (e.g., bacterial ClpX) is used to bind to individual substrate proteins, unfold them, and translocate them through its nano-channel. In some embodiments, the translocase is labeled with a donor dye, and FRET occurs between the donor on the translocase and two or more distinct acceptor dyes on a substrate when the substrate passes through the nano-channel. The order of the labeled amino acids can then be determined from the FRET signal. In some embodiments, one or more of the following non-limiting labeled ATP analogues shown in Table 5 can be used.

TABLE-US-00005 TABLE 5 Non-limiting examples of labeled ATP analogues. Phosphate-labeled ATP: [00001]  (γ-[(6-Amino)hexyl]-ATP) [00002]  (γ-[(6-Amino)hexyl]imido-ATP) [00003]  (γ-[(6-Amino)hexyl]imido-ATP-Cy3) [00004]  (γ-[(6-Amino)hexyl]imido-ATP-Cy3) [00005]  (BODIPY FL ATPγS) Ribose-labeled ATP: [00006]  (EDA-ATP) [00007]  (EDA-ATP-Cy3) [00008]  (EDA-ATP-Cy3) Base-labeled ATP: [00009]  (N.sup.6-(6-Amino)hexyl-ATP) [00010]  (N.sup.6-(6-Amino)hexyl-ATP-Cy3)

Preparation of Samples for Sequencing

[0236] A polypeptide sample can be modified prior to sequencing. In some embodiments, the N-terminal amino acid or the C-terminal amino acid of a polypeptide is modified. FIG. **10A** illustrates a non-limiting example of terminal end modification for preparing terminally modified polypeptides from a protein sample. In step (1), protein sample **1000** is fragmented to produce polypeptide fragments **1002**. A polypeptide can be fragmented by cleaving (e.g., chemically) and/or digesting (e.g., enzymatically, for example using a peptidase, for example trypsin) a polypeptide of interest. Fragmentation can be performed before or after labeling. In some embodiments, fragmentation is performed after labeling of whole proteins. One or more amino acids can be labeled before or after cleavage to produce labeled polypeptides. In some embodiments, polypeptides are size selected after chemical or enzymatic fragmentation. In some embodiments, smaller polypeptides (e.g., <2 kDa) are removed and larger polypeptides are retained for sequence analysis. Size selection can be achieved using a technique such as gel filtration, SEC, dialysis, PAGE gel extraction, microfluidic tension flow, or any other suitable technique. In step (2), the N-termini or C-termini of polypeptide fragments **1002** are modified to produce terminally modified polypeptides **1004**. In some embodiments, modification comprises adding an immobilization moiety. In some embodiments, modification comprises adding a coupling moiety.

[0237] Accordingly, provided herein are methods of modifying terminal ends of proteins and polypeptides with moieties that enable immobilization to a surface (e.g., a surface of a sample well on a chip used for protein analysis). In some embodiments, such methods comprise modifying a terminal end of a labeled polypeptide to be analyzed in accordance with the application. In yet

other embodiments, such methods comprise modifying a terminal end of a protein or enzyme that degrades or translocates a protein or polypeptide substrate in accordance with the application. [0238] In some embodiments, a carboxy-terminus of a protein or polypeptide is modified in a method comprising: (i) blocking free carboxylate groups of the protein or polypeptide; (ii) denaturing the protein or polypeptide (e.g., by heat and/or chemical means); (iii) blocking free thiol groups of the protein or polypeptide; (iv) digesting the protein or polypeptide to produce at least one polypeptide fragment comprising a free C-terminal carboxylate group; and (v) conjugating (e.g., chemically) a functional moiety to the free C-terminal carboxylate group. In some embodiments, the method further comprises, after (i) and before (ii), dialyzing a sample comprising the protein or polypeptide.

[0239] In some embodiments, a carboxy-terminus of a protein or polypeptide is modified in a method comprising: (i) denaturing the protein or polypeptide (e.g., by heat and/or chemical means); (ii) blocking free thiol groups of the protein or polypeptide; (iii) digesting the protein or polypeptide to produce at least one polypeptide fragment comprising a free C-terminal carboxylate group; (iv) blocking the free C-terminal carboxylate group to produce at least one polypeptide fragment comprising a blocked C-terminal carboxylate group; and (v) conjugating (e.g., enzymatically) a functional moiety to the blocked C-terminal carboxylate group. In some embodiments, the method further comprises, after (iv) and before (v), dialyzing a sample comprising the protein or polypeptide.

[0240] In some embodiments, blocking free carboxylate groups refers to a chemical modification of these groups which alters chemical reactivity relative to an unmodified carboxylate. Suitable carboxylate blocking methods are known in the art and should modify side-chain carboxylate groups to be chemically different from a carboxy-terminal carboxylate group of a polypeptide to be functionalized. In some embodiments, blocking free carboxylate groups comprises esterification or amidation of free carboxylate groups of a polypeptide. In some embodiments, blocking free carboxylate groups comprises methyl esterification of free carboxylate groups of a polypeptide, e.g., by reacting the polypeptide with methanolic HCl. Additional examples of reagents and techniques useful for blocking free carboxylate groups include, without limitation, 4-sulfo-2,3,5,6-tetrafluorophenol (STP) and/or a carbodiimide such as N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride (EDAC), uronium reagents, diazomethane, alcohols and acid for Fischer esterification, the use of N-hydroxysuccinimide (NHS) to form NHS esters (potentially as an intermediate to subsequent ester or amine formation), or reaction with carbonyldiimidazole (CDI) or the formation of mixed anhydrides, or any other method of modifying or blocking carboxylic acids, potentially through the formation of either esters or amides.

[0241] In some embodiments, blocking free thiol groups refers to a chemical modification of these groups which alters chemical reactivity relative to an unmodified thiol. In some embodiments, blocking free thiol groups comprises reducing and alkylating free thiol groups of a protein or polypeptide. In some embodiments, reduction and alkylation is carried out by contacting a polypeptide with dithiothreitol (DTT) and one or both of iodoacetamide and iodoacetic acid. Examples of additional and alternative cysteine-reducing reagents which may be used are well known and include, without limitation, 2-mercaptoethanol, Tris(2-carboxyethyl) phosphine hydrochloride (TCEP), tributylphosphine, dithiobutylamine (DTBA), or any reagent capable of reducing a thiol group. Examples of additional and alternative cysteine-blocking (e.g., cysteine-alkylating) reagents which may be used are well known and include, without limitation, acrylamide, 4-vinylpyridine, N-Ethylmaleimide (NEM), N-8-maleimidocaproic acid (EMCA), or any reagent that modifies cysteines so as to prevent disulfide bond formation.

[0242] In some embodiments, digestion comprises enzymatic digestion. In some embodiments, digestion is carried out by contacting a protein or polypeptide with an endopeptidase (e.g., trypsin) under digestion conditions. In some embodiments, digestion comprises chemical digestion. Examples of suitable reagents for chemical and enzymatic digestion are known in the art and

include, without limitation, trypsin, chymotrypsin, Lys-C, Arg-C, Asp-N, Lys-N, BNPS-Skatole, CNBr, caspase, formic acid, glutamyl endopeptidase, hydroxylamine, iodosobenzoic acid, neutrophil elastase, pepsin, proline-endopeptidase, proteinase K, staphylococcal peptidase I, thermolysin, and thrombin.

[0243] In some embodiments, the functional moiety comprises a biotin molecule. In some embodiments, the functional moiety comprises a reactive chemical moiety, such as an alkynyl. In some embodiments, conjugating a functional moiety comprises biotinylation of carboxy-terminal carboxy-methyl ester groups by carboxypeptidase Y, as known in the art.

[0244] In some embodiments, a solubilizing moiety is added to a polypeptide. FIG. 10B illustrates a non-limiting example of a solubilizing moiety added to a terminal amino acid of a polypeptide, for example using a process of conjugating a solubilizing linker to the polypeptide.

[0245] In some embodiments, a terminally modified polypeptide **1010** comprising a linker conjugating moiety **1012** is conjugated to a solubilizing linker **1020** comprising a polypeptide conjugating moiety **1022**. In some embodiments, the solubilizing linker comprises a solubilizing polymer, such as a biomolecule (e.g., shown as stippled shape). In some embodiments, a resulting linker-conjugated polypeptide **1030** comprising a linkage **1032** formed between 1012 and 1022 further comprises a surface conjugating moiety **1034**. Accordingly, in some embodiments methods and compositions provided herein are useful for modifying terminal ends of polypeptides with moieties that increase their solubility. In some embodiments, a solubilizing moiety is useful for small polypeptides that result from fragmentation (e.g., enzymatic fragmentation, for example using trypsin) and that are relatively insoluble. For example, in some embodiments, short polypeptides in a polypeptide pool can be solubilized by conjugating a polymer (e.g., a short oligo, a sugar, or other charged polymer) to the polypeptides.

[0246] In some embodiments, one or more surfaces of a sample well (e.g., sidewalls of a sample well) can be modified. A non-limiting example of passivation and/or antifouling of a sample well sidewall is shown in FIG. 10C where an example schematic of a sample well is illustrated with modified surfaces which may be used to promote single molecule immobilization to a bottom surface. In some embodiments, **1040** is S.sub.102. In some embodiments, **1042** is a polypeptide conjugating moiety (e.g., TCO, tetrazine, N₃, alkyne, aldehyde, NCO, NHS, thiol, alkene, DBCO, BCN, TPP, biotin, or other suitable conjugating moiety). In some embodiments, **1050** is TiO.sub.2 or Al.sub.2O.sub.3. In some embodiments, **1052** is a hydrophobic C.sub.4-18 molecule, a polytetrafluoroethylene compound (e.g., (CF.sub.2).sub.4-12), a polyol, such as a polyethylene glycol (e.g., PEG3-100), polypropylene glycol, polyoxyethylene glycol, or combinations or variations thereof, or a zwitterion, such as sulfobetaine. In some embodiments, **1060** is Si. In some embodiments, **1070** is Al. In some embodiments, **1080** is TiN.

Luminescent Labels

[0247] As used herein, a luminescent label is a molecule that absorbs one or more photons and may subsequently emit one or more photons after one or more time durations. In some embodiments, the term is used interchangeably with “label” or “luminescent molecule” depending on context. A luminescent label in accordance with certain embodiments described herein may refer to a luminescent label of a labeled affinity reagent, a luminescent label of a labeled peptidase (e.g., a labeled exopeptidase, a labeled non-specific exopeptidase), a luminescent label of a labeled peptide, a luminescent label of a labeled cofactor, or another labeled composition described herein. In some embodiments, a luminescent label in accordance with the application refers to a labeled amino acid of a labeled polypeptide comprising one or more labeled amino acids.

[0248] In some embodiments, a luminescent label may comprise a first and second chromophore. In some embodiments, an excited state of the first chromophore is capable of relaxation via an energy transfer to the second chromophore. In some embodiments, the energy transfer is a Förster resonance energy transfer (FRET). Such a FRET pair may be useful for providing a luminescent label with properties that make the label easier to differentiate from amongst a plurality of

luminescent labels in a mixture—e.g., as illustrated and described herein for labeled aptamer **106** of FIG. **1C**. In yet other embodiments, a FRET pair comprises a first chromophore of a first luminescent label and a second chromophore of a second luminescent label—e.g., as illustrated and described herein for sequencing of labeled peptides using a labeled cofactor (see, e.g., FIG. **9**). In certain embodiments, the FRET pair may absorb excitation energy in a first spectral range and emit luminescence in a second spectral range.

[0249] In some embodiments, a luminescent label refers to a fluorophore or a dye. Typically, a luminescent label comprises an aromatic or heteroaromatic compound and can be a pyrene, anthracene, naphthalene, naphthylamine, acridine, stilbene, indole, benzindole, oxazole, carbazole, thiazole, benzothiazole, benzoxazole, phenanthridine, phenoxazine, porphyrin, quinoline, ethidium, benzamide, cyanine, carbocyanine, salicylate, anthranilate, coumarin, fluorescein, rhodamine, xanthene, or other like compound.

[0250] In some embodiments, a luminescent label comprises a dye selected from one or more of the following: 5/6-Carboxyrhodamine 6G, 5-Carboxyrhodamine 6G, 6-Carboxyrhodamine 6G, 6-TAMRA, Abberior® STAR 440SXP, Abberior® STAR 470SXP, Abberior® STAR 488, Abberior® STAR 512, Abberior® STAR 520SXP, Abberior® STAR 580, Abberior® STAR 600, Abberior® STAR 635, Abberior® STAR 635P, Abberior® STAR RED, Alexa Fluor® 350, Alexa Fluor® 405, Alexa Fluor® 430, Alexa Fluor® 480, Alexa Fluor® 488, Alexa Fluor® 514, Alexa Fluor® 532, Alexa Fluor® 546, Alexa Fluor® 555, Alexa Fluor® 568, Alexa Fluor® 594, Alexa Fluor® 610-X, Alexa Fluor® 633, Alexa Fluor® 647, Alexa Fluor® 660, Alexa Fluor® 680, Alexa Fluor® 700, Alexa Fluor® 750, Alexa Fluor® 790, AMCA, ATTO 390, ATTO 425, ATTO 465, ATTO 488, ATTO 495, ATTO 514, ATTO 520, ATTO 532, ATTO 542, ATTO 550, ATTO 565, ATTO 590, ATTO 610, ATTO 620, ATTO 633, ATTO 647, ATTO 647N, ATTO 655, ATTO 665, ATTO 680, ATTO 700, ATTO 725, ATTO 740, ATTO Oxal2, ATTO Rho101, ATTO Rho11, ATTO Rho12, ATTO Rho13, ATTO Rho14, ATTO Rho3B, ATTO Rho6G, ATTO Thio12, BD Horizon™ V450, BODIPY® 493/501, BODIPY® 530/550, BODIPY® 558/568, BODIPY® 564/570, BODIPY® 576/589, BODIPY® 581/591, BODIPY® 630/650, BODIPY® 650/665, BODIPY® FL, BODIPY® FL-X, BODIPY® R6G, BODIPY® TMR, BODIPY® TR, CAL Fluor® Gold 540, CAL Fluor® Green 510, CAL Fluor® Orange 560, CAL Fluor® Red 590, CAL Fluor® Red 610, CAL Fluor® Red 615, CAL Fluor® Red 635, Cascade® Blue, CF™350, CF™405M, CF™405S, CF™488A, CF™514, CF™532, CF™543, CF™546, CF™555, CF™568, CF™594, CF™620R, CF™633, CF™633-V1, CF™640R, CF™640R-V1, CF™640R-V2, CF™660C, CF™660R, CF™680, CF™680R, CF™680R-V1, CF™750, CF™770, CF™790, Chromeo™ 642, Chromis 425N, Chromis 500N, Chromis 515N, Chromis 530N, Chromis 550A, Chromis 550C, Chromis 550Z, Chromis 560N, Chromis 570N, Chromis 577N, Chromis 600N, Chromis 630N, Chromis 645A, Chromis 645C, Chromis 645Z, Chromis 678A, Chromis 678C, Chromis 678Z, Chromis 770A, Chromis 770C, Chromis 800A, Chromis 800C, Chromis 830A, Chromis 830C, Cy®3, Cy®3.5, Cy®3B, Cy®5, Cy®5.5, Cy®7, DyLight® 350, DyLight® 405, DyLight® 415-Col, DyLight® 425Q, DyLight® 485-LS, DyLight® 488, DyLight® 504Q, DyLight® 510-LS, DyLight® 515-LS, DyLight® 521-LS, DyLight® 530-R2, DyLight® 543Q, DyLight® 550, DyLight® 554-R0, DyLight® 554-R1, DyLight® 590-R2, DyLight® 594, DyLight® 610-B1, DyLight® 615-B2, DyLight® 633, DyLight® 633-B1, DyLight® 633-B2, DyLight® 650, DyLight® 655-B1, DyLight® 655-B2, DyLight® 655-B3, DyLight® 655-B4, DyLight® 662Q, DyLight® 675-B1, DyLight® 675-B2, DyLight® 675-B3, DyLight® 675-B4, DyLight® 679-C5, DyLight® 680, DyLight® 683Q, DyLight® 690-B1, DyLight® 690-B2, DyLight® 696Q, DyLight® 700-B1, DyLight® 700-B1, DyLight® 730-B1, DyLight® 730-B2, DyLight® 730-B3, DyLight® 730-B4, DyLight® 747, DyLight® 747-B1, DyLight® 747-B2, DyLight® 747-B3, DyLight® 747-B4, DyLight® 755, DyLight® 766Q, DyLight® 775-B2, DyLight® 775-B3, DyLight® 775-B4, DyLight® 780-B1, DyLight® 780-B2, DyLight® 780-B3, DyLight® 800, DyLight® 830-B2, Dyomics-350, Dyomics-350XL, Dyomics-360XL, Dyomics-370XL, Dyomics-

375XL, Dyomics-380XL, Dyomics-390XL, Dyomics-405, Dyomics-415, Dyomics-430, Dyomics-431, Dyomics-478, Dyomics-480XL, Dyomics-481XL, Dyomics-485XL, Dyomics-490, Dyomics-495, Dyomics-505, Dyomics-510XL, Dyomics-511XL, Dyomics-520XL, Dyomics-521XL, Dyomics-530, Dyomics-547, Dyomics-547P1, Dyomics-548, Dyomics-549, Dyomics-549P1, Dyomics-550, Dyomics-554, Dyomics-555, Dyomics-556, Dyomics-560, Dyomics-590, Dyomics-591, Dyomics-594, Dyomics-601XL, Dyomics-605, Dyomics-610, Dyomics-615, Dyomics-630, Dyomics-631, Dyomics-632, Dyomics-633, Dyomics-634, Dyomics-635, Dyomics-636, Dyomics-647, Dyomics-647P1, Dyomics-648, Dyomics-648P1, Dyomics-649, Dyomics-649P1, Dyomics-650, Dyomics-651, Dyomics-652, Dyomics-654, Dyomics-675, Dyomics-676, Dyomics-677, Dyomics-678, Dyomics-679P1, Dyomics-680, Dyomics-681, Dyomics-682, Dyomics-700, Dyomics-701, Dyomics-703, Dyomics-704, Dyomics-730, Dyomics-731, Dyomics-732, Dyomics-734, Dyomics-749, Dyomics-749P1, Dyomics-750, Dyomics-751, Dyomics-752, Dyomics-754, Dyomics-776, Dyomics-777, Dyomics-778, Dyomics-780, Dyomics-781, Dyomics-782, Dyomics-800, Dyomics-831, eFluor® 450, Eosin, FITC, Fluorescein, HiLyte™ Fluor 405, HiLyte™ Fluor 488, HiLyte™ Fluor 532, HiLyte™ Fluor 555, HiLyte™ Fluor 594, HiLyte™ Fluor 647, HiLyte™ Fluor 680, HiLyte™ Fluor 750, IRDye® 680LT, IRDye® 750, IRDye® 800CW, JOE, LightCycler® 640R, LightCycler® Red 610, LightCycler® Red 640, LightCycler® Red 670, LightCycler® Red 705, Lissamine Rhodamine B, Naphthofluorescein, Oregon Green® 488, Oregon Green® 514, Pacific Blue™, Pacific Green™, Pacific Orange™, PET, PF350, PF405, PF415, PF488, PF505, PF532, PF546, PF555P, PF568, PF594, PF610, PF633P, PF647P, Quasar® 570, Quasar® 670, Quasar® 705, Rhodamine 123, Rhodamine 6G, Rhodamine B, Rhodamine Green, Rhodamine Green-X, Rhodamine Red, ROX, Seta™ 375, Seta™ 470, Seta™ 555, Seta™ 632, Seta™ 633, Seta™ 650, Seta™ 660, Seta™ 670, Seta™ 680, Seta™ 700, Seta™ 750, Seta™ 780, Seta™ APC-780, Seta™ PerCP-680, Seta™ R-PE-670, Seta™ 646, SeTau 380, SeTau 425, SeTau 647, SeTau 405, Square 635, Square 650, Square 660, Square 672, Square 680, Sulforhodamine 101, TAMRA, TET, Texas Red®, TMR, TRITC, Yakima Yellow™, Zenon®, Zy3, Zy5, Zy5.5, and Zy7.

Luminescence

[0251] In some aspects, the application relates to polypeptide sequencing and/or identification based on one or more luminescence properties of a luminescent label. In some embodiments, a luminescent label is identified based on luminescence lifetime, luminescence intensity, brightness, absorption spectra, emission spectra, luminescence quantum yield, or a combination of two or more thereof. In some embodiments, a plurality of types of luminescent labels can be distinguished from each other based on different luminescence lifetimes, luminescence intensities, brightnesses, absorption spectra, emission spectra, luminescence quantum yields, or combinations of two or more thereof. Identifying may mean assigning the exact identity and/or quantity of one type of amino acid (e.g., a single type or a subset of types) associated with a luminescent label, and may also mean assigning an amino acid location in a polypeptide relative to other types of amino acids.

[0252] In some embodiments, luminescence is detected by exposing a luminescent label to a series of separate light pulses and evaluating the timing or other properties of each photon that is emitted from the label. In some embodiments, information for a plurality of photons emitted sequentially from a label is aggregated and evaluated to identify the label and thereby identify an associated type of amino acid. In some embodiments, a luminescence lifetime of a label is determined from a plurality of photons that are emitted sequentially from the label, and the luminescence lifetime can be used to identify the label. In some embodiments, a luminescence intensity of a label is determined from a plurality of photons that are emitted sequentially from the label, and the luminescence intensity can be used to identify the label. In some embodiments, a luminescence lifetime and luminescence intensity of a label is determined from a plurality of photons that are emitted sequentially from the label, and the luminescence lifetime and luminescence intensity can be used to identify the label.

[0253] In some aspects of the application, a single polypeptide molecule is exposed to a plurality of separate light pulses and a series of emitted photons are detected and analyzed. In some embodiments, the series of emitted photons provides information about the single polypeptide molecule that is present and that does not change in the reaction sample over the time of the experiment. However, in some embodiments, the series of emitted photons provides information about a series of different molecules that are present at different times in the reaction sample (e.g., as a reaction or process progresses). By way of example and not limitation, such information may be used to sequence and/or identify a polypeptide subjected to chemical or enzymatic degradation in accordance with the application.

[0254] In certain embodiments, a luminescent label absorbs one photon and emits one photon after a time duration. In some embodiments, the luminescence lifetime of a label can be determined or estimated by measuring the time duration. In some embodiments, the luminescence lifetime of a label can be determined or estimated by measuring a plurality of time durations for multiple pulse events and emission events. In some embodiments, the luminescence lifetime of a label can be differentiated amongst the luminescence lifetimes of a plurality of types of labels by measuring the time duration. In some embodiments, the luminescence lifetime of a label can be differentiated amongst the luminescence lifetimes of a plurality of types of labels by measuring a plurality of time durations for multiple pulse events and emission events. In certain embodiments, a label is identified or differentiated amongst a plurality of types of labels by determining or estimating the luminescence lifetime of the label. In certain embodiments, a label is identified or differentiated amongst a plurality of types of labels by differentiating the luminescence lifetime of the label amongst a plurality of the luminescence lifetimes of a plurality of types of labels.

[0255] Determination of a luminescence lifetime of a luminescent label can be performed using any suitable method (e.g., by measuring the lifetime using a suitable technique or by determining time-dependent characteristics of emission). In some embodiments, determining the luminescence lifetime of one label comprises determining the lifetime relative to another label. In some embodiments, determining the luminescence lifetime of a label comprises determining the lifetime relative to a reference. In some embodiments, determining the luminescence lifetime of a label comprises measuring the lifetime (e.g., fluorescence lifetime). In some embodiments, determining the luminescence lifetime of a label comprises determining one or more temporal characteristics that are indicative of lifetime. In some embodiments, the luminescence lifetime of a label can be determined based on a distribution of a plurality of emission events (e.g., 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 30, 40, 50, 60, 70, 80, 90, 100, or more emission events) occurring across one or more time-gated windows relative to an excitation pulse. For example, a luminescence lifetime of a label can be distinguished from a plurality of labels having different luminescence lifetimes based on the distribution of photon arrival times measured with respect to an excitation pulse.

[0256] It should be appreciated that a luminescence lifetime of a luminescent label is indicative of the timing of photons emitted after the label reaches an excited state and the label can be distinguished by information indicative of the timing of the photons. Some embodiments may include distinguishing a label from a plurality of labels based on the luminescence lifetime of the label by measuring times associated with photons emitted by the label. The distribution of times may provide an indication of the luminescence lifetime which may be determined from the distribution. In some embodiments, the label is distinguishable from the plurality of labels based on the distribution of times, such as by comparing the distribution of times to a reference distribution corresponding to a known label. In some embodiments, a value for the luminescence lifetime is determined from the distribution of times.

[0257] As used herein, in some embodiments, luminescence intensity refers to the number of emitted photons per unit time that are emitted by a luminescent label which is being excited by delivery of a pulsed excitation energy. In some embodiments, the luminescence intensity refers to

the detected number of emitted photons per unit time that are emitted by a label which is being excited by delivery of a pulsed excitation energy, and are detected by a particular sensor or set of sensors.

[0258] As used herein, in some embodiments, brightness refers to a parameter that reports on the average emission intensity per luminescent label. Thus, in some embodiments, “emission intensity” may be used to generally refer to brightness of a composition comprising one or more labels. In some embodiments, brightness of a label is equal to the product of its quantum yield and extinction coefficient.

[0259] As used herein, in some embodiments, luminescence quantum yield refers to the fraction of excitation events at a given wavelength or within a given spectral range that lead to an emission event, and is typically less than 1. In some embodiments, the luminescence quantum yield of a luminescent label described herein is between 0 and about 0.001, between about 0.001 and about 0.01, between about 0.01 and about 0.1, between about 0.1 and about 0.5, between about 0.5 and 0.9, or between about 0.9 and 1. In some embodiments, a label is identified by determining or estimating the luminescence quantum yield.

[0260] As used herein, in some embodiments, an excitation energy is a pulse of light from a light source. In some embodiments, an excitation energy is in the visible spectrum. In some embodiments, an excitation energy is in the ultraviolet spectrum. In some embodiments, an excitation energy is in the infrared spectrum. In some embodiments, an excitation energy is at or near the absorption maximum of a luminescent label from which a plurality of emitted photons are to be detected. In certain embodiments, the excitation energy is between about 500 nm and about 700 nm (e.g., between about 500 nm and about 600 nm, between about 600 nm and about 700 nm, between about 500 nm and about 550 nm, between about 550 nm and about 600 nm, between about 600 nm and about 650 nm, or between about 650 nm and about 700 nm). In certain embodiments, an excitation energy may be monochromatic or confined to a spectral range. In some embodiments, a spectral range has a range of between about 0.1 nm and about 1 nm, between about 1 nm and about 2 nm, or between about 2 nm and about 5 nm. In some embodiments, a spectral range has a range of between about 5 nm and about 10 nm, between about 10 nm and about 50 nm, or between about 50 nm and about 100 nm.

Sequencing

[0261] Aspects of the application relate to sequencing biological polymers, such as polypeptides and proteins. As used herein, “sequencing,” “sequence determination,” “determining a sequence,” and like terms, in reference to a polypeptide or protein includes determination of partial sequence information as well as full sequence information of the polypeptide or protein. That is, the terminology includes sequence comparisons, fingerprinting, probabilistic fingerprinting, and like levels of information about a target molecule, as well as the express identification and ordering of each amino acid of the target molecule within a region of interest. In some embodiments, the terminology includes identifying a single amino acid of a polypeptide. In yet other embodiments, more than one amino acid of a polypeptide is identified. As used herein, in some embodiments, “identifying,” “determining the identity,” and like terms, in reference to an amino acid includes determination of an express identity of an amino acid as well as determination of a probability of an express identity of an amino acid. For example, in some embodiments, an amino acid is identified by determining a probability (e.g., from 0% to 100%) that the amino acid is of a specific type, or by determining a probability for each of a plurality of specific types. Accordingly, in some embodiments, the terms “amino acid sequence,” “polypeptide sequence,” and “protein sequence” as used herein may refer to the polypeptide or protein material itself and is not restricted to the specific sequence information (e.g., the succession of letters representing the order of amino acids from one terminus to another terminus) that biochemically characterizes a specific polypeptide or protein.

[0262] In some embodiments, sequencing of a polypeptide molecule comprises identifying at least

two (e.g., at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, at least 25, at least 30, at least 35, at least 40, at least 45, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, or more) amino acids in the polypeptide molecule. In some embodiments, the at least two amino acids are contiguous amino acids. In some embodiments, the at least two amino acids are non-contiguous amino acids.

[0263] In some embodiments, sequencing of a polypeptide molecule comprises identification of less than 100% (e.g., less than 99%, less than 95%, less than 90%, less than 85%, less than 80%, less than 75%, less than 70%, less than 65%, less than 60%, less than 55%, less than 50%, less than 45%, less than 40%, less than 35%, less than 30%, less than 25%, less than 20%, less than 15%, less than 10%, less than 5%, less than 1% or less) of all amino acids in the polypeptide molecule. For example, in some embodiments, sequencing of a polypeptide molecule comprises identification of less than 100% of one type of amino acid in the polypeptide molecule (e.g., identification of a portion of all amino acids of one type in the polypeptide molecule). In some embodiments, sequencing of a polypeptide molecule comprises identification of less than 100% of each type of amino acid in the polypeptide molecule.

[0264] In some embodiments, sequencing of a polypeptide molecule comprises identification of at least 1, at least 5, at least 10, at least 15, at least 20, at least 25, at least 30, at least 35, at least 40, at least 45, at least 50, at least 55, at least 60, at least 65, at least 70, at least 75, at least 80, at least 85, at least 90, at least 95, at least 100 or more types of amino acids in the polypeptide.

[0265] In some embodiments, the application provides compositions and methods for sequencing a polypeptide by identifying a series of amino acids that are present at a terminus of a polypeptide over time (e.g., by iterative detection and cleavage of amino acids at the terminus). In yet other embodiments, the application provides compositions and methods for sequencing a polypeptide by identifying labeled amino content of the polypeptide and comparing to a reference sequence database.

[0266] In some embodiments, the application provides compositions and methods for sequencing a polypeptide by sequencing a plurality of fragments of the polypeptide. In some embodiments, sequencing a polypeptide comprises combining sequence information for a plurality of polypeptide fragments to identify and/or determine a sequence for the polypeptide. In some embodiments, combining sequence information may be performed by computer hardware and software. The methods described herein may allow for a set of related polypeptides, such as an entire proteome of an organism, to be sequenced. In some embodiments, a plurality of single molecule sequencing reactions are performed in parallel (e.g., on a single chip) according to aspects of the present application. For example, in some embodiments, a plurality of single molecule sequencing reactions are each performed in separate sample wells on a single chip.

[0267] In some embodiments, methods provided herein may be used for the sequencing and identification of an individual protein in a sample comprising a complex mixture of proteins. In some embodiments, the application provides methods of uniquely identifying an individual protein in a complex mixture of proteins. In some embodiments, an individual protein is detected in a mixed sample by determining a partial amino acid sequence of the protein. In some embodiments, the partial amino acid sequence of the protein is within a contiguous stretch of approximately 5 to 50 amino acids.

[0268] Without wishing to be bound by any particular theory, it is believed that most human proteins can be identified using incomplete sequence information with reference to proteomic databases. For example, simple modeling of the human proteome has shown that approximately 98% of proteins can be uniquely identified by detecting just four types of amino acids within a stretch of 6 to 40 amino acids (see, e.g., Swaminathan, et al. PLOS Comput Biol. 2015, 11 (2): e1004080; and Yao, et al. Phys. Biol. 2015, 12 (5): 055003). Therefore, a complex mixture of proteins can be degraded (e.g., chemically degraded, enzymatically degraded) into short

polypeptide fragments of approximately 6 to 40 amino acids, and sequencing of this polypeptide library would reveal the identity and abundance of each of the proteins present in the original complex mixture. Compositions and methods for selective amino acid labeling and identifying polypeptides by determining partial sequence information are described in detail in U.S. patent application Ser. No. 15/510,962, filed Sep. 15, 2015, titled “SINGLE MOLECULE PEPTIDE SEQUENCING,” which is incorporated by reference in its entirety.

[0269] Embodiments are capable of sequencing single polypeptide molecules with high accuracy, such as an accuracy of at least about 50%, 60%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, 99%, 99.9%, 99.99%, 99.999%, or 99.9999%. In some embodiments, the target molecule used in single molecule sequencing is a polypeptide that is immobilized to a surface of a solid support such as a bottom surface or a sidewall surface of a sample well. The sample well also can contain any other reagents needed for a sequencing reaction in accordance with the application, such as one or more suitable buffers, co-factors, labeled affinity reagents, and enzymes (e.g., catalytically active or inactive exopeptidase enzymes, which may be luminescently labeled or unlabeled).

[0270] As described above, in some embodiments, sequencing in accordance with the application comprises identifying an amino acid by determining a probability that the amino acid is of a specific type. Conventional protein identification systems require identification of each amino acid in a polypeptide to identify the polypeptide. However, it is difficult to accurately identify each amino acid in a polypeptide. For example, data collected from an interaction in which a first recognition molecule associates with a first amino acid may not be sufficiently different from data collected from an interaction in which a second recognition molecule associates with a second amino acid to differentiate between the two amino acids. In some embodiments, sequencing in accordance with the application avoids this problem by using a protein identification system that, unlike conventional protein identification systems, does not require (but does not preclude) identification of each amino acid in the protein.

[0271] Accordingly, in some embodiments, sequencing in accordance with the application may be carried out using a protein identification system that uses machine learning techniques to identify proteins. In some embodiments, the system operates by: (1) collecting data about a polypeptide of a protein using a real-time protein sequencing device; (2) using a machine learning model and the collected data to identify probabilities that certain amino acids are part of the polypeptide at respective locations; and (3) using the identified probabilities, as a “probabilistic fingerprint” to identify the protein. In some embodiments, data about the polypeptide of the protein may be obtained using reagents that selectively bind amino acids. As an example, the reagents and/or amino acids may be labeled with luminescent labels that emit light in response to application of excitation energy. In this example, a protein sequencing device may apply excitation energy to a sample of a protein (e.g., a polypeptide) during binding interactions of reagents with amino acids in the sample. In some embodiments, one or more sensors in the sequencing device (e.g., a photodetector, an electrical sensor, and/or any other suitable type of sensor) may detect binding interactions. In turn, the data collected and/or derived from the detected light emissions may be provided to the machine learning model. Machine learning models and associated systems and methods are described in detail in U.S. Provisional Patent Appl. No. 62/860,750, filed Jun. 12, 2019, titled “MACHINE LEARNING ENABLED PROTEIN IDENTIFICATION,” which is incorporated by reference in its entirety.

[0272] Sequencing in accordance with the application, in some aspects, may involve immobilizing a polypeptide on a surface of a substrate (e.g., of a solid support, for example a chip, for example an integrated device as described herein). In some embodiments, a polypeptide may be immobilized on a surface of a sample well (e.g., on a bottom surface of a sample well) on a substrate. In some embodiments, the N-terminal amino acid of the polypeptide is immobilized (e.g., attached to the surface). In some embodiments, the C-terminal amino acid of the polypeptide

is immobilized (e.g., attached to the surface). In some embodiments, one or more non-terminal amino acids are immobilized (e.g., attached to the surface). The immobilized amino acid(s) can be attached using any suitable covalent or non-covalent linkage, for example as described in this application. In some embodiments, a plurality of polypeptides are attached to a plurality of sample wells (e.g., with one polypeptide attached to a surface, for example a bottom surface, of each sample well), for example in an array of sample wells on a substrate.

[0273] Sequencing in accordance with the application, in some aspects, may be performed using a system that permits single molecule analysis. The system may include an integrated device and an instrument configured to interface with the integrated device. The integrated device may include an array of pixels, where individual pixels include a sample well and at least one photodetector. The sample wells of the integrated device may be formed on or through a surface of the integrated device and be configured to receive a sample placed on the surface of the integrated device. Collectively, the sample wells may be considered as an array of sample wells. The plurality of sample wells may have a suitable size and shape such that at least a portion of the sample wells receive a single sample (e.g., a single molecule, such as a polypeptide). In some embodiments, the number of samples within a sample well may be distributed among the sample wells of the integrated device such that some sample wells contain one sample while others contain zero, two or more samples.

[0274] Excitation light is provided to the integrated device from one or more light source external to the integrated device. Optical components of the integrated device may receive the excitation light from the light source and direct the light towards the array of sample wells of the integrated device and illuminate an illumination region within the sample well. In some embodiments, a sample well may have a configuration that allows for the sample to be retained in proximity to a surface of the sample well, which may ease delivery of excitation light to the sample and detection of emission light from the sample. A sample positioned within the illumination region may emit emission light in response to being illuminated by the excitation light. For example, the sample may be labeled with a fluorescent marker, which emits light in response to achieving an excited state through the illumination of excitation light. Emission light emitted by a sample may then be detected by one or more photodetectors within a pixel corresponding to the sample well with the sample being analyzed. When performed across the array of sample wells, which may range in number between approximately 10,000 pixels to 1,000,000 pixels according to some embodiments, multiple samples can be analyzed in parallel.

[0275] The integrated device may include an optical system for receiving excitation light and directing the excitation light among the sample well array. The optical system may include one or more grating couplers configured to couple excitation light to the integrated device and direct the excitation light to other optical components. The optical system may include optical components that direct the excitation light from a grating coupler towards the sample well array. Such optical components may include optical splitters, optical combiners, and waveguides. In some embodiments, one or more optical splitters may couple excitation light from a grating coupler and deliver excitation light to at least one of the waveguides.

[0276] According to some embodiments, the optical splitter may have a configuration that allows for delivery of excitation light to be substantially uniform across all the waveguides such that each of the waveguides receives a substantially similar amount of excitation light. Such embodiments may improve performance of the integrated device by improving the uniformity of excitation light received by sample wells of the integrated device. Examples of suitable components, e.g., for coupling excitation light to a sample well and/or directing emission light to a photodetector, to include in an integrated device are described in U.S. patent application Ser. No. 14/821,688, filed Aug. 7, 2015, titled "INTEGRATED DEVICE FOR PROBING, DETECTING AND ANALYZING MOLECULES," and U.S. patent application Ser. No. 14/543,865, filed Nov. 17, 2014, titled "INTEGRATED DEVICE WITH EXTERNAL LIGHT SOURCE FOR PROBING, DETECTING,

AND ANALYZING MOLECULES,” both of which are incorporated by reference in their entirety. Examples of suitable grating couplers and waveguides that may be implemented in the integrated device are described in U.S. patent application Ser. No. 15/844,403, filed Dec. 15, 2017, titled “OPTICAL COUPLER AND WAVEGUIDE SYSTEM,” which is incorporated by reference in its entirety.

[0277] Additional photonic structures may be positioned between the sample wells and the photodetectors and configured to reduce or prevent excitation light from reaching the photodetectors, which may otherwise contribute to signal noise in detecting emission light. In some embodiments, metal layers which may act as a circuitry for the integrated device, may also act as a spatial filter. Examples of suitable photonic structures may include spectral filters, a polarization filters, and spatial filters and are described in U.S. patent application Ser. No. 16/042,968, filed Jul. 23, 2018, titled “OPTICAL REJECTION PHOTONIC STRUCTURES,” which is incorporated by reference in its entirety.

[0278] Components located off of the integrated device may be used to position and align an excitation source to the integrated device. Such components may include optical components including lenses, mirrors, prisms, windows, apertures, attenuators, and/or optical fibers. Additional mechanical components may be included in the instrument to allow for control of one or more alignment components. Such mechanical components may include actuators, stepper motors, and/or knobs. Examples of suitable excitation sources and alignment mechanisms are described in U.S. patent application Ser. No. 15/161,088, filed May 20, 2016, titled “PULSED LASER AND SYSTEM,” which is incorporated by reference in its entirety. Another example of a beam-steering module is described in U.S. patent application Ser. No. 15/842,720, filed Dec. 14, 2017, titled “COMPACT BEAM SHAPING AND STEERING ASSEMBLY,” which is incorporated herein by reference. Additional examples of suitable excitation sources are described in U.S. patent application Ser. No. 14/821,688, filed Aug. 7, 2015, titled “INTEGRATED DEVICE FOR PROBING, DETECTING AND ANALYZING MOLECULES,” which is incorporated by reference in its entirety.

[0279] The photodetector(s) positioned with individual pixels of the integrated device may be configured and positioned to detect emission light from the pixel's corresponding sample well. Examples of suitable photodetectors are described in U.S. patent application Ser. No. 14/821,656, filed Aug. 7, 2015, titled “INTEGRATED DEVICE FOR TEMPORAL BINNING OF RECEIVED PHOTONS,” which is incorporated by reference in its entirety. In some embodiments, a sample well and its respective photodetector(s) may be aligned along a common axis. In this manner, the photodetector(s) may overlap with the sample well within the pixel.

[0280] Characteristics of the detected emission light may provide an indication for identifying the marker associated with the emission light. Such characteristics may include any suitable type of characteristic, including an arrival time of photons detected by a photodetector, an amount of photons accumulated over time by a photodetector, and/or a distribution of photons across two or more photodetectors. In some embodiments, a photodetector may have a configuration that allows for the detection of one or more timing characteristics associated with a sample's emission light (e.g., luminescence lifetime). The photodetector may detect a distribution of photon arrival times after a pulse of excitation light propagates through the integrated device, and the distribution of arrival times may provide an indication of a timing characteristic of the sample's emission light (e.g., a proxy for luminescence lifetime). In some embodiments, the one or more photodetectors provide an indication of the probability of emission light emitted by the marker (e.g., luminescence intensity). In some embodiments, a plurality of photodetectors may be sized and arranged to capture a spatial distribution of the emission light. Output signals from the one or more photodetectors may then be used to distinguish a marker from among a plurality of markers, where the plurality of markers may be used to identify a sample within the sample. In some embodiments, a sample may be excited by multiple excitation energies, and emission light and/or timing

characteristics of the emission light emitted by the sample in response to the multiple excitation energies may distinguish a marker from a plurality of markers.

[0281] In operation, parallel analyses of samples within the sample wells are carried out by exciting some or all of the samples within the wells using excitation light and detecting signals from sample emission with the photodetectors. Emission light from a sample may be detected by a corresponding photodetector and converted to at least one electrical signal. The electrical signals may be transmitted along conducting lines in the circuitry of the integrated device, which may be connected to an instrument interfaced with the integrated device. The electrical signals may be subsequently processed and/or analyzed. Processing or analyzing of electrical signals may occur on a suitable computing device either located on or off the instrument.

[0282] The instrument may include a user interface for controlling operation of the instrument and/or the integrated device. The user interface may be configured to allow a user to input information into the instrument, such as commands and/or settings used to control the functioning of the instrument. In some embodiments, the user interface may include buttons, switches, dials, and a microphone for voice commands. The user interface may allow a user to receive feedback on the performance of the instrument and/or integrated device, such as proper alignment and/or information obtained by readout signals from the photodetectors on the integrated device. In some embodiments, the user interface may provide feedback using a speaker to provide audible feedback. In some embodiments, the user interface may include indicator lights and/or a display screen for providing visual feedback to a user.

[0283] In some embodiments, the instrument may include a computer interface configured to connect with a computing device. The computer interface may be a USB interface, a FireWire interface, or any other suitable computer interface. A computing device may be any general purpose computer, such as a laptop or desktop computer. In some embodiments, a computing device may be a server (e.g., cloud-based server) accessible over a wireless network via a suitable computer interface. The computer interface may facilitate communication of information between the instrument and the computing device. Input information for controlling and/or configuring the instrument may be provided to the computing device and transmitted to the instrument via the computer interface. Output information generated by the instrument may be received by the computing device via the computer interface. Output information may include feedback about performance of the instrument, performance of the integrated device, and/or data generated from the readout signals of the photodetector.

[0284] In some embodiments, the instrument may include a processing device configured to analyze data received from one or more photodetectors of the integrated device and/or transmit control signals to the excitation source(s). In some embodiments, the processing device may comprise a general purpose processor, a specially-adapted processor (e.g., a central processing unit (CPU) such as one or more microprocessor or microcontroller cores, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), a custom integrated circuit, a digital signal processor (DSP), or a combination thereof). In some embodiments, the processing of data from one or more photodetectors may be performed by both a processing device of the instrument and an external computing device. In other embodiments, an external computing device may be omitted and processing of data from one or more photodetectors may be performed solely by a processing device of the integrated device.

[0285] According to some embodiments, the instrument that is configured to analyze samples based on luminescence emission characteristics may detect differences in luminescence lifetimes and/or intensities between different luminescent molecules, and/or differences between lifetimes and/or intensities of the same luminescent molecules in different environments. The inventors have recognized and appreciated that differences in luminescence emission lifetimes can be used to discern between the presence or absence of different luminescent molecules and/or to discern between different environments or conditions to which a luminescent molecule is subjected. In

some cases, discerning luminescent molecules based on lifetime (rather than emission wavelength, for example) can simplify aspects of the system. As an example, wavelength-discriminating optics (such as wavelength filters, dedicated detectors for each wavelength, dedicated pulsed optical sources at different wavelengths, and/or diffractive optics) may be reduced in number or eliminated when discerning luminescent molecules based on lifetime. In some cases, a single pulsed optical source operating at a single characteristic wavelength may be used to excite different luminescent molecules that emit within a same wavelength region of the optical spectrum but have measurably different lifetimes. An analytic system that uses a single pulsed optical source, rather than multiple sources operating at different wavelengths, to excite and discern different luminescent molecules emitting in a same wavelength region can be less complex to operate and maintain, more compact, and may be manufactured at lower cost.

[0286] Although analytic systems based on luminescence lifetime analysis may have certain benefits, the amount of information obtained by an analytic system and/or detection accuracy may be increased by allowing for additional detection techniques. For example, some embodiments of the systems may additionally be configured to discern one or more properties of a sample based on luminescence wavelength and/or luminescence intensity. In some implementations, luminescence intensity may be used additionally or alternatively to distinguish between different luminescent labels. For example, some luminescent labels may emit at significantly different intensities or have a significant difference in their probabilities of excitation (e.g., at least a difference of about 35%) even though their decay rates may be similar. By referencing binned signals to measured excitation light, it may be possible to distinguish different luminescent labels based on intensity levels.

[0287] According to some embodiments, different luminescence lifetimes may be distinguished with a photodetector that is configured to time-bin luminescence emission events following excitation of a luminescent label. The time binning may occur during a single charge-accumulation cycle for the photodetector. A charge-accumulation cycle is an interval between read-out events during which photo-generated carriers are accumulated in bins of the time-binning photodetector. Examples of a time-binning photodetector are described in U.S. patent application Ser. No. 14/821,656, filed Aug. 7, 2015, titled "INTEGRATED DEVICE FOR TEMPORAL BINNING OF RECEIVED PHOTONS," which is incorporated herein by reference. In some embodiments, a time-binning photodetector may generate charge carriers in a photon absorption/carrier generation region and directly transfer charge carriers to a charge carrier storage bin in a charge carrier storage region. In such embodiments, the time-binning photodetector may not include a carrier travel/capture region. Such a time-binning photodetector may be referred to as a "direct binning pixel." Examples of time-binning photodetectors, including direct binning pixels, are described in U.S. patent application Ser. No. 15/852,571, filed Dec. 22, 2017, titled "INTEGRATED PHOTODETECTOR WITH DIRECT BINNING PIXEL," which is incorporated herein by reference.

[0288] In some embodiments, different numbers of fluorophores of the same type may be linked to different reagents in a sample, so that each reagent may be identified based on luminescence intensity. For example, two fluorophores may be linked to a first labeled affinity reagent and four or more fluorophores may be linked to a second labeled affinity reagent. Because of the different numbers of fluorophores, there may be different excitation and fluorophore emission probabilities associated with the different affinity reagents. For example, there may be more emission events for the second labeled affinity reagent during a signal accumulation interval, so that the apparent intensity of the bins is significantly higher than for the first labeled affinity reagent.

[0289] The inventors have recognized and appreciated that distinguishing nucleotides or any other biological or chemical samples based on fluorophore decay rates and/or fluorophore intensities may enable a simplification of the optical excitation and detection systems. For example, optical excitation may be performed with a single-wavelength source (e.g., a source producing one characteristic wavelength rather than multiple sources or a source operating at multiple different

characteristic wavelengths). Additionally, wavelength discriminating optics and filters may not be needed in the detection system. Also, a single photodetector may be used for each sample well to detect emission from different fluorophores. The phrase “characteristic wavelength” or “wavelength” is used to refer to a central or predominant wavelength within a limited bandwidth of radiation (e.g., a central or peak wavelength within a 20 nm bandwidth output by a pulsed optical source). In some cases, “characteristic wavelength” or “wavelength” may be used to refer to a peak wavelength within a total bandwidth of radiation output by a source.

Computational Techniques

[0290] Aspects of the present application relate to computational techniques for analyzing the data generated by the polypeptide sequencing techniques described herein. As discussed above, for example in connection with FIGS. 1A and 1B, the data generated by using these sequencing techniques may include a series of signal pulses indicative of instances where an amino acid recognition molecule is associated with an amino acid exposed at the terminus of the polypeptide being sequenced. The series of signal pulses may have varying one or more features (e.g., pulse duration, interpulse duration, change in magnitude), depending on the type of amino acid presently at the terminus, over time as the degradation process proceeds in removing successive amino acids. The resulting signal trace may include characteristic patterns, which arise from the varying one or more features, associated with respective amino acids. The computational techniques described herein may be implemented as part of analyzing such data obtained using these sequencing techniques to identify an amino acid sequence.

[0291] Some embodiments may involve obtaining data during a degradation process of a polypeptide, analyzing the data to determine portions of the data corresponding to amino acids that are sequentially exposed at a terminus of the polypeptide during the degradation process, and outputting an amino acid sequence representative of the polypeptide. FIG. 11 is a diagram of an illustrative processing pipeline **1100** for identifying an amino acid sequence by analyzing data obtained using the polypeptide sequencing techniques described herein. As shown in FIG. 11, analyzing sequencing data **1102** may involve using association event identification technique **1104** and amino acid identification technique **1106** to output amino acid sequence(s) **1108**.

[0292] As discussed herein, sequencing data **1102** may be obtained during a degradation process of a polypeptide. In some embodiments, the sequencing data **1102** is indicative of amino acid identity at the terminus of the polypeptide during the degradation process. In some embodiments, the sequencing data **1102** is indicative of a signal produced by one or more amino acid recognition molecules binding to different types of terminal amino acids at the terminus during the degradation process. Exemplary sequencing data is shown in FIGS. 1A and 1B, which are discussed above.

[0293] Depending on how signals are generated during the degradation process, sequencing data **1102** may be indicative of one or more different types of signals. In some embodiments, sequencing data **1102** is indicative of a luminescent signal generated during the degradation process. For example, a luminescent label may be used to label an amino acid recognition molecule, and luminescence emitted by the luminescent label may be detected as the amino acid recognition molecule associates with a particular amino acid, resulting in a luminescent signal. In some embodiments, sequencing data **1102** is indicative of an electrical signal generated during the degradation process. For example, a polypeptide molecule being sequenced may be immobilized to a nanopore, and an electrical signal (e.g., changes in conductance) may be detected as an amino acid recognition molecule associates with a particular amino acid.

[0294] Some embodiments involve analyzing sequencing data **1102** to determine portions of sequencing data **1102** corresponding to amino acids that are sequentially exposed at a terminus of the polypeptide during the degradation process. As shown in FIG. 11, association event identification technique **1104** may access sequencing data **1102** and analyze sequencing data to identify portions of sequencing data **1102** that correspond to association events. The association events may correspond to characteristic patterns, such as CP1 and CP2 shown in FIG. 1B, in the

data. In some embodiments, association event identification technique **1104** may involve detecting a series of cleavage events and determining portions of sequencing data **1102** between successive cleavage events. As an example, a cleavage event between CP1 and CP2 shown in FIG. **1B** may be detected such that a first portion of the data corresponding to CP1 may be identified as a first association event and a second portion of the data corresponding CP2 may be identified as a second association event.

[0295] Some embodiments involve identifying a type of amino acid for one or more of the determined portions of sequencing data **1102**. As shown in FIG. **11**, amino acid identification technique **1106** may be used to determine a type of amino acid for one or more of the association events identified by association event identification technique **1104**. In some embodiments, the individual portions of data identified by association event identification technique **1104** may include a pulse pattern, and amino acid identification technique **1106** may determine a type of amino acid for one or more of the portions based on its respective pulse pattern. Referring to FIG. **1B**, amino acid identification technique **1106** may identify a first type of amino acid for CP1 and a second type of amino acid for CP2. In some embodiments, determining the type of amino acid may include identifying an amount of time within a portion of data, such as a portion identified using association event identification technique **1104**, when the data is above a threshold value and comparing the amount of time to a duration of time for the portion of data. For example, identifying a type of amino acid for CP1 may include determining an amount of time within CP1 where the signal is above a threshold value, such as time period, pd, where the signal is above ML, and comparing it to a total duration of time for CP1. In some embodiments, determining the type of amino acid may involve identifying one or more pulse durations for one or more portions of data identified by association event identification technique **1102**. For example, identifying a type of amino acid for CP1 may include determining a pulse duration for CP1, such as time period, pd. In some embodiments, determining the type of amino acid may involve identifying one or more interpulse durations for one or more portions of the data identified using association event identification technique **1104**. For example, identifying a type of amino acid for CP1 may include identifying an interpulse duration, such as ipd.

[0296] By identifying a type of amino acid for successive portions of sequencing data **1102**, amino acid identification technique **1106** may output amino acid sequence(s) **1108** representative of the polypeptide. In some embodiments, the amino acid sequence includes a series of amino acids corresponding to the portions of data identified using association event identification technique **1104**.

[0297] FIG. **12** is a flow chart of an illustrative process **1200** for determining an amino acid sequence of a polypeptide molecule, in accordance with some embodiments of the technology described herein. Process **1200** may be performed on any suitable computing device(s) (e.g., a single computing device, multiple computing devices co-located in a single physical location or located in multiple physical locations remote from one another, one or more computing devices part of a cloud computing system, etc.), as aspects of the technology described herein are not limited in this respect. In some embodiments, association event identification technique **1104** and amino acid identification technique **1106** may perform some or all of process **1200** to determine amino acid sequence(s).

[0298] Process **1200** begins at act **1202**, which involves contacting a single polypeptide molecule with one or more terminal amino acid recognition molecules. Next, process **1200** proceeds to act **1104**, which involves detecting a series of signal pulses indicative of association of the one or more terminal amino acid recognition molecules with successive amino acids exposed at a terminus of the single polypeptide while the single polypeptide is being degraded. The series of pulses may allow for sequencing of the single polypeptide molecule, such as by using association event identification technique **1104** and amino acid identification technique **1106**.

[0299] In some embodiments, process **1200** may include act **1206**, which involves identifying a

first type of amino acid in the single polypeptide molecule based on a first characteristic pattern in the series of signal pulses, such as by using amino acid identification technique **1106**.

[0300] FIG. **13** is a flow chart of an illustrative process **1300** for determining an amino acid sequence representative of a polypeptide, in accordance with some embodiments of the technology described herein. Process **1300** may be performed on any suitable computing device(s) (e.g., a single computing device, multiple computing devices co-located in a single physical location or located in multiple physical locations remote from one another, one or more computing devices part of a cloud computing system, etc.), as aspects of the technology described herein are not limited in this respect. In some embodiments, association event identification technique **1104** and amino acid identification technique **1106** may perform some or all of process **1300** to determine amino acid sequence(s).

[0301] Process **1300** begins at act **1302**, where data during a degradation process of a polypeptide is obtained. In some embodiments, the data is indicative of amino acid identity at the terminus of the polypeptide during the degradation process. In some embodiments, the data is indicative of a signal produced by one or more amino acid recognition molecules binding to different types of terminal amino acids at the terminus during the degradation process. In some embodiments, the data is indicative of a luminescent signal generated during the degradation process. In some embodiments, the data is indicative of an electrical signal generated during the degradation process.

[0302] Next, process **1300** proceeds to act **1304**, where the data is analyzed to determine portions of the data corresponding to amino acids that are sequentially exposed at a terminus of the polypeptide during the degradation process, such as by using association event identification technique **1104** and amino acid identification technique **1106**. In some embodiments, analyzing the data further comprises detecting a series of cleavage events and determining the portions of the data between successive cleavage events, such as by using association event identification technique **1104**.

[0303] In some embodiments, analyzing the data further comprises determining a type of amino acid for each of the individual portions, such as by using amino acid identification technique **1106**. In some embodiments, each of the individual portions comprises a pulse pattern, and analyzing the data further comprises determining a type of amino acid for one or more of the portions based on its respective pulse pattern. In some embodiments, determining the type of amino acid further comprises identifying an amount of time within a portion when the data is above a threshold value and comparing the amount of time to a duration of time for the portion. In some embodiments, determining the type of amino acid further comprises identifying at least one pulse duration for each of the one or more portions. In some embodiments, determining the type of amino acid further comprises identifying at least one interpulse duration for each of the one or more portions.

[0304] Next, process **1300** proceeds to act **1306**, where an amino acid sequence representative of the polypeptide is outputted, such as via a user interface. In some embodiments, the amino acid sequence includes a series of amino acids corresponding to the portions.

[0305] An illustrative implementation of a computer system **1400** that may be used in connection with any of the embodiments of the technology described herein is shown in FIG. **14**. The computer system **1400** includes one or more processors **1410** and one or more articles of manufacture that comprise non-transitory computer-readable storage media (e.g., memory **1420** and one or more non-volatile storage media **1430**). The processor **1410** may control writing data to and reading data from the memory **1420** and the non-volatile storage device **1430** in any suitable manner, as the aspects of the technology described herein are not limited in this respect. To perform any of the functionality described herein, the processor **1410** may execute one or more processor-executable instructions stored in one or more non-transitory computer-readable storage media (e.g., the memory **1420**), which may serve as non-transitory computer-readable storage media storing processor-executable instructions for execution by the processor **1410**.

[0306] Computing device **1400** may also include a network input/output (I/O) interface **1440** via

which the computing device may communicate with other computing devices (e.g., over a network), and may also include one or more user I/O interfaces **1450**, via which the computing device may provide output to and receive input from a user. The user I/O interfaces may include devices such as a keyboard, a mouse, a microphone, a display device (e.g., a monitor or touch screen), speakers, a camera, and/or various other types of I/O devices.

[0307] The above-described embodiments can be implemented in any of numerous ways. For example, the embodiments may be implemented using hardware, software or a combination thereof. When implemented in software, the software code can be executed on any suitable processor (e.g., a microprocessor) or collection of processors, whether provided in a single computing device or distributed among multiple computing devices. It should be appreciated that any component or collection of components that perform the functions described above can be generically considered as one or more controllers that control the above-discussed functions. The one or more controllers can be implemented in numerous ways, such as with dedicated hardware, or with general purpose hardware (e.g., one or more processors) that is programmed using microcode or software to perform the functions recited above.

[0308] In this respect, it should be appreciated that one implementation of the embodiments described herein comprises at least one computer-readable storage medium (e.g., RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other tangible, non-transitory computer-readable storage medium) encoded with a computer program (i.e., a plurality of executable instructions) that, when executed on one or more processors, performs the above-discussed functions of one or more embodiments. The computer-readable medium may be transportable such that the program stored thereon can be loaded onto any computing device to implement aspects of the techniques discussed herein. In addition, it should be appreciated that the reference to a computer program which, when executed, performs any of the above-discussed functions, is not limited to an application program running on a host computer. Rather, the terms computer program and software are used herein in a generic sense to reference any type of computer code (e.g., application software, firmware, microcode, or any other form of computer instruction) that can be employed to program one or more processors to implement aspects of the techniques discussed herein.

EXAMPLES

Example 1. Edman Degradation by Chemical Cleavage

##STR00011##

[0309] A surface-attached oligopeptide of approximately 3 to approximately 30 amino acids ($n=3-30$) is provided, where amino acid residues R.sub.1-R.sub.3 can be any of the common 20 amino acids or an endogenously modified amino acid (e.g., modified by a post-translational modification). In the isothiocyanate N-terminal reaction, Step 1, an isothiocyanate X-NCS is added to a vessel containing the surface-attached oligopeptide, where X is phenyl (Ph), 4-NO.sub.2Ph, 4-SO.sub.3Ph, naphthyl, benzyl, alkyl, or a derivative thereof. Step 1 is carried out under the following Conditions to afford the X-NCS derivatized N-terminal amino acid: aqueous buffer pH 4-10, MeOH or EtOH or IPA alcoholic co-solvents, trialkylamines in organic solvents (DCM, THF, MeCN, DMF, and the like), 20° C. to 50° C. In the thiourea cleavage reaction, Step 2, an Acid or a Base is added to a vessel containing the X-NCS derivatized N-terminal amino acid, where the Acid is acetic acid, formic acid, trichloroacetic acid, trifluoroacetic acid, phosphoric acid, or hydrochloric acid, as neat or aqueous solutions, or where the Base is a trialkylamine or a buffered trialkylamine (e.g., Et.sub.3NH.sup.+ AcO.sup.-). Step 2 is carried out under the following Conditions to afford the n-1 oligopeptide and thiohydantoin byproduct: neat acid or with aqueous/organic co-solvents of any ratio, 20° C. to 50° C.

Example 2. Solubilizing Linkers for Peptide Surface Immobilization

[0310] Seeking to improve oligopeptide solubility in aqueous buffer, it was determined that peptide

fragments could be conjugated with oligonucleotide linkers to both improve aqueous solubility and provide a functional moiety for surface immobilization of peptides at the single molecule level. Different peptide-linker conjugates were synthesized, with example structures depicted in FIG. 15A for a peptide-DNA conjugate and a peptide-PEG conjugate. Linker conjugation was observed to greatly enhance peptide solubility in aqueous solution for each of the different peptide-linker conjugates evaluated.

[0311] The peptide-linker conjugates were evaluated for amino acid cleavage at peptide N-termini by N-terminal aminopeptidases (Table 6, below).

TABLE-US-00006 TABLE 6 Terminal amino acid cleavage of peptide-linker conjugates. Cleav- Cleav- SEQ ed by ed En- ID Rat by try Peptide NO. Class Linker APN PIP

1	KF 70	positive oligo	No	2	KKMKKM 71	positive oligo	No	{LYS(N3)}	3	KKMKKM 71	positive oligo	No	{LYS(N3)}	PEG	4	KKMKKM 71	positive PEG4	Yes	{LYS(N3)}	5	DDMDDM 72	negative oligo	Yes	{LYS(N3)}	6	FFMFFM 73	aromatic oligo	Yes	{LYS(N3)}	7	AAMAAM 74	hydro- oligo	Yes	{LYS(N3)}	phobic	8	FPFPFP 75	aromatic oligo	Yes	{LYS(N3)}	9	DPDPDP 76	negative oligo	Yes	{LYS(N3)}	10	KPKPKP 77	positive oligo	No	{LYS(N3)}	11	KPKPKP 77	positive PEG4	Yes	{LYS(N3)}
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[0312] The peptide-linker conjugates shown in Table 6 were incubated with either proline iminopeptidase (“PIP”) or rat aminopeptidase N (“Rat APN”), and peptide cleavage was monitored by LCMS. An example of an LCMS demonstrating cleavage of Entry 5 from Table 6 is shown in FIG. 15B. All other cleavage reactions were measured in a similar manner. As shown in Table 6, while positively charged peptide-DNA conjugates (“oligo” and “oligo-PEG” linkers) were not cleaved by the aminopeptidases tested, all other conjugate classes (negatively charged, aromatic, hydrophobic) with DNA oligonucleotide linkers were cleaved. By comparison, the positively charged peptide-PEG conjugates were shown to be cleaved by at least one of the aminopeptidases.

[0313] Using labeled peptide-linker conjugates, it was shown that peptides of different compositions could be immobilized to individual sample well surfaces for single molecule analysis. For these experiments, the DNA linker was labeled with a dye (e.g., as depicted in FIG. 15A for the peptide-DNA conjugate), and loading of different peptide-DNA conjugates into individual sample wells was measured by dye fluorescence. An example loading experiment is shown in FIG. 15C. By measuring fluorescence emission of a labeled peptide-DNA conjugate (50 pM), it was determined that at least 18% of sample wells on a chip were loaded at single occupancy per sample well with a surface-immobilized conjugate. These experiments demonstrated that peptide-linker conjugates display enhanced aqueous solubility compared to non-conjugated peptide counterpart, that conjugated linkers do not prevent terminal amino acid cleavage of peptides by different aminopeptidases, and that peptide-linker conjugates of different compositions can be immobilized to chip surfaces at the single molecule level.

Example 3. Exopeptidase Cleavage of Polypeptide Substrates

[0314] The cleavage capabilities of various aminopeptidases were tested. The conditions and results for a set of cleavage assay experiments are shown in Table 7, including concentration of peptide substrate, concentration of enzyme, buffer conditions, temperature, and incubation time. Cleavage of peptide substrates by the indicated enzymes was assayed using High Performance Liquid Chromatography (HPLC). The “HPLC assay conv” value in Table 7 indicates the percentage of the peptide substrate that was converted to cleavage product. To determine the “HPLC assay conv” value, two solutions were prepared containing the same starting concentration of peptide. One solution was subjected to enzymatic digestion, while the other solution did not contain any enzyme, but was diluted with an equivalent amount of buffer used to store the enzyme. The reactions were quenched at the time indicated. The amount of reactant converted to product was determined by dividing the area of the peak obtained by HPLC of the starting material remaining after enzymatic digestion by the peak area of the control solution of undigested peptide, and then multiplying this ratio by 100. In Table 7, “NH2” indicates an amine group, “yPIP” refers

NO: 96) NO: 97) Rat APN AAPAAP 1 mM peptide, 50 nM 30° C./ 100%
 AAPAAP {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr {LYSN3}NH2 (SEQ ID
 25 mM Tris pH 8.0 (SEQ ID NO: 98) NO: 99) Rat APN KKPKKP 1 mM
 peptide, 50 nM 30° C./ 85% KPKKP {LYSN3}NH2 enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr {LYSN3}NH2 (SEQ ID 25 mM Tris pH 8.0 (SEQ ID
 NO: 100) NO: 101) Rat APN KMKKKM 1 mM peptide, 50 nM 30° C./ 50%
 KMKKKM {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr {LYSN3}NH2 (SEQ
 ID 25 mM Tris pH 8.0 (SEQ ID NO: 71) NO: 102) VPr RRPRRP 1 mM
 peptide, 2 mM 30° C./ 100% {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1
 hr (SEQ ID 25 mM Tris pH 8.0 NO: 96) VPr AAPAAP 1 mM peptide, 2 mM
 30° C./ 100% AAPAAP {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr
 {LYSN3}NH2 (SEQ ID 25 mM Tris pH 8.0 (SEQ ID NO: 98) NO: 99) VPr
 KKPKKP 1 mM peptide, 2 mM 30° C./ 100% KPKKP {LYSN3}NH2 enzyme,
 2.5 mM ZnCl.sub.2, 1 hr {LYSN3}NH2 (SEQ ID 25 mM Tris pH 8.0
 (SEQ ID NO: 100) NO: 101) VPr YYPYYP 1 mM peptide, 2 mM 30° C./
 50% YPYYP {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr {LYSN3}NH2 (SEQ
 ID 25 mM Tris pH 8.0 (SEQ ID NO: 92) NO: 93) VPr FPFPPF 1 mM
 peptide, 2 mM 30° C./ 100% FPFPPF {LYSN3}NH2 enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr {LYSN3}NH2 (SEQ ID 25 mM Tris pH 8.0 (SEQ ID
 NO: 94) NO: 95) VPr AAMAAM 1 mM peptide, 2 mM 30° C./ 100% multiple
 {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr pds (SEQ ID 25 mM Tris
 pH 8.0 NO: 74) VPr KMKKKM 1 mM peptide, 2 mM 30° C./ 100% multiple
 {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr pds (SEQ ID 25 mM Tris
 pH 8.0 NO: 89) VPr YYMYYM 1 mM peptide, 2 mM 30° C./ >90% multiple
 {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr pds (SEQ ID 25 mM Tris
 pH 8.0 NO: 103) yPIP Attor6g- 44 mM Peptide, 7 mM 30° C./ <10% PAAAFK-
 1kPEG- enzyme, 200 mM Bis- 0.5 hr 1kPEG- DPAAAFK tris Propane, 30 mM
 Biotin {LysN3}- KOAc, 25 mM Mg(OAc).sub.2, (SEQ ID 1kPEG- 32 mM 3,4
 NO: 105) Biotin Dihydroxy-benzoic (SEQ ID acid and 12 mM NO: 104)
 Nitrobenzoic acid + PCD + TXV PfuPIP FPFPPF 1 mM Peptide, 1 μM 80°
 C./ 100% PPFPPF {LYSN3}NH2 enzyme, 1 mM 0.5 hr {LYSN3}NH2 (SEQ ID
 CoCl.sub.2, 50 mM (SEQ ID NO: 75) HEPES, 50 mM KCl NO: 88) PfuPIP
 Attor6g- 1 mM Peptide, 1 μM 80° C./ 40% PAAAFK- 1kPEG- enzyme, 1 mM
 0.5 hr 1kPEG- DPAAAFK CoCl.sub.2, 50 mM Biotin {LysN3}- HEPES, 50 mM
 KCl (SEQ ID 1kPEG- NO: 105) Biotin (SEQ ID NO: 104) yPIP Attor6g- 20 μM
 Q24 conjugate, 30° C./ 100% PAAAFK- 1kPEG- 30 μM ypip, 0.3 hr 1kPEG-
 ODN-DD60- 1x Mg buffer Biotin Biotin (SEQ ID NO: 105) yPIP FPFPPF 1 mM
 Peptide, 7 μM 37° C./ 100% PPFPPF {LYSN3}NH2 Enzyme, 50 mM MOPS,
 0.5 hr {LYSN3}NH2 (SEQ ID 10 mM Mg(OAc).sub.2 pH (SEQ ID NO: 75)
 8.0 NO: 88) yPIP Attor6g- 10 μM Peptide, 7 μM 37° C./ 100% PAAAFK- 1kPEG-
 Enzyme, 50 mM MOPS, 0.5 hr 1kPEG- ODN-DD60- 10 mM Mg(OAc).sub.2
 pH Biotin Biotin 8.0 (SEQ ID NO: 105) PfuPIP FPFPPF 1 mM Peptide, 1 μM
 80° C./ 40% PPFPPF {LYSN3}NH2 Enzyme, 50 mM MOPS, 0.5 hr
 {LYSN3}NH2 (SEQ ID 10 mM Mg(OAc).sub.2 pH (SEQ ID NO: 75) 8.0
 NO: 88) yPIP- FPFPPF 1 mM Peptide, 2.1 μM 37° C./ 100% PPFPPF Q24-
 {LYSN3}NH2 Enzyme, 50 mM MOPS, 0.5 hr {LYSN3}NH2 Cy3B (SEQ ID
 10 mM Mg(OAc).sub.2 pH (SEQ ID NO: 75) 8.0 NO: 88) yPIP- YPYYPY 1
 mM Peptide, 100 nM 37° C./ 100%+ YPYYPY Q24- {LYSN3}NH2 Enzyme, 1X CB2
 0.5 hr {LYSN3}NH2 Rho6G (SEQ ID (SEQ ID NO: 86) NO: 93) yPIP- YPYYPY
 1 mM Peptide, ~5 μM 37° C./ <15% YPYYPY Q24Dark {LYSN3}NH2 Enzyme,

1X CB2 0.5 hr {LYSN3}NH2 (SEQ ID (SEQ ID NO: 86) Rat APN
 QP5-Atto649N 37 μ M Peptide, 100 37° C./ >95% (KAAAAAAAFK nM Enzyme,
 2.5 mM 0.5 hr {LYSN3}NH2) ZnCl.sub.2, 25 mM Tris (SEQ ID pH 8.0
 NO: 106) VPr QP5-Atto649N 37 μ M Peptide, 8 μ M 37° C./ 100% (KAAAAAAAFK
 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr {LYSN3}NH2) 25 mM Tris pH 8.0
 (SEQ ID NO: 106) K287pA YPYPYP 1 mM Peptide, 1 μ M 37° C./ 100%
 PYPYPK zF-Cy3 {LYSN3}NH2 Enzyme, 50 mM MOPS, 1 hr (SEQ ID yPIP
 (SEQ ID 10 mM Mg(OAc).sub.2 pH NO: 83) NO: 86) 8.0 V. AAPAAP 1 mM
 Peptide, 1 μ M 37° C./ ~5% *cholera* {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr APT (SEQ ID 25 mM Tris pH 8.0 NO: 98) Bst M28
 AAPAAP 1 mM Peptide, 1 μ M 37° C./ 100% {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr (SEQ ID 25 mM Tris pH 8.0 NO: 98) Taq APT AAPAAP
 1 mM Peptide, 1 μ M 37° C./ >90% {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr (SEQ ID 25 mM Tris pH 8.0 NO: 98) V. YYPYYP 1 mM
 Peptide, 1 μ M 37° C./ 5% *cholera* {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr APT (SEQ ID 25 mM Tris pH 8.0 NO: 92) Bst M28
 YYPYYP 1 mM Peptide, 1 μ M 37° C./ 10% {LYSN3}NH2 Enzyme, 2.5
 mM ZnCl.sub.2, 1 hr (SEQ ID 25 mM Tris pH 8.0 NO: 92) Taq APT
 YYPYYP 1 mM Peptide, 1 μ M 37° C./ 30% {LYSN3}NH2 Enzyme, 2.5
 mM ZnCl.sub.2, 1 hr (SEQ ID 25 mM Tris pH 8.0 NO: 92) V. FFPFFP 1
 mM Peptide, 1 μ M 37° C./ >95% *cholera* {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr APT (SEQ ID 25 mM Tris pH 8.0 NO: 94) Bst M28
 FFPFFP 1 mM Peptide, 1 μ M 37° C./ 30% {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr (SEQ ID 25 mM Tris pH 8.0 NO: 94) Taq APT FFPFFP
 1 mM Peptide, 1 μ M 37° C./ 60% {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr (SEQ ID 25 mM Tris pH 8.0 NO: 94) V. YYMYYM 1
 mM Peptide, 1 μ M 37° C./ 30% *cholera* {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr APT (SEQ ID 25 mM Tris pH 8.0 NO: 103) Bst M28
 YYMYYM 1 mM Peptide, 1 μ M 37° C./ >50% multiple {LYSN3}NH2
 Enzyme, 2.5 mM ZnCl.sub.2, 1 hr (N-1, -2, (SEQ ID 25 mM Tris pH
 8.0 -3, etc.) NO: 103) Taq APT YYMYYM 1 mM Peptide, 1 μ M 37° C./
 85% multiple {LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 1 hr (N-1, -2,
 (SEQ ID 25 mM Tris pH 8.0 -3, etc.) NO: 103) Cy3B- YPYPYP 1 mM
 Peptide, 7 μ M 37° C./ 100% Q24- {LYSN3}NH2 Enzyme, 10 mM MgCl.sub.2,
 0.5 hr pAzF- (SEQ ID 50 mM MOPS yPIP NO: 86) Cy3B- FFFFP 1 mM
 Peptide, 10 μ M 37° C./ 50% BstTET {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 0.5 hr (SEQ ID 25 mM Tris pH 8.0 NO: 94) Cy3B- FFPFFP 1
 mM Peptide, 20 μ M 37° C./ 100% taqAPT {LYSN3}NH2 Enzyme, 2.5 mM
 ZnCl.sub.2, 0.5 hr 2nd peak (SEQ ID 25 mM Tris pH 8.0 NO: 94) Cy3B-
 FFPFFP 1 mM Peptide, 20 μ M 37° C./ 100% taqAPT {LYSN3}NH2 Enzyme,
 2.5 mM ZnCl.sub.2, 0.5 hr 4th peak (SEQ ID 25 mM Tris pH 8.0 NO:
 94) PhaloM28 RRPRRP 1 mM Peptide, 1 μ M 37° C./ 30% multiple,
 {LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr even (SEQ ID 25 mM
 Tris pH 8.0 higher NO: 96) mass yPAP RRPRRP 1 mM Peptide, 1 μ M 37° C./
 100% {LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr (SEQ ID 25 mM
 Tris pH 8.0 NO: 96) yPAP AAPAAP 1 mM Peptide, 1 μ M 37° C./ 100%
 {LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr (SEQ ID 25 mM Tris
 pH 8.0 NO: 98) yPAP KKPKKP 1 mM Peptide, 1 μ M 37° C./ 100%
 {LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr (SEQ ID 25 mM Tris
 pH 8.0 NO: 100) yPAP YYPYYP 1 mM Peptide, 1 μ M 37° C./ 100%

{LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr (SEQ ID 25 mM Tris
 pH 8.0 NO: 92) PhaloM28 FFPFFP 1 mM Peptide, 1 μ M 37° C./ >80%
 {LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr (SEQ ID 25 mM Tris
 pH 8.0 NO: 94) yPAP FFPFFP 1 mM Peptide, 1 μ M 37° C./ >80%
 {LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr (SEQ ID 25 mM Tris
 pH 8.0 NO: 94) V. QP5-Atto649N 1 mM Peptide, 5 μ M 37° C./ 100% *cholera*
 (KAAAAAAAFK Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr APT {LYSN3}NH2) 25 mM
 Tris pH 8.0 (SEQ ID NO: 106) yPAP YYPYYP 1 mM Peptide, 2 μ M 37° C./
 10% {LYSN3}NH2 Enzyme, 2.5 mM ZnCl.sub.2, 0.5 hr (SEQ ID 25
 mM Tris pH 8.0 NO: 92) V. RRRRRP 1 mM peptide, 2 μ M 30° C./ >90%
anguil- {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr *larum* (SEQ ID 25
 mM Tris pH 8.0 APN NO: 96) V. AAPAAP 1 mM peptide, 2 μ M 30° C./
 50% *anguil*- {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr *larum* (SEQ
 ID 25 mM Tris pH 8.0 APN NO: 98) V. KKPKKP 1 mM peptide, 2 μ M
 30° C./ <5% *anguil*- {LYSN3}NH2 enzyme, 2.5 mM ZnCl.sub.2, 1 hr *larum*
 (SEQ ID 25 mM Tris pH 8.0 APN NO: 100) VPr FYPLPWPDDDY 300 μ M
 Peptide, 4 μ M 37° C./ 100% {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 0.5
 hr (SEQ ID 50 mM MOPS pH 8.0 NO: 107) yPIP YPLPWPDDDY 300 μ M
 Peptide, 7 μ M 37° C./ 100% {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 0.5
 hr (SEQ ID 50 mM MOPS pH 8.0 NO: 108) VPr PLPWPDDDY 300 μ M
 Peptide, 4 μ M 37° C./ 100% {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 0.5
 hr (SEQ ID 50 mM MOPS pH 8.0 NO: 109) yPIP LPWPDDDY 300 μ M
 Peptide, 7 μ M 37° C./ 100% {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 0.5
 hr (SEQ ID 50 mM MOPS pH 8.0 NO: 110) hTET FYPLPWPDDDY 200 μ M
 Peptide, 2 μ M 37° C./ 55% YPLPWPDDDY {LYSN3}NH2 enzyme, 2.5 mM
 ZnCl.sub.2, 1 hr {LYSN3}NH2 (SEQ ID 25 mM Tris pH 8.0 (SEQ ID
 NO: 107) NO: 108) hTET FYPLPWPDDDY 200 μ M Peptide, 2 μ M 37° C./
 55% YPLPWPDDDY {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 1 hr
 {LYSN3}NH2 (SEQ ID 50 mM MOPS pH 8.0 (SEQ ID NO: 107) NO: 108)
 Pro VPrA FYPLPWPDDDY 200 μ M Peptide, 2.2 37° C./ 6%
 YPLPWPDDDY mbr {LYSN3}NH2 μ M enzyme, 10 mM 1 hr {LYSN3}NH2
 (SEQ ID MgCl.sub.2, 50 mM MOPS (SEQ ID NO: 107) pH 8.0 NO: 108)
 ThrCut- FYPLPWPDDDY 200 μ M Peptide, 2.1 37° C./ 40% YPLPWPDDDY
 ProVPrA {LYSN3}NH2 μ M enzyme, 10 mM 1 hr {LYSN3}NH2 mbr (SEQ ID
 MgCl.sub.2, 50 mM MOPS (SEQ ID NO: 107) pH 8.0 NO: 108) VPr
 FYPLPWPDDDY 200 μ M Peptide, 4 μ M 37° C./ 100% YPLPWPDDDY
 {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 0.5 hr {LYSN3}NH2 (SEQ ID 50
 mM MOPS pH 8.0 (SEQ ID NO: 107) NO: 108) ProVPrA FYPLPWPDDDY 200
 μ M Peptide, 4 μ M 37° C./ 50% YPLPWPDDDY mbr {LYSN3}NH2 enzyme, 10
 mM MgCl.sub.2, 0.5 hr {LYSN3}NH2 (SEQ ID 50 mM MOPS pH 8.0
 (SEQ ID NO: 107) NO: 108) ThrCut- FYPLPWPDDDY 200 μ M Peptide, 4 μ M
 37° C./ 40% YPLPWPDDDY ProVPrA {LYSN3}NH2 enzyme, 10 mM
 MgCl.sub.2, 0.5 hr {LYSN3}NH2 mbr (SEQ ID 50 mM MOPS pH 8.0 (SEQ
 ID NO: 107) NO: 108) VPr FWPLPWPDDDY 200 μ M Peptide, 4 μ M 37° C./
 100% {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 1 hr (SEQ ID 50 mM
 MOPS pH 8.0 NO: 107) VPr FWPLPWPDDDY 200 μ M Peptide, 4 μ M 37° C./
 96% pAzF {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 1 hr (SEQ ID 50
 mM MOPS pH 8.0 NO: 107) ThrCut- FWPLPWPDDDY 200 μ M Peptide, 6.4
 37° C./ 100% ProVPrA {LYSN3}NH2 μ M enzyme, 10 mM 1 hr mbr Cy3
 (SEQ ID MgCl.sub.2, 50 mM MOPS clicked NO: 107) pH 8.0 hTET

PLPWPDDDY 200 μ M Peptide, 4 μ M 37° C./ 100% {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 1 hr (SEQ ID 50 mM MOPS pH 8.0 NO: 109) VPr FWPLPWPDDDY 400 μ M Peptide, 8 μ M 37° C./ 100% {LYSN3}NH2 enzyme, 50 mM HEPES 1 hr (SEQ ID pH 8.0, 300 mM NaCl, NO: 107) 1 mM DTT, 5% Glycerol, 32 mM PCA VPr FYPLPWPDDDY 200 μ M Peptide, 8 μ M RT/ 100% YPLPWPDDDY {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 1 hr {LYSN3}NH2 (SEQ ID 50 mM MOPS pH 5.0 (SEQ ID NO: 107) NO: 108) VPr FYPLPWPDDDY 200 μ M Peptide, 8 μ M RT/ 100% YPLPWPDDDY {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 0.5 hr {LYSN3}NH2 (SEQ ID 50 mM MOPS pH 5.0 (SEQ ID NO: 107) NO: 108) VPr FYPLPWPDDDY 200 μ M Peptide, 8 μ M RT/ 100% YPLPWPDDDY {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 1 hr {LYSN3}NH2 (SEQ ID 50 mM MOPS pH 5.0 (SEQ ID NO: 107) NO: 108) VPr FYPLPWPDDDY 200 μ M Peptide, 8 μ M RT/ 100% YPLPWPDDDY {LYSN3}NH2 enzyme, 10 mM MgCl.sub.2, 0.5 hr {LYSN3}NH2 (SEQ ID 50 mM MOPS pH 5.0 (SEQ ID NO: 107) NO: 108) VPr FYPLPWPDDDY 600 μ M Peptide, 0.8 RT/ 100% YPLPWPDDDY {LYSN3}NH2 μ M enzyme, 10 mM 0.5 hr {LYSN3}NH2 (SEQ ID Mg(OAc).sub.2, 50 mM MOPS (SEQ ID NO: 107) pH 5.5 NO: 108) VPr FYPLPWPDDDY 600 μ M Peptide, 0.8 RT/ 100% YPLPWPDDDY {LYSN3}NH2 μ M enzyme, 10 mM 0.5 hr {LYSN3}NH2 (SEQ ID Mg(OAc).sub.2, 50 mM MOPS (SEQ ID NO: 107) pH 6.5 NO: 108) VPr FYPLPWPDDDY 600 μ M Peptide, 0.8 RT/ 100% YPLPWPDDDY {LYSN3}NH2 μ M enzyme, 10 mM 0.5 hr {LYSN3}NH2 (SEQ ID Mg(OAc).sub.2, 50 mM MOPS (SEQ ID NO: 107) pH 7.5 NO: 108) VPr FYPLPWPDDDY 600 μ M Peptide, 0.8 RT/ 100% YPLPWPDDDY {LYSN3}NH2 μ M enzyme, 10 mM 0.5 hr {LYSN3}NH2 (SEQ ID Mg(OAc).sub.2, 50 mM MOPS (SEQ ID NO: 107) pH 8.5 NO: 108) VPr FYPLPWPDDDY 1200 μ M Peptide, RT/ 50% YPLPWPDDDY {LYSN3}NH2 0.08 μ M enzyme, 10 0.5 hr {LYSN3}NH2 (SEQ ID mM Mg(OAc).sub.2, 50 mM (SEQ ID NO: 107) MOPS pH 5.5 NO: 108) VPr FYPLPWPDDDY 1200 μ M Peptide, RT/ 100% YPLPWPDDDY {LYSN3}NH2 0.08 μ M enzyme, 10 0.5 hr {LYSN3}NH2 (SEQ ID mM Mg(OAc).sub.2, 50 mM (SEQ ID NO: 107) MOPS pH 6.5 NO: 108) VPr FYPLPWPDDDY 1200 μ M Peptide, RT/ 100% YPLPWPDDDY {LYSN3}NH2 0.08 μ M enzyme, 10 0.5 hr {LYSN3}NH2 (SEQ ID mM Mg(OAc).sub.2, 50 mM (SEQ ID NO: 107) MOPS pH 7.5 NO: 108) VPr FYPLPWPDDDY 1200 μ M Peptide, RT/ 100% YPLPWPDDDY {LYSN3}NH2 0.08 μ M enzyme, 10 0.5 hr {LYSN3}NH2 (SEQ ID mM Mg(OAc).sub.2, 50 mM (SEQ ID NO: X) MOPS pH 8.5 NO: 108) VPr QP15 1200 μ M Peptide, RT/ 1.40% YPLPWPDDDY FYPLPWPDDDY 0.008 μ M enzyme, 10 0.5 hr {LYSN3}NH2 {LYSN3}NH2 mM Mg(OAc).sub.2, 50 mM (SEQ ID (SEQ ID MOPS pH 5.5 NO: 108) NO: 107) VPr QP15 1200 μ M Peptide, RT/ 56% YPLPWPDDDY FYPLPWPDDDY 0.008 μ M enzyme, 10 0.5 hr {LYSN3}NH2 {LYSN3}NH2 mM Mg(OAc).sub.2, 50 mM (SEQ ID (SEQ ID MOPS pH 6.5 NO: 108) NO: 107) VPr FYPLPWPDDDY 1200 μ M Peptide, RT/ 100% YPLPWPDDDY {LYSN3}NH2 0.008 μ M enzyme, 10 0.5 hr {LYSN3}NH2 (SEQ ID mM Mg(OAc).sub.2, 50 mM (SEQ ID NO: 107) MOPS pH 7.5 NO: 108) VPr FYPLPWPDDDY 1200 μ M Peptide, RT/ 100% YPLPWPDDDY {LYSN3}NH2 0.008 μ M enzyme, 10 0.5 hr {LYSN3}NH2 (SEQ ID mM Mg(OAc).sub.2, 50 mM (SEQ ID NO: 107) MOPS pH 8.5 NO: 108) VPr FYPLPWPDDDY 1200 μ M Peptide, 800 RT/ 2.70% YPLPWPDDDY {LYSN3}NH2 pM enzyme, 10 mM 0.5 hr {LYSN3}NH2 (SEQ ID

Mg(OAc).sub.2, 50 mM MOPS (SEQ ID NO: 107) pH 7.5 NO: 108) VPr
FYPLPWPDDDDY 1200 μ M Peptide, 800 RT/ 6.80% YPLPWPDDDDY {LYSN3}NH2
pM enzyme, 10 mM 0.5 hr {LYSN3}NH2 (SEQ ID Mg(OAc).sub.2, 50
mM MOPS (SEQ ID NO: 107) pH 8.5 NO: 108) VPr FAAAWPDDDDF1 600 μ M
Peptide, 8 RT/ 100% WPDDF1 (SEQ ID μ M enzyme, 10 mM 0.5 hr
(SEQ ID NO: 11) Mg(OAc).sub.2, 50 mM MOPS NO: 112 pH 8 VPr
WAAAFPDDDDF1 600 μ M Peptide, 8 RT/ 100% FPDDF1 (SEQ ID μ M
enzyme, 10 mM 0.5 hr (SEQ ID NO: 13) Mg(OAc).sub.2, 50 mM MOPS
NO: 114) pH 8 VPr WAAAFPDDDDF1 300 μ M Peptide, 8 RT/ 100% FPDDF1
(SEQ ID μ M enzyme, 10 mM 0.5 hr (SEQ ID NO: 13) Mg(OAc).sub.2,
50 mM MOPS NO: 114) pH 8 VPr WAAAFPDDDDF1 300 μ M Peptide, 8 RT/
100% FPDDF1 (SEQ ID μ M enzyme, 10 mM 0.5 hr (SEQ ID NO: 13)
Mg(OAc).sub.2, 50 mM MOPS NO: 114) pH 8, 0.2% Tween20 VPr
WAAAFPDDDDF1 300 μ M Peptide, 8 RT/ 30% 70% -3, (SEQ ID μ M
enzyme, 50/50 0.5 hr 30% -4 NO: 13) MOPS Mg buffer/RB1 VPr
WAAAFPDDDDF1 300 μ M Peptide, 8 RT/ 5% Almost all (SEQ ID
 μ M enzyme, RB2 + 0.5 hr -1, -2 and NO: 13) Mg -3 products VPr
WAAAFPDDDDF1 300 μ M Peptide, 8 RT/ 100% FPDDF1 (SEQ ID μ M
enzyme, RB4 0.5 hr (SEQ ID NO: 13) NO: 114) VPr WAAAFPDDDDF1 300 μ M
Peptide, 8 RT/ 70% 30% -3, (SEQ ID μ M enzyme, 10 mM 0.5 hr
70% -4 NO: 13) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc VPr
WAAAFPDDDDF1 300 μ M Peptide, 8 RT/ 100% FPDDF1 (SEQ ID μ M
enzyme, 10 mM 2 hr (SEQ ID NO: 13) Mg(OAc).sub.2, 50 mM MOPS
NO: 114) pH 8, 60 mM KOAc VPr WAAAFPDDDDF1 300 μ M Peptide, 8 37°
C./ 100% FPDDF1 (SEQ ID μ M enzyme, 10 mM 2 hr (SEQ ID NO: 13)
Mg(OAc).sub.2, 50 mM MOPS NO: 114) pH 8, 60 mM KOAc VPr
FAAAYPDDDDF1 600 μ M Peptide, 8 RT/ 100% YPDDF1 (SEQ ID μ M
enzyme, 10 mM 0.5 hr (SEQ ID NO: 11) Mg(OAc).sub.2, 50 mM MOPS
NO: 115) pH 8, 60 mM KOAc VPr FAAAYPDDDDF1 600 μ M Peptide, 0.8 RT/
50% YPDDF1 (SEQ ID μ M enzyme, 10 mM 0.5 hr (SEQ ID NO: 11)
Mg(OAc).sub.2, 50 mM MOPS NO: 115) pH 8, 60 mM KOAc VPr
FAAAYPDDDDF1 600 μ M Peptide, 0.08 RT/ 5% YPDDF1 (SEQ ID μ M
enzyme, 10 mM 0.5 hr (SEQ ID NO: 11) 5Mg(OAc).sub.2, 0 mM MOPS
NO: 115) pH 8, 60 mM KOAc VPr RRPFFQQ 1 mM Peptide, 1 μ M RT/
79% RPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 116)
Mg(OAc).sub.2, 50 mM MOPS NO: 117) pH 8, 60 mM KOAc VPr AAPFQQ
1 mM Peptide, 1 μ M RT/ 100% APFQQ (SEQ ID enzyme, 10 mM
0.5 hr (SEQ ID NO: 118) Mg(OAc).sub.2, 50 mM MOPS NO: 119) pH 8,
60 mM KOAc VPr KKPFFQQ 1 mM Peptide, 1 μ M RT/ 32% KPFFQQ (SEQ
ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 120) Mg(OAc).sub.2, 50 mM
MOPS NO: 121) pH 8, 60 mM KOAc VPr YYPFFQQ 1 mM Peptide, 1 μ M
RT/ 56% YPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 122)
Mg(OAc).sub.2, 50 mM MOPS NO: 123) pH 8, 60 mM KOAc VPr FFPFQQ 1
mM Peptide, 1 μ M RT/ 100% FPFQQ (SEQ ID enzyme, 10 mM 0.5 hr
(SEQ ID NO: 124) Mg(OAc).sub.2, 50 mM MOPS NO: 125) pH 8, 60
mM KOAc VPr DDPFFQQ 1 mM Peptide, 1 μ M RT/ 0% DPFQQ (SEQ
ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 126) Mg(OAc).sub.2, 50 mM
MOPS NO: 127) pH 8, 60 mM KOAc VPr EEPFQQ 1 mM Peptide, 1 μ M
RT/ 0% EPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO:
128) Mg(OAc).sub.2, 50 mM MOPS NO: 129) pH 8, 60 mM KOAc VPr

NNPFQQ 1 mM Peptide, 1 μM RT/ 15% NPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 130) Mg(OAc).sub.2, 50 mM MOPS NO: 131) pH 8, 60 mM KOAc VPr QPFQQ 1 mM Peptide, 1 μM RT/ 66% QPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 132) Mg(OAc).sub.2, 50 mM MOPS NO: 133) pH 8, 60 mM KOAc VPr VVPFQQ 1 mM Peptide, 1 μM RT/ 100% VPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 134) Mg(OAc).sub.2, 50 mM MOPS NO: 135) pH 8, 60 mM KOAc VPr IIPFQQ 1 mM Peptide, 1 μM RT/ 100% IPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 136) Mg(OAc).sub.2, 50 mM MOPS NO: 137) pH 8, 60 mM KOAc VPr LLPFQQ 1 mM Peptide, 1 μM RT/ 100% LPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 138) Mg(OAc).sub.2, 50 mM MOPS NO: 139) pH 8, 60 mM KOAc VPr SSPFQQ 1 mM Peptide, 1 μM RT/ 48% SPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 140) Mg(OAc).sub.2, 50 mM MOPS NO: 141) pH 8, 60 mM KOAc VPr TTPFQQ 1 mM Peptide, 1 μM RT/ 100% TPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 142) Mg(OAc).sub.2, 50 mM MOPS NO: 143) pH 8, 60 mM KOAc VPr CCPFQQ 1 mM Peptide, 1 μM RT/ 70% CPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 144) Mg(OAc).sub.2, 50 mM MOPS NO: 145) pH 8, 60 mM KOAc VPr WWPFQQ 1 mM Peptide, 1 μM RT/ 82% WPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 146) Mg(OAc).sub.2, 50 mM MOPS NO: 147) pH 8, 60 mM KOAc VPr MMPFQQ 1 mM Peptide, 1 μM RT/ 100% MPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 148) Mg(OAc).sub.2, 50 mM MOPS NO: 149) pH 8, 60 mM KOAc VPr PPPFQQ 1 mM Peptide, 1 uM RT/ 0% PPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 150) Mg(OAc).sub.2, 50 mM MOPS NO: 151) pH 8, 60 mM KOAc VPr GGPFQQ 1 mM Peptide, 1 μM RT/ 8% GPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 152) Mg(OAc).sub.2, 50 mM MOPS NO: 153) pH 8, 60 mM KOAc VPr HHPFQQ 1 mM Peptide, 1 μM RT/ 12% HPFQQ (SEQ ID enzyme, 10 mM 0.5 hr (SEQ ID NO: 154) Mg(OAc).sub.2, 50 mM MOPS NO: 155) pH 8, 60 mM KOAc yPIP RRPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 116) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP AAPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 118) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP KKPFFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 120) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP YYPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 122) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP FFPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 124) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP DDPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 126) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP EEPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 128) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP NNPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 130) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP QPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 132) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP VVPFQQ

1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM
 0.5 hr NO: 134) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP
 IIPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10
 mM 0.5 hr NO: 136) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc
 yPIP LLPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID
 enzyme, 10 mM 0.5 hr NO: 138) Mg(OAc).sub.2, 50 mM MOPS pH 8,
 60 mM KOAc yPIP SSPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 140) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc yPIP TTPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A
 pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 142) Mg(OAc).sub.2, 50 mM
 MOPS pH 8, 60 mM KOAc yPIP CCPFQQ 1 mM Peptide, 1 uM RT/
 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr NO: 144)
 Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP WWPFFQQ 1 mM
 Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM 0.5 hr
 NO: 146) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP MMPFQQ
 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID enzyme, 10 mM
 0.5 hr NO: 148) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc yPIP
 PPPFQQ 1 mM Peptide, 1 uM RT/ 26% multiple pAzF (SEQ ID enzyme,
 10 mM 0.5 hr NO: 150) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM
 KOAc yPIP GGPFFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF (SEQ ID
 enzyme, 10 mM 0.5 hr NO: 152) Mg(OAc).sub.2, 50 mM MOPS pH 8,
 60 mM KOAc yPIP HHPFQQ 1 mM Peptide, 1 uM RT/ 0% N/A pAzF
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 154) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET RRPFFQQ 1 mM Peptide, 1 uM RT/ 0%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 116) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET AAPFFQQ 1 mM Peptide, 1 uM RT/ 100%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 118) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET KKPFFQQ 1 mM Peptide, 1 uM RT/ 0%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 120) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET YYPFFQQ 1 mM Peptide, 1 uM RT/ 99%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 122) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET FFPFFQQ 1 mM Peptide, 1 uM RT/ 100%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 124) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET DDPFFQQ 1 mM Peptide, 1 uM RT/ 4%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 126) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET EEPFFQQ 1 mM Peptide, 1 uM RT/ 0%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 128) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET NNPFQQ 1 mM Peptide, 1 uM RT/ 69%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 130) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET QQPFFQQ 1 mM Peptide, 1 uM RT/ 63%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 132) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET VVPFFQQ 1 mM Peptide, 1 uM RT/ 100%
 (SEQ ID enzyme, 10 mM 0.5 hr NO: 134) Mg(OAc).sub.2, 50 mM MOPS
 pH 8, 60 mM KOAc hTET IIPFFQQ 1 mM Peptide, 1 uM RT/ 100% (SEQ
 ID enzyme, 10 mM 0.5 hr NO: 136) Mg(OAc).sub.2, 50 mM MOPS pH
 8, 60 mM KOAc hTET LLPFQQ 1 mM Peptide, 1 uM RT/ 100% (SEQ
 ID enzyme, 10 mM 0.5 hr NO: 138) Mg(OAc).sub.2, 50 mM MOPS pH
 8, 60 mM KOAc hTET SSPFQQ 1 mM Peptide, 1 uM RT/ 100% (SEQ
 ID enzyme, 10 mM 0.5 hr NO: 140) Mg(OAc).sub.2, 50 mM MOPS pH
 8, 60 mM KOAc hTET TTPFQQ 1 mM Peptide, 1 uM RT/ 100% (SEQ

ID	enzyme,	10	mM	0.5	hr	NO: 142)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	hTET	CCPFQQ	1	mM	Peptide,	1	μM	RT/ 32% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 144)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	hTET	WWPFQQ	1	mM	Peptide,	1	μM	RT/ 4% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 146)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	hTET	MMPFQQ	1	mM	Peptide,	1	uM	RT/ 0% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 148)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	hTET	PPPFQQ	1	mM	Peptide,	1	uM	RT/ 0% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 150)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	hTET	GGPFQQ	1	mM	Peptide,	1	μM	RT/ 33% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 152)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	hTET	HHPFQQ	1	mM	Peptide,	1	μM	RT/ 26% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 154)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	RRPFQQ	1	mM	Peptide,	1	μM	RT/ 100% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 116)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	AAPFQQ	1	mM	Peptide,	1	μM	RT/ 100% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 118)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	KKPFQQ	1	mM	Peptide,	1	μM	RT/ 100% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 120)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	YYPFQQ	1	mM	Peptide,	1	μM	RT/ 100% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 122)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	FFPFQQ	1	mM	Peptide,	1	μM	RT/ 65% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 124)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	DDPFQQ	1	mM	Peptide,	1	μM	RT/ 86% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 126)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	EEPFQQ	1	mM	Peptide,	1	μM	RT/ 93% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 128)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	NNPFQQ	1	mM	Peptide,	1	μM	RT/ 81% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 130)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	QQPFQQ	1	mM	Peptide,	1	μM	RT/ 100% (SEQ	
ID	enzyme,	10	mM	0.5	hr	NO: 132)	Mg(OAc).sub.2,	50	mM	MOPS	pH	
8,	60	mM	KOAc	PfuTET	VVPFQQ	1	mM	Peptide,	1	uM	RT/ 0%	
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 134)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	IIPFQQ	1	mM	Peptide,	1	uM	RT/ 0%
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 136)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	LLPFQQ	1	mM	Peptide,	1	μM	RT/ 99%
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 138)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	SSPFQQ	1	mM	Peptide,	1	μM	RT/ 90%
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 140)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	TTPFQQ	1	mM	Peptide,	1	μM	RT/ 100%
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 142)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	CCPFQQ	1	mM	Peptide,	1	μM	RT/ 72%
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 144)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	WWPFQQ	1	mM	Peptide,	1	μM	RT/ 37%
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 146)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	MMPFQQ	1	mM	Peptide,	1	μM	RT/ 100%
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 148)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	PPPFQQ	1	mM	Peptide,	1	uM	RT/ 0%
(SEQ	ID	enzyme,	10	mM	0.5	hr	NO: 150)	Mg(OAc).sub.2,	50	mM	MOPS	
pH	8,	60	mM	KOAc	PfuTET	GGPFQQ	1	mM	Peptide,	1	uM	RT/ 0%

(SEQ ID enzyme, 10 mM 0.5 hr NO: 152) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc PfuTET HHPFQQ 1 mM Peptide, 1 μM RT/ 19% (SEQ ID enzyme, 10 mM 0.5 hr NO: 154) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN RRPFFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 116) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN AAPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 118) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN KKPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 120) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN YYPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 122) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN FFPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 124) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN DDPFQQ 1 mM Peptide, 1.3 RT/ 21% (Glu/Asp (SEQ ID μM enzyme, 10 mM 0.5 hr APN) NO: 126) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN EEPFQQ 1 mM Peptide, 1.3 RT/ 31% (Glu/Asp (SEQ ID μM enzyme, 10 mM 0.5 hr APN) NO: 128) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN NNPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 130) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN QQPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 132) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN VVPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 134) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN IIPFQQ 1 mM Peptide, 1.3 RT/ 42% (Glu/Asp (SEQ ID μM enzyme, 10 mM 0.5 hr APN) NO: 136) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN LLPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 138) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN SSPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 140) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN TTPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 142) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN CCPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 144) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc EDAPN WWPFQQ 1 mM Peptide, 1.3 RT/ 0% (Glu/Asp (SEQ ID uM enzyme, 10 mM 0.5 hr APN) NO: 146) Mg(OAc).sub.2, 50 mM MOPS pH 8, 60 mM KOAc PfuTET YAAFAAWADDDW1 1 mM Peptide, 1.0 RT/ Distri- Mainly (SEQ ID μM enzyme, 10 mM 0.5 bution AD DDWK NO: 156) Mg(OAc).sub.2, 50 mM MOPS hr (SEQ ID pH 8, 60 mM KOAc NO: 157) hTET YAAFAAWADDDW1 1 mM Peptide, 1.0 RT/ 100% WAD DDWK (SEQ ID μM enzyme, 10 mM 0.5 (SEQ ID NO: 156) Mg(OAc).sub.2, 50 mM MOPS hr NO: 158) pH 8, 60 mM KOAc VPr YAAFAAWADDDW1 1 mM Peptide, 1.2 RT/ Distri- Mainly (SEQ ID μM enzyme, 10 mM 0.5 bution AD DDWK NO: 156) Mg(OAc).sub.2, 50 mM MOPS hr (SEQ ID pH 8, 60 mM KOAc NO: 157)

[0315] A summary of amino acid cleavage activities for select exopeptidases of Table 7 is shown in FIG. 16. Specific cleavage activities are shown for the following enzymes: “cVPr” (*V. proteolyticus*

aminopeptidase), “yPIP” (*Y. pestis* proline iminopeptidase), “D/E APN” (*L. pneumophila* M1 Aminopeptidase), hTET (*Pyrococcus horikoshii* TET aminopeptidase), and Pfu API (“PfuTET” in Table 7). Specific activities with respect to terminal amino acids are classified as shown, with single-letter abbreviations used for amino acids (“XP-” represents any terminal amino acid (X) having an adjacent, or penultimate, proline (P) residue).

Example 4. Terminal Amino Acid Cleavage of Immobilized Peptides at Single Molecule Level

[0316] Assays for on-chip amino acid cleavage of immobilized peptides were developed using labeled peptide conjugates. The assays were designed to provide a method for determining enzymatic recognition and cleavage activity of exopeptidases toward immobilized peptides, which could permit measurement of kinetic binding parameters and general binding affinities.

[0317] To evaluate N-terminal amino acid cleavage of a peptide, a dye labeled peptide was designed and synthesized which contained an N-terminal aspartate that was attached to the dye by way of a PEG spacer. This peptide also contained a proline residue adjacent to the modified aspartate that is recognized specifically by the enzyme proline iminopeptidase (from *Yersinia pestis*, known elsewhere and referred to herein as “yPIP”). The enzyme yPIP should cleave only an N-terminal amino acid upstream from a proline residue.

[0318] After showing that this and other labeled peptides were efficiently cut by yPIP in bulk (e.g., as described in Example 2), an on-chip dye/peptide conjugate assay was developed to observe N-terminal amino acid cleavage at the single molecule level. FIG. 17A shows a general scheme for the dye/peptide conjugate assay (inset panel). As shown, a peptide having a label attached to an N-terminal amino acid via a spacer is immobilized to a surface by way of a linker. After being exposed to peptidase, N-terminal amino acid cleavage results in the removal of the labeled residue from a detectable observation volume and is measured by a concomitant loss in signal from the label. The enzyme-peptide complex to the right of the inset panel generically depicts the N-terminal cleavage site.

[0319] FIG. 17A shows a labeled peptide construct (at bottom) that was designed and synthesized for use in the dye/peptide conjugate assay. In these experiments, a rhodamine dye (ATTO Rho6G) was attached to an N-terminal aspartate residue of a peptide having a penultimate proline residue at the N-terminus. As shown, the peptide was further conjugated to a solubilizing DNA linker with a biotin moiety for surface immobilization.

[0320] The labeled peptide conjugate was loaded onto a glass chip having an array of sample wells. Images of the chip were acquired before and after loading to determine the percent loading of sample wells at single occupancy by rhodamine fluorescence. The enzyme yPIP was then introduced onto the loaded chip and allowed to incubate for two hours at 37° C. An image of the chip following the introduction of yPIP was taken and the percentage of green dyes lost were calculated to evaluate N-terminal amino acid cleavage. FIG. 17B shows imaging results from an experiment which displayed 6-7% loading in the loading stage and 91% loss of signal in previously loaded wells after incubation with yPIP, which was indicative of N-terminal amino acid cleavage. FIG. 17C shows representative signal traces from these experiments, which demonstrate a detected increase in dye signal upon loading of labeled peptide and a detected loss in dye signal following exposure to yPIP.

[0321] As further confirmation of N-terminal amino acid cleavage at the single molecule level, on-chip FRET assays were developed to evaluate exopeptidase recognition and cleavage activity. FIG. 18A generically depicts a FRET peptide conjugate assay (panel A) and a FRET enzyme conjugate assay (panel B). In the FRET peptide conjugate assay (panel A), an immobilized peptide construct includes a FRET donor label attached to the linker and a FRET acceptor label attached at the N-terminus. N-terminal amino acid cleavage is detected by a loss in signal from the FRET acceptor label when exposed to peptidase. Additionally, this design permits monitoring loading of the peptide conjugate throughout an experiment by following emission from the FRET donor label.

[0322] In the FRET enzyme conjugate assay (panel B), an immobilized peptide construct includes

a first label of a FRET pair attached to the linker and a peptidase is labeled with a second label of the FRET pair. N-terminal amino acid cleavage is detected by an enhancement in fluorescence attributable to FRET interactions, which would occur with sufficient proximity of peptidase to peptide and with sufficient residence time at the N-terminus. Additionally, this assay permits evaluating processive amino acid cleavage by a processive exopeptidase by detecting an increasing FRET signal over time with processive cleavage.

[0323] FIG. **18A** also shows a FRET peptide construct under panel A that was designed and synthesized for use in the FRET peptide conjugate assay of panel A. As shown, the FRET peptide construct included a rhodamine dye (ATTO 647N) attached to an N-terminal aspartate residue of a peptide having a penultimate proline residue at the N-terminus. The peptide was further conjugated to a solubilizing DNA linker which was attached to a cyanine dye (Cy3B) for FRET and a biotin moiety for surface immobilization.

[0324] In this experiment, the FRET peptide construct was loaded onto a glass chip having an array of sample wells, and collected light was filtered first by a green filter and then a red filter. Loading of the FRET peptide construct was detected by measuring a signal passing through both the green and red filters. Terminal amino acid cleavage was detected when the signal was measurable only in the green filter, which indicated that the red dye conjugated N-terminal amino acid from the FRET peptide construct was cleaved by yPIP. This detection pattern is illustrated in panel C. As shown, if both dyes are detectable before the addition of yPIP, and only the green dye is visible after incubation with yPIP, it can be reasonably concluded this change in detection pattern is due to cleavage of the peptide and not photobleaching or loss of the peptide as a whole. Additionally, an increase in fluorescence from the lone green dye would be expected, as its emissions are no longer absorbed by the red dye.

[0325] Following loading of the FRET peptide construct onto the chip, which had been modified by surface passivation using phosphonic acid and silane, yPIP was introduced and images were obtained at several time points. To assess the overall cleaving trend, the ratio of (green)/(green+red) was computed for each experiment. This ratio increases with the extent of cleaving that occurs. FIG. **18B** is a plot of FRET emission ratio across all apertures at different time points of incubation with yPIP. As shown, the green dye contribution to the ratio of fluorescence emissions increases over time during incubation with yPIP, indicating that more N-terminal aspartate residues have been cut, leaving behind the truncated peptide with just the green dye.

[0326] Cutting efficiency was then evaluated at different time points by determining at which time points dye fluorescence was observed. This was done with simple thresholding—e.g., if the average dye emission signal was >2.5 during excitation, the dyes were determined to be present (when each corresponding filter was applied). Apertures exhibiting cutting would then display both green and red dyes during the loading phase of the experiment, but only green dye at time points exposed to yPIP. As shown in FIG. **18C**, progressively more cutting was observed as the chip was exposed to longer incubation times with yPIP. Example signal traces showing cutting displayed at each of the three yPIP-treated time points are shown in FIG. **18D**.

[0327] Additional experiments were performed with yPIP and other peptidases using chips that had been modified by surface passivation using dextran, which produced similar results showing an increase in terminal amino acid cleavage over time following introduction of peptidase onto chips. FIG. **18E** is a plot of FRET emission ratio across loaded apertures at different time points of incubation with yPIP. FIG. **18F** is a plot of FRET emission ratio across loaded apertures at different time points of incubation with an aminopeptidase. Overall, the experiments here demonstrate that N-terminal amino acid cleavage is detectable in real-time at the single molecule level using different exopeptidases and different labeling strategies.

Example 5. Terminal Amino Acid Discrimination by Labeled Affinity Reagent

[0328] An adaptor protein involved in proteolytic pathways was identified as a potential candidate for use as a labeled affinity reagent for detecting N-terminal aromatic residues. The adaptor protein,

ClpS2 from an *a*-proteobacterium (*A. tumefaciens*), was expressed and labeled at an exposed cysteine residue. FIG. 19A shows a crystal structure of the ClpS2 protein, with the exposed cysteine residue shown as sticks. The exposed cysteine residue was labeled with a rhodamine dye (ATTO 532).

[0329] Peptides having different N-terminal aromatic residues were prepared to test whether the labeled ClpS2 was capable of N-terminal amino acid discrimination at the single molecule level. Example single molecule intensity traces from these experiments are shown in FIG. 19B. As shown, the signal traces demonstrate residue-specific on-off binding patterns corresponding to the labeled affinity reagent reversibly binding the N-terminus of peptides having either: an N-terminal phenylalanine residue (F, top signal trace), an N-terminal tyrosine residue (Y, middle signal trace), or an N-terminal tryptophan residue (W, bottom signal trace).

[0330] Further analyses of the single molecule trajectories were carried out, with the results shown in FIGS. 19C-19E. FIG. 19C is a plot showing discriminant pulse durations (time duration of signal peaks) among the three N-terminal residues when reversibly bound by labeled ClpS2. FIG. 19D is a plot showing discriminant interpulse durations (time duration between signal pulses) among the three N-terminal residues. FIG. 19E shows plots which further illustrate the discriminant pulse durations among phenylalanine, tyrosine, and tryptophan at peptide N-termini. Mean pulse duration for the different N-terminal residues is visualized by histograms (A)-(B) and layered histogram (C).

[0331] Another adaptor protein, ClpS from *Thermosynochoccus elongatus* (teClpS) was evaluated for use as a labeled affinity reagent for leucine recognition. The data obtained from dwell time analysis, shown in FIGS. 19F-19H, demonstrated that the labeled teClpS protein produces detectable binding interactions with a terminal leucine residue of polypeptides with a mean pulse duration of 0.71 seconds. The amino acid sequence of the teClpS protein used in these experiments is shown in Table 1.

[0332] Similar experiments were carried out to evaluate *A. tumefaciens* ClpS1 and *S. elongatus* ClpS2 as potential reagents for leucine recognition, and GID4 as a potential reagent for proline recognition. FIG. 19I shows example results from dwell time analysis which showed differentiable recognition of phenylalanine, leucine, tryptophan, and tyrosine by *A. tumefaciens* ClpS1. FIG. 19J shows example results from dwell time analysis demonstrating leucine recognition by *S. elongatus* ClpS2. FIGS. 19K-19L show example results from dwell time analysis demonstrating proline recognition by GID4.

Example 6. Polypeptide Sequencing by Recognition During Degradation

[0333] Experiments were conducted to evaluate peptide sequencing by N-terminal amino acid recognition during an ongoing degradation reaction. Example results from these experiments are shown in FIGS. 20A-20D, which show single molecule intensity traces obtained over two independent polypeptide sequencing reactions conducted in real-time using a labeled ClpS2 protein and an aminopeptidase in the same reaction mixture. In each reaction, a polypeptide of sequence YAAWAAFADDDWK (SEQ ID NO: 78) was immobilized to a chip surface through the C-terminal lysine residue by loading the peptide composition (10 pM) onto chips for 20 minutes, and the immobilized peptide was monitored in the presence of a labeled affinity reagent (ATTO 542-labeled *A. Tumefaciens* ClpS2-V1 at 500 nM) and an aminopeptidase cleaving reagent (VPr at 8 μ M).

[0334] FIGS. 20A and 20C show signal trace data for two different sequencing runs, with the top panel (panel 1 in FIG. 20A, panel 2 in FIG. 20C) showing a full trace, and the bottom panels (Y, W, F) showing zoomed-in regions corresponding to each of the highlighted regions in the full trace. FIGS. 20B and 20D show pulse duration statistics in histograms for the trace data of the corresponding panels as labeled in FIGS. 20A and 20C, respectively. As shown in the full signal trace of each sequencing run (panels 1, 2), three separate time intervals of signal pulses were observed over the course of the reaction. As highlighted by the zoomed-in regions (panels Y, W, F), the three intervals are visually distinguishable from one another based on an observable difference

in pattern of signal pulses.

[0335] To further analyze the signal pulse data, pulse duration statistics were determined for each time interval (FIGS. 20B and 20D). The differences in pulse duration distribution were determined to correspond to those observed for these amino acids individually in steady-state on-chip binding assays with ClpS2, and the signal pulse information was phenotypically consistent between intervals from sequencing runs and the individual amino acid binding assays.

[0336] As confirmed by the analysis of signal pulse information, the three time intervals of signal pulses observed over the progression of each sequencing run correspond to recognition patterns of Y, W, and F, respectively (panels 1, 2). The intervening time period between signal pulse patterns is due to the selectivity of ClpS2-V1, which does not bind to N-terminal alanine residues. As illustrated by the full signal trace, the first interval corresponds to Y recognition, which is followed by a pause as VPr peptidase cuts Y and two alanine residues, followed by the second interval corresponding to W recognition, which is followed by another pause as VPr peptidase cuts W and two alanine residues, and finally the third interval corresponding to F recognition before VPr peptidase cuts off the F and stops at the remaining ADDDWK peptide. These results show that pulse duration information, which was obtained by terminal amino acid recognition during an ongoing degradation reaction, can be used to determine characteristic patterns that discriminate between different types of terminal amino acids.

Example 7. Terminal Amino Acid Identification and Cleavage by Labeled Exopeptidase

[0337] Studies were performed to investigate the potential for a single reagent that is capable of both identifying a terminal amino acid of a peptide and cleaving the terminal amino acid from the peptide. As a single reagent, an exopeptidase must be able to bind to the peptide while retaining cleavage activity toward a terminal residue. Accordingly, an initial approach employing traditional labeling strategies was carried out by targeting the native surface-exposed amino acids of different exopeptidases. In these experiments, surface-exposed cysteine (—SH) or lysine (—NH₂) residues were labeled with fluorescent dyes, which proved to be a robust methodology for exopeptidase labeling. In certain cases, however, this approach produced a heterogeneous population of proteins that are labeled with one or more dyes.

[0338] In order to more precisely control where labeling occurs on exopeptidases and ensure that each exopeptidase molecule is labeled with a single fluorescent dye (as well as eliminate off-target reactivity of the dye), a new labeling strategy was investigated. In these experiments, labeled exopeptidases were prepared using a site-specific labeling strategy in which an unnatural amino acid containing a reactive functional group is introduced into the exopeptidase (see, e.g., Chin, J. W., et al. J Am Chem Soc. 2002 Aug. 7; 124 (31): 9026-9027).

[0339] The proline iminopeptidase from *Yersinia pestis* (yPIP) was modified by mutation of a lysine residue at position 287 to a residue having a para-azidophenylalanine (pAzF) side chain. FIG. 21A shows a crystal structure of yPIP, with the mutation indicated by the chemical structure of pAzF shown with the K287 sidechain shown as sticks. This mutation site was selected based on the stability provided by the alpha helix at this position and to ensure that the new azido functional group is solvent exposed.

[0340] A pEVOL plasmid containing the mutant amino tRNA synthetase and the mutant tRNA necessary to incorporate pAzF into the amino acid chain was obtained. The amber stop codon (TAG), which is necessary for the specific incorporation of pAzF, was then introduced into the cDNA using the QuickChange II mutagenesis kit. The cDNA was then sequenced and the TAG codon position was confirmed. This was followed by co-transfection of both the pET21b+ plasmid containing the yPIP amber mutant and the pEVOL plasmid containing the cellular machinery to charge the tRNA for the amber codon with pAzF. The co-transfected cells were then grown to 0.8 ODU, induced with 0.02% arabinose and 1 mM IPTG in the presence of 2 mM pAzF in 2 L of LB, and harvested using chemolysis. Purification was carried out using a 5 mL affinity chromatography column, and the protein was eluted in 100 mM imidazole. The resulting protein was then dialyzed

and concentrated into 50 mM HEPES pH 8.0 and 0.5 M KCl, aliquoted, and flash frozen prior to storage at -20°C .

[0341] To confirm the presence of the azido group in the purified protein, DBCO-Cy3 (2 mM) was reacted with the pAzF-yPIP variant (220 μM) (Reaction Conditions: 50 mM HEPES pH 8.0, 0.5 mM KCl, 20% DMSO; 10 hours at 37°C ., 48 hours at room temperature). The protein reaction product was purified by size-exclusion chromatography, and it was determined that the resulting protein was 100% labeled with the azide-reactive DBCO-Cy3 reagent (FIG. 21B), indicating robust incorporation of the unnatural amino acid.

[0342] Protein labeling and purity of the final product was confirmed by SDS-PAGE analysis of the unlabeled and labeled pAzF variant. FIG. 21C shows a picture of SDS-PAGE gel confirming Cy3-labeling of pAzF-yPIP (overexposed image of gel shown in FIG. 21D to show ladder). FIG. 21E shows a picture of Coomassie-stained gel confirming that both dye and protein co-migrate and are pure.

[0343] The dye-labeled pAzF-yPIP variant was used in an activity assay to confirm that the enzyme was still active after labeling and purification. As shown in FIG. 21F, Cy3-pAzF-yPIP was able to hydrolyze 100% of the peptide substrate in 1 hour using 1000-fold excess substrate, as measured by HPLC. These experiments demonstrate a methodology which allows site-specific modification and labeling of an exopeptidase with minimal perturbation of the native protein structure/function.

Example 8. Recognition of Modified Amino Acids in Polypeptide Sequencing

[0344] Experiments were performed to evaluate recognition of amino acids containing specific post-translation modifications. A triple-mutant variant (T8V, S10A, K15L) of the Src Homology 2 (SH2) domain from Fyn, a tyrosine kinase, was tested as a potential recognition molecule for phosphorylated tyrosine residues in peptide sequencing. The variant protein was immobilized to the bottom of sample wells, and single-molecule signal traces were collected upon addition of a fluorescently-labeled peptide containing N-terminal phospho-tyrosine. Peptide binding by the immobilized protein was detected during these experiments, as shown by the representative traces in FIG. 22A. Pulse duration data collected during these experiments is shown in FIG. 22B (top, middle, and bottom plots corresponding to the top, middle, and bottom traces of FIG. 22A, respectively). Pulse duration and interpulse duration statistics are shown in FIG. 22C (top and bottom panels, respectively).

[0345] Control experiments were performed to confirm that the Fyn protein was specific for the phosphorylated tyrosine. The experiments were repeated for each of three different peptides: a first peptide containing N-terminal unmodified tyrosine (Y; FIG. 22D), a second peptide containing N-terminal and penultimate unmodified tyrosines (YY; FIG. 22E), and a third peptide containing N-terminal phospho-serine (FIG. 22F). As shown, binding was not detected with any of the peptides used in the negative control experiments.

Example 9. Recognition of Penultimate Amino Acids in Polypeptide Sequencing

[0346] Experiments were performed to determine the effects of penultimate amino acids on pulse duration for *A. Tumefaciens* ClpS2-V1. Forty-nine different fluorescently-labeled peptides were prepared containing unique dipeptide sequences at the N-terminus, where the N-terminal amino acid was F, W, or Y, and the penultimate position was one of the **20** natural amino acids. For each experiment, ClpS2-V1 was immobilized at the bottom of sample wells, and single-molecule signal traces were collected for 10.sup.–20 minutes upon addition of one of the fluorescently-labeled peptides. Pulse duration data was collected for a minimum of **50** sample wells for each peptide.

[0347] FIG. 23 shows the median pulse duration for each of the **50** peptides, with data points grouped by penultimate amino acid (x-axis) and N-terminal amino acids represented with different symbols.

Example 10. Simultaneous Amino Acid Recognition with Multiple Recognition Molecules

[0348] Single-molecule peptide recognition experiments were performed to demonstrate terminal amino acid recognition of an immobilized peptide by more than one labeled recognition molecule.

Single peptide molecules containing N-terminal phenylalanine (FYPLPDDDDY (SEQ ID NO: 79)) were immobilized in sample wells of a chip. Buffer containing 500 nM each of at ClpS1 (*Agrobacterium tumifaciens* ClpS1; sequence provided in Table 1) and at ClpS2-V1 (*Agrobacterium tumifaciens* ClpS2 variant 1; sequence provided in Table 1) was added, where at ClpS1 and at ClpS2-V1 were labeled with Cy3 and Cy3B, respectively. Since the intensity of Cy3B is higher than Cy3, at ClpS2-V1 binding events were readily distinguishable from at ClpS1 binding events.

[0349] FIGS. 24A-24C shows the results of the experiments showing single-molecule peptide recognition with differentially labeled recognition molecules. A representative trace is displayed in FIG. 24A. The pulse duration distributions were distinct for each binder (FIG. 24B) and corresponded to their kinetic profiles as observed in single-binder experiments. Mean pulse duration was 1.3 seconds for at ClpS1 and 1.0 seconds for at ClpS2-V1 (FIG. 24C). Pulse rate was also distinct: 8.1 pulses/min for at ClpS1 and 14.1 pulses/min for at Clp2-V1 (FIG. 24C). Thus, when more than one recognition molecule is included for dynamic recognition of immobilized peptides, the binding characteristics of each recognition molecule (including pulse duration, interpulse duration, and pulse rate) can simultaneously provide information about peptide sequence.

Example 11. Enhancing Photostability with Recognition Molecule Linkers

[0350] Experiments were performed to evaluate the photostability of immobilized peptides during single-molecule sequencing. The dye-labeled at ClpS2-V1 described in Example 5 was added to sample wells containing immobilized peptide substrates in the presence of excitation light at 532 nm to monitor recognition by emission from ATTO 532. A representative trace is shown in FIG. 25A. As shown in the top panel, recognition was observed to cease at approximately 600 seconds into the experiment. The bottom panel is a zoomed view showing signal pulses at approximately 180-430 seconds into the reaction.

[0351] FIG. 25B shows a visualization of the crystal structure of the ClpS2 protein used in these experiments. As shown, the cysteine residue that serves as the dye conjugation site is approximately 2 nm from the terminal amino acid binding site. It was hypothesized that photodamage to the peptide was caused by proximity of the dye to the N-terminus of peptide during binding. To mitigate the potential photodamaging effects of dye proximity, the ClpS2 protein was dye-labeled through a linker that increased distance between the dye and N-terminus of peptide by more than 10 nm. The linker included streptavidin and a double-stranded nucleic acid; the double-stranded nucleic acid was labeled with two Cy3B dye molecules and attached to streptavidin through a bis-biotin moiety, and a ClpS2 protein was attached to each of the remaining two binding sites on streptavidin through a biotin moiety. A representative trace using this dye-shielded ClpS2 molecule is shown in FIG. 25C. As shown in the top panel, recognition time was extended to approximately 6,000 seconds into the experiment. The bottom panel is a zoomed view showing signal pulses at approximately 750-930 seconds into the reaction.

[0352] A DNA-streptavidin recognition molecule was generated with a linker containing a double-stranded nucleic acid labeled with two Cy3B dye molecules and attached to streptavidin through a bis-biotin moiety, and a single ClpS2 protein attached to the remaining two binding sites on streptavidin through a bis-biotin moiety. This construct was used in a single-molecule peptide sequencing reaction, and representative traces from these experiments are shown in FIGS. 26A-26D.

[0353] The sequencing experiments described in example 6 were repeated, with the reaction conditions changed as follows: the DNA-streptavidin ClpS2 recognition molecule was used in combination with hTET amino acid cleaving reagent. A representative signal trace is shown in FIG. 27.

Example 12. Sequencing by Recognition During Degradation by Multiple Exopeptidases

[0354] Experiments were performed to evaluate the use of multiple types of exopeptidases with

differential cleavage specificities in a single-molecule peptide sequencing reaction mixture. Single peptide molecules (YAAWAAFADDDWK (SEQ ID NO: 78)) were immobilized through a C-terminal lysine residue in sample wells of a chip. Buffer containing at ClpS2-V1 for amino acid recognition and hTET for amino acid cleavage was added. A representative trace is displayed in FIG. 28A, with expanded views of pulse pattern regions shown in FIG. 28B.

[0355] An experiment was carried out to evaluate sequencing reactions in the presence of two types of exopeptidases with differential specificities. Single peptide molecules (FYPLPWPDDDYK (SEQ ID NO: 80)) were immobilized through a C-terminal lysine residue in sample wells of a chip. Buffer containing at ClpS2-V1 for amino acid recognition, and both hTET and yPIP for amino acid cleavage was added. A representative trace is displayed in FIG. 28C, with expanded views of pulse pattern regions shown in FIG. 28D. Additional representative traces from these reaction conditions are shown in FIG. 28E.

[0356] Further experiments were carried out to evaluate sequencing reactions in the presence of two types of exopeptidases with differential specificities. Single peptide molecules (YPLPWPDDDYK (SEQ ID NO: 81)) were immobilized through a C-terminal lysine residue in sample wells of a chip. In one experiment, buffer containing at ClpS2-V1 for amino acid recognition, and both hTET and yPIP for amino acid cleavage was added. A representative trace is displayed in FIG. 28F, with expanded views of pulse pattern regions shown in FIG. 28G. Additional representative traces from these reaction conditions are shown in FIG. 28H. In a further experiment, buffer (50 mM MOPS, 60 mM KOAc, 200 μ M Co (OAc) 2) containing at ClpS2-V1 for amino acid recognition, and both PfuTET and yPIP for amino acid cleavage was added. A representative trace is displayed in FIG. 28I, with expanded views of pulse pattern regions shown in FIG. 28J.

EQUIVALENTS AND SCOPE

[0357] In the claims articles such as “a,” “an,” and “the” may mean one or more than one unless indicated to the contrary or otherwise evident from the context. Claims or descriptions that include “or” between one or more members of a group are considered satisfied if one, more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process unless indicated to the contrary or otherwise evident from the context. The invention includes embodiments in which exactly one member of the group is present in, employed in, or otherwise relevant to a given product or process. The invention includes embodiments in which more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process.

[0358] Furthermore, the invention encompasses all variations, combinations, and permutations in which one or more limitations, elements, clauses, and descriptive terms from one or more of the listed claims is introduced into another claim. For example, any claim that is dependent on another claim can be modified to include one or more limitations found in any other claim that is dependent on the same base claim. Where elements are presented as lists, e.g., in Markush group format, each subgroup of the elements is also disclosed, and any element(s) can be removed from the group. It should be understood that, in general, where the invention, or aspects of the invention, is/are referred to as comprising particular elements and/or features, certain embodiments of the invention or aspects of the invention consist, or consist essentially of, such elements and/or features. For purposes of simplicity, those embodiments have not been specifically set forth in haec verba herein.

[0359] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended

language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0360] As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0361] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0362] It should also be understood that, unless clearly indicated to the contrary, in any methods claimed herein that include more than one step or act, the order of the steps or acts of the method is not necessarily limited to the order in which the steps or acts of the method are recited.

[0363] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03. It should be appreciated that embodiments described in this document using an open-ended transitional phrase (e.g., “comprising”) are also contemplated, in alternative embodiments, as “consisting of” and “consisting essentially of” the feature described by the open-ended transitional phrase. For example, if the application describes “a composition comprising A and B,” the application also contemplates the alternative embodiments “a composition consisting of A and B” and “a composition consisting essentially of A and B.”

[0364] Where ranges are given, endpoints are included. Furthermore, unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or sub-range within the stated ranges in different embodiments of the invention, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.

[0365] This application refers to various issued patents, published patent applications, journal articles, and other publications, all of which are incorporated herein by reference. If there is a conflict between any of the incorporated references and the instant specification, the specification

shall control. In addition, any particular embodiment of the present invention that falls within the prior art may be explicitly excluded from any one or more of the claims. Because such embodiments are deemed to be known to one of ordinary skill in the art, they may be excluded even if the exclusion is not set forth explicitly herein. Any particular embodiment of the invention can be excluded from any claim, for any reason, whether or not related to the existence of prior art. [0366] Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation many equivalents to the specific embodiments described herein. The scope of the present embodiments described herein is not intended to be limited to the above Description, but rather is as set forth in the appended claims. Those of ordinary skill in the art will appreciate that various changes and modifications to this description may be made without departing from the spirit or scope of the present invention, as defined in the following claims.

[0367] The recitation of a listing of chemical groups in any definition of a variable herein includes definitions of that variable as any single group or combination of listed groups. The recitation of an embodiment for a variable herein includes that embodiment as any single embodiment or in combination with any other embodiments or portions thereof. The recitation of an embodiment herein includes that embodiment as any single embodiment or in combination with any other embodiments or portions thereof.

Claims

1-283. (canceled)

284. A method of sequencing a polypeptide, the method comprising: a) contacting a single polypeptide molecule with a composition comprising one or more terminal amino acid recognition molecules; b) detecting signal pulses indicative of association of the one or more terminal amino acid recognition molecules with a terminus of the single polypeptide molecule; c) contacting the single polypeptide molecule with a composition comprising one or more cleaving reagents; and d) repeating (a)-(c) one or more times, wherein the detected signal pulses form a series of signal pulses that is indicative of a series of amino acids exposed at the terminus over time as a result of terminal amino acid cleavage by the one or more cleaving reagents, and wherein the single polypeptide molecule is immobilized to a surface through a linkage group comprising an oligonucleotide.

285. The method of claim 284, wherein association of the one or more terminal amino acid recognition molecules with each type of amino acid exposed at the terminus produces a characteristic pattern in the series of signal pulses that is different from other types of amino acids exposed at the terminus.

286. The method of claim 285, wherein the characteristic pattern comprises a portion of the series of signal pulses.

287. The method of claim 285, wherein a signal pulse of the characteristic pattern corresponds to an individual association event between a terminal amino acid recognition molecule and an amino acid exposed at the terminus.

288. The method of claim 287, wherein the characteristic pattern is indicative of the amino acid exposed at the terminus of the single polypeptide molecule and an amino acid at a contiguous position.

289. The method of claim 285, wherein signal pulses of the characteristic pattern comprise a mean pulse duration of between about 10 milliseconds and about 100 milliseconds or between about 100 milliseconds and about 500 milliseconds.

290. The method of claim 284, wherein at least one of the one or more terminal amino acid recognition molecules comprises a degradation pathway protein, a peptidase, an antibody, an aminotransferase, a tRNA synthetase, or an SH2 domain-containing protein or fragment thereof.

291. The method of claim 284, wherein at least one of the one or more terminal amino acid

recognition molecules comprises a detectable label.

292. The method of claim 291, wherein the detectable label is a luminescent label.

293. The method of claim 284, wherein sequencing comprises identifying at least a portion of all types of successive amino acids exposed at the terminus of the single polypeptide molecule while the single polypeptide molecule is being degraded by the one or more cleaving reagents.

294. The method of claim 284, wherein sequencing comprises identifying that an amino acid of the single polypeptide molecule comprises a post-translational modification.

295. The method of claim 294, wherein the post-translational modification is selected from acetylation, ADP-ribosylation, caspase cleavage, citrullination, formylation, N-linked glycosylation, O-linked glycosylation, hydroxylation, methylation, myristoylation, neddylation, nitration, oxidation, palmitoylation, phosphorylation, prenylation, S-nitrosylation, sulfation, sumoylation, and ubiquitination.

296. The method of claim 294, wherein the amino acid of the single polypeptide molecule comprises phospho-tyrosine or phospho-serine.

297. The method of claim 284, wherein the linkage group comprises a biotin molecule and an avidin protein.

298. The method of claim 297, wherein the avidin protein comprises streptavidin, traptavidin, tamavidin, bradavidin, or xenavidin.

299. The method of claim 298, wherein the avidin protein comprises streptavidin.

300. The method of claim 284, wherein the surface comprises a surface of a substrate.

301. The method of claim 300, wherein the substrate comprises an array of sample wells.

302. The method of claim 301, wherein the single polypeptide molecule is immobilized within a sample well of the array.

303. The method of claim 301, wherein the substrate comprises a plurality of polypeptide sequencing reactions, each polypeptide sequencing reaction occurring in an individual sample well of the array.

304-312. (canceled)

313. The method of claim 300, further comprising, prior to (a): contacting the substrate with the single polypeptide molecule, wherein the surface of the substrate comprises a polypeptide conjugating moiety and the single polypeptide molecule comprises a surface conjugating moiety, and wherein the polypeptide conjugating moiety attaches to the surface conjugating moiety to form the linkage group, thereby immobilizing the single polypeptide molecule to the surface of the substrate.

314. The method of claim 313, wherein the polypeptide conjugating moiety is attached to the surface of the substrate through the oligonucleotide prior to the immobilizing.

315. The method of claim 313, wherein the surface conjugating moiety is attached to the single polypeptide molecule through the oligonucleotide prior to the immobilizing.

316. The method of claim 313, wherein the polypeptide conjugating moiety forms a non-covalent attachment with the surface conjugating moiety.

317. The method of claim 313, wherein the polypeptide conjugating moiety comprises a biotin molecule and the surface conjugating moiety comprises an avidin protein.

318. The method of claim 313, wherein the polypeptide conjugating moiety comprises an avidin protein and the surface conjugating moiety comprises a biotin molecule.

319. The method of claim 313, wherein the polypeptide conjugating moiety forms a covalent attachment with the surface conjugating moiety.

320. The method of claim 313, wherein the surface conjugating moiety is a reactive moiety selected from an amine, azide, alkyne, nitron, alkene, tetrazine, and tetrazole.

321. The method of claim 313, wherein the polypeptide conjugating moiety is a reactive moiety selected from an amine, azide, alkyne, nitron, alkene, tetrazine, and tetrazole.

322. The method of claim 284, wherein the method comprises contacting the single polypeptide

molecule with a single composition comprising the one or more terminal amino acid recognition molecules and the one or more cleaving reagents.
